你说：

这是之前咱们讨论的技能组合https://chatgpt.com/c/6ad0c31f-1f94-83ee-b4c2-862bfc89ec03，现在你需要调研所有相关的文献，并进行分析，近两年的，我想发TRO或者PAMI，science robotics，给出建议和可操作的方法，要优先开源的。给出文件链接，可以供我搜索，总结要生成md文件，我可以查看

思考了 15m 46s >

ChatGPT 说：

机器人技能组合与长时序操作：2024–2026 年文献调研及投稿路线

这次调研以 **2024 年 10 月至 2026 年 10 月** 为主要范围，重点研究：

机器人如何从少量独立学到的操作技能出发，在没有见过的新任务中，判断技能能否使用、如何组合、如何处理技能之间的状态不匹配，以及如何在失败后继续完成任务。

研究范围限定为机械臂操作，而不是人形机器人全身运动。优先考虑单卡 RTX 4090、无触觉传感器、MuJoCo/ManiSkill 仿真和开源实现，不把频繁调用大模型 API 作为核心依赖。

一、初步结论：技能组合已经是一个竞争激烈的方向

目前发现三篇与你的原始想法高度重合的代表性论文：

论文	发表	已解决的关键问题
DeCo ↗	RA-L 2026	原子技能提取、动态调度、空间感知的技能衔接
SPIN ↗	CoRL 2025	技能适用性检查、Skill-RRT 组合规划、连接策略
Compose by Focus ↗	ICRA 2026	利用场景图学习原子技能，提高新场景下的组合泛化

这些论文直接覆盖了我们之前讨论的若干思路。 arXiv +2

因此，**单纯提出“学习几个技能，再设计一个调度器实现长序列任务”，已经不足以形成有竞争力的 TRO、TPAMI 或 Science Robotics 创新点。**

尤其需要警惕两个方向：一是简单地学习前置条件、判断技能兼容性；二是在两个技能之间添加过渡策略。这些都已经有较直接的先例。

真正值得继续挖掘的是一个更深的问题：

独立学到的技能，即使各自成功率很高，在新环境中组合时仍可能失败。能否通过少量有针对性的物理交互，发现尚未知晓的技能组合失效边界，并把发现的规律迁移到未测试过的任务组合？

这是我目前优先考虑的研究主线，但它仍需与最新论文进行严格的新颖性对照，不能现在就认定它一定达到了顶刊创新水平。

二、完整调研报告已整理完成

机器人技能组合：2024–2026 文献调研与顶刊选题论证
Markdown · 约 55 KB · 715 行 · 包含论文链接、GitHub 仓库、创新性分析、算法设计和实验规划

下载完整 Markdown 调研报告 ↗

报告包含约 36 项相关研究及基础平台资源，并对最可能与你的选题发生创新冲突的 12 篇论文进行了重点分析。它是主题式深度调研，而非宣称穷尽所有相关文献。

三、进一步排重后的关键判断

需要特别注意 2026 年几项新工作：BATON ↗ 已关注相邻技能之间的影响和交接恢复；Recursia ↗ 涉及技能接口修复和经验复用；Deliberate Practice ↗ 已研究有限交互预算下的技能练习。

因此，建议避免把"技能契约""动态过渡""失败重试"或"主动试错"单独作为论文核心创新。

更值得验证的科学问题

机器人能否通过少量有针对性的物理交互，发现独立技能之间可迁移的失效规律，从而完成从未训练过的长序列任务？

这里的关键不是训练更强的单技能，而是解释并预测：为什么两个分别成功的技能，连续执行时却会失败？

四、最建议先使用的开源平台

资源	链接	建议用途
ManiSkill 3	GitHub ↗	单卡 4090 上构造和验证技能交接问题
RoboHiMan	GitHub ↗	长序列组合任务和外部评测
Compose by Focus	代码 ↗	直接相关的开源方法对照
LIBERO-Pro	GitHub ↗	新环境、新组合的泛化验证
ManipGen	GitHub ↗	单技能训练与复用参考

部分仓库虽公开，但不代表提供完整训练数据、权重或一键复现环境；报告中已单独标注这类风险。

五、我的投稿建议

优先以 T-RO 为目标，而不是同时围绕三个期刊设计论文。

最值得先做的是一个小规模但严格的验证：固定 `OpenDrawer`、`PickFromDrawer`、`PlaceInDrawer`、`CloseDrawer` 等单技能，系统改变前驱技能的执行终态，统计后继技能失败的原因。然后检验这些失败关系能否通过少量试探迁移到新对象、新技能组合。

如果简单的几何规则或者同等数据量的 MLP 就能解决这一问题，那么这条创新路线需要及时调整，而不是继续增加网络模块。

只有验证出具有普遍性的组合规律，并在公开基准和真实机器人上证明方法有效，才适合进一步组织 T-RO 级别的完整工作。TPAMI 更适合能够提炼成一般性组合表示学习方法的成果；Science Robotics 则需要更强的真实机器人能力突破及实证支持。

建议先阅读报告第 3、6、7、8、10 节。 这几节分别回答了最重要的五个问题：别人已经做了什么、哪些创新点不能再用、还能研究什么、怎么证明不是简单改进，以及应该先做哪个实验。

机器人技能组合_近两年文献调研与顶刊选题_2024-2026....
文档

你说：

我该用什么baseline呢，怎么开展

思考了 6m 37s

ChatGPT 说：

一、我的决定：先用 Compose by Focus，暂时不碰大型 VLA

结合你希望发 T-RO、只有一张 RTX 4090、没有触觉传感器、不想从头训练大型模型的条件，我建议采用：

主论文基线：Compose by Focus（ICRA 2026）

主实验环境：其官方 Skill Composition Benchmark（基于 ManiSkill2）

第二阶段外部验证：DeCo / RoboHiMan

这里需要区分三个东西：**实验平台、已有论文方法、你自己的改进模块。**不能只下载一个仿真器就称为复现了 baseline。

层次	选择	作用
主 baseline	Compose by Focus ↗	现有原子技能学习与组合方法
代码	作者代码 ↗	复现原子技能策略
环境	作者 Benchmark ↗	原子任务、组合任务、运动规划示范
扩展对照	RoboHiMan ↗	更长任务及不同规划架构的验证

为什么选择它?

它已经有原子技能环境和长序列组合环境，可以直接研究**技能独立训练后，组合执行为什么失败**，不需要先构建一套大型数据集。其 Benchmark 包含 Cube Out and In、Sort by Color、Blocks Stacking Game、Tools Usage 和 Obstacle Avoidance 五类任务，并公开了运动规划脚本。 GitHub +1

但要明确一个复现风险：作者策略仓库目前只有简要的训练和评估入口，完整安装说明、数据准备和预训练权重信息并不充分。我不能保证它可以直接一键复现。 GitHub

因此，我建议**先跑通官方 Benchmark 的脚本技能，确认能执行组合任务，再决定是否训练作者的 Diffusion Policy**。这样即使作者模型不好复现，也不会把整个研究卡住。

还有一个重要细节：不要直接把 ManiSkill3 的 `PickCube-v1` 和 `OpenCabinetDrawer-v1` 串接。前者可用 Panda，后者官方任务限定 Fetch，而且两者是不同的独立环境，不能天然共享连续物理状态。要研究抽屉取物，必须使用同一机器人、同一场景的完整组合任务。

ManiSkill 3.... +1

二、你的第一批实验应该怎么做?

不要先训练 GNN、Diffusion Policy 或 Transformer。先验证一个基本假设:

> 单独执行成功的技能，在前驱技能执行后是否会出现新的、可重复的失败？这些失败是否包含可以跨技能组合迁移的规律?

第一步：让两个技能连续执行

例如从官方 Benchmark 中找出能在同一场景串接的两个技能:



在技能 A 达成目标的前提下，改变它留下的物体位置、末端退场位置等终态，再执行固定的技能 B。

需要确认：**A 成功，但 B 因为 A 留下的物理状态而失败。**

第一轮只需要输出三类结果：技能独立成功率、真实连续执行成功率，以及失败随物体几何关系和操作参数变化的分布。

第二步：实现四种简单对照

Baseline	具体实现	目的
B0：Direct Chaining	直接按任务顺序调用技能	最基础对照
B1：Rule-based	IK、碰撞检测、已知几何约束	排除简单规则已足够
B2：Pair-MLP	输入技能 ID、状态、参数，预测组合成功率	排除简单监督学习
B3：Shared GNN/MLP	多技能对共享物体关系特征	检验结构迁移的价值

暂时不用完全复现 SPIN。它的 Skill-RRT、适用性检查和连接策略必须在论文中讨论，但目前没有核实到足够完整的官方代码，不能把它设为第一阶段的实施依赖。 Proceedings of...

第三步：通过两个判断门槛再做创新算法

- **G1：物理交接问题确实存在**
  单技能表现良好，但链式执行存在显著失败，而且不只是路径长度或随机执行误差造成。
- **G2：简单方法解决不了跨组合泛化**
  规则和普通 MLP 在新物体、新物理关系或未见技能对上仍有可量化的缺口。

只有两个门槛都通过，我才建议开始做"结构化失效规律学习 + 有针对性的物理试探 + 后继约束驱动的技能参数选择"。

三、第一周的实际操作

先在 Linux 服务器上下载：

```
mkdir -p ~/skill_compose
cd ~/skill_compose

git clone https://github.com/han20192019/skill-composition-benchmark.git
```

```
git clone https://github.com/han20192019/skill-composition-code.git

cd skill-composition-benchmark

git rev-parse HEAD
sed -n '1,200p' README.md
sed -n '1,200p' scripts/run_all.sh
```

先按作者说明核查环境依赖，再执行 `scripts/run_all.sh`。由于 Benchmark 是 ManiSkill2，而主方法评估脚本包含自定义模块和权重路径，两者都需要环境检查，不能假定简单执行就会成功。

第一周的目标不是训练出模型，而是**成功运行一条可以保存视频、状态和成功标记的组合轨迹。**

四、我给你制定了实施文档

技能组合 Baseline 选择与分阶段实验实施方案

Markdown · 包含代码入口、阶段 G0–G4、实验数据字段、对照方法、消融和止损标准

下载实施方案 Markdown ↗

最后，我认为这项工作能否冲击 T-RO，决定因素不是训练出多复杂的模型，而是能否证明：**从少量技能组合的物理失败中学到的知识，能够帮助机器人执行从未试过的其他技能组合。**

所以先复现 Benchmark，拿到真实的交接失败数据，再设计算法。这是目前风险最低、逻辑最清晰的研究起点。

技能组合_Baseline选择与分阶段实验实施方案_2026-10-...
文档

你说：

我是想要明确的路线，一定可以发TRO或者PAMI的

思考了 6m 44s

ChatGPT 说：

TRO技能组合_锁定研究路线与实验清单_2026-10-26.md
文档

你说：

这个怎么安装环境呀：mkdir -p ~/skill_future_viability && cd ~/skill_future_viability
git clone https://github.com/han20192019/skill-composition-benchmark.git
git clone https://github.com/han20192019/skill-composition-code.git

思考了 2m 51s >

ChatGPT 说：

你说：

hub, mplib, arm-pytorch-utilities, pytorch_kinematics, mani_skill
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.4
    Uninstalling numpy-1.26.4:
      Successfully uninstalled numpy-1.26.4
Successfully installed GitPython-3.2.1 IPython-8.39.0 absl-py-2.5.1 anyio-4.15.1 arm-pytorch-utilities-0.5.0 asttokens-3.0.2 certifi-2026.10.23 charset_normalizer-3.5.2 click-8.5.0 cloudpickle-3.1.2 contourpy-1.3.2 cycler-0.12.1 dacite-1.9.2 decorator-5.3.1 docstring-parser-0.18.0 exceptiongroup-1.3.1 executing-2.3.0 farama-notifications-0.0.6 fast_kinematics-0.2.2 fonttools-4.65.0 gitdb-4.0.12 gymnasium-1.4.0 h11-0.16.0 h5py-3.16.0 hf-xet-1.7.0 httpcore2-2.14.0 httpx2-2.14.0 huggingface_hub-2.3.0 idna-3.20 imageio-2.38.0 imageio-ffmpeg-0.6.0 importlib_resources-7.1.0 jedi-0.20.1 kiwisolver-1.5.1 lxml-6.1.3 mani_skill-3.0.0b22 matplotlib-3.10.9 matplotlib-inline-0.2.2 mplib-0.1.1 numpy-2.2.6 nvidia-ml-py-13.615.71 opencv-python-5.0.0.93 parso-0.8.7 pexpect-4.9.0 prompt_toolkit-3.0.53 psutil-7.2.2 ptyprocess-0.7.0 pure-eval-0.2.4 pygments-2.21.0 pyparsing-3.3.3 pyperclip-1.11.0 python-dateutil-3.0.0 pytorch-seed-0.2.0 pytorch_kinematics-0.7.6 pyyaml-6.0.3 requests-2.34.2 sapien-3.0.3 scipy-1.15.3 six-1.17.0 smmap-5.0.3 stack_data-0.6.3 tabulate-0.10.0 tomli-2.5.0 toppra-0.6.10 tqdm-4.70.1 traitlets-5.16.1 transforms3d-0.4.2 trimesh-5.2.0 truststore-0.10.5 typeguard-4.6.0 tyro-1.0.16 tzdata-2026.5 urllib3-2.8.0 wcwidth-0.9.4
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ python - <<'PY'
> import mani_skill

```
> import sapien
> import gymnasium
>
> print("ManiSkill:", mani_skill.__version__)
> print("ManiSkill path:", mani_skill.__file__)
> print("SAPIEN:", sapien.__version__)
> print("Gymnasium:", gymnasium.__version__)
> PY
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/home/users/lemon/skill_future_viability/skill-
composition-benchmark/mani_skill/__init__.py", line 70, in
<module>
    from .envs import *
  File "/home/users/lemon/skill_future_viability/skill-
composition-benchmark/mani_skill/envs/__init__.py", line 2, in
<module>
    from .tasks import *
  File "/home/users/lemon/skill_future_viability/skill-
composition-benchmark/mani_skill/envs/tasks/__init__.py",
line 1, in <module>
    from .tabletop_skill_compose import *
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/mani_skill/envs/tasks/tabletop_skill_compose/__ini
t__.py", line 2, in <module>
    from .cube_out_and_in_compose import
Cube_out_and_in_compose_Env
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/mani_skill/envs/tasks/tabletop_skill_compose/cub
e_out_and_in_compose.py", line 21, in <module>
    import clip
ModuleNotFoundError: No module named 'clip'
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$ pip install numpy==1.26.4
Looking in indexes:
https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
Collecting numpy==1.26.4
  Using cached numpy-1.26.4-cp310-cp310-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.2 MB)
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 2.2.6
    Uninstalling numpy-2.2.6:
```

```
    Successfully uninstalled numpy-2.2.6

ERROR: pip's dependency resolver does not currently take into
account all the packages that are installed. This behaviour is
the source of the following dependency conflicts.
opencv-python 5.0.0.93 requires numpy>=2; python_version
>= "3.9", but you have numpy 1.26.4 which is incompatible.
Successfully installed numpy-1.26.4
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
```

思考了 1m 58s >

ChatGPT 说：

你说：

```
on -m mani_skill.examples.motionplanning.panda.run \
>     -e cube_out_and_in_skill_1 \
>     -o none \
>     -n 1 \
>     -b cpu
100%|
████████████████████████████████████████████████|
338M/338M [00:50<00:00, 7.00MiB/s]
Traceback (most recent call last):
  File
"/home/users/lemon/miniconda3/envs/skillcompose/lib/pyth
on3.10/runpy.py", line 196, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File
"/home/users/lemon/miniconda3/envs/skillcompose/lib/pyth
on3.10/runpy.py", line 86, in _run_code
    exec(code, run_globals)
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/mani_skill/examples/motionplanning/panda/run.p
y", line 175, in <module>
    mp.set_start_method("spawn")
  File
"/home/users/lemon/miniconda3/envs/skillcompose/lib/pyth
on3.10/multiprocessing/context.py", line 247, in
set_start_method
```

```
    raise RuntimeError('context has already been set')
RuntimeError: context has already been set
```

ChatGPT 说：

你说：

```
>     -b cpu
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
Xlib:  extension "GLX" missing on display "localhost:10.0".
段错误 (核心已转储)
```

思考了 1m 59s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ python -m mani_skill.examples.motionplanning.panda.run \
>     -e cube_out_and_in_skill_1 \
>     -o none \
>     -n 1 \
>     -b cpu

2026-10-26 04:48:49,421 - mani_skill  - WARNING - No initial pose set for actor builder of cube_in, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 04:48:49,422 - mani_skill  - WARNING - No initial
```

pose set for actor builder of bin, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
Motion Planning Running on cube_out_and_in_skill_1
proc_id: 0: 100%|█| 1/1 [00:00<00:00, 5.64it/s, success_rate=0, failed_motion_plan_r
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$

思考了 2m 5s >

ChatGPT 说：

你说：

e:
>   print("Planner result:", res)
>
> env.close()
> PY

2026-10-26 04:52:45,084 - mani_skill - WARNING - The chosen render mode is "rgb_array", but selected rendering device "None" does not support rendering
2026-10-26 04:52:45,350 - mani_skill - WARNING - No initial pose set for actor builder of cube_in, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 04:52:45,684 - mani_skill - WARNING - No initial pose set for actor builder of bin, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
Initial cube: [0.04962566 0.05364437 0.025   ]
Initial bin: [0.04962566 0.05364437 0.0025   ]

Final cube: [0.04877913 0.1568082 0.01999993]
Final bin: [0.04962566 0.05364437 0.0025   ]
Cube displacement: 0.1032884
XY distance from bin: 0.1031673

```
Object static: tensor([True])
Object grasped: tensor([False])
Elapsed steps: tensor([105], dtype=torch.int32)
Reported success: tensor([False])
```

思考了 1m 17s >

ChatGPT 说：

你说：

```
proc_id: 0: 100%|█| 1/1 [00:00<00:00,  4.48it/s, success_rate=1,
failed_motion_plan_r
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$ python -m
mani_skill.examples.motionplanning.panda.run \
>     -e cube_out_and_in_skill_1 \
>     -o none \
>     -n 20 \
>     -b cpu
2026-10-26 04:58:09,707 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_in, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 04:58:09,707 - mani_skill  - WARNING - No initial
pose set for actor builder of bin, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
Motion Planning Running on cube_out_and_in_skill_1
proc_id: 0: 100%|█| 20/20 [00:02<00:00,  6.81it/s,
success_rate=1, failed_motion_plan
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
```

ChatGPT 说：

你说：

```
Planner returned -1: False

Object: cube_out
Initial: [-0.07518717  0.05364437  0.012     ]
```

Final: [0.00771538 -0.07738492 0.01699992]
Displacement: 0.15513381
XY distance to bin: 0.0039565237
Z relative to bin: 0.014499918
Static: tensor([True])
Grasped: tensor([False])

Elapsed steps: tensor([132], dtype=torch.int32)
Original reported success: tensor([False])

==========
==========
Task: cube_out_and_in_compose
2026-10-26 05:32:26,394 - mani_skill - WARNING - The chosen render mode is "rgb_array", but selected rendering device "None" does not support rendering
2026-10-26 05:32:26,697 - mani_skill - WARNING - No initial pose set for actor builder of cube_in, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 05:32:27,010 - mani_skill - WARNING - No initial pose set for actor builder of cube_out, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 05:32:27,324 - mani_skill - WARNING - No initial pose set for actor builder of bin, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
Planner returned -1: False

Object: cube_in
Initial: [0.00884774 -0.0735939   0.025   ]
Final: [0.00928336 0.02711186 0.01999991]
Displacement: 0.10083076
XY distance to bin: 0.10070671
Z relative to bin: 0.017499909
Static: tensor([True])
Grasped: tensor([False])

Object: cube_out
Initial: [-0.07518717 0.05364437 0.012   ]
Final: [0.00538138 -0.08157449 0.01699991]

Displacement: 0.15748152
XY distance to bin: 0.0087008905
Z relative to bin: 0.014499914
Static: tensor([True])
Grasped: tensor([False])

Elapsed steps: tensor([195], dtype=torch.int32)
Original reported success: tensor([False])

思考了 4m 43s >

ChatGPT 说：

你说：

你直接生成代码，并且保留视频，我看看场景

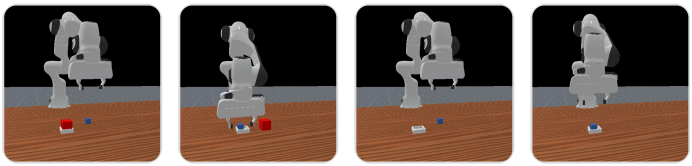
思考了 5m 47s >

ChatGPT 说：

audit_cube_video.py
代码

README.md
文档

你说：



cube_out_and_in...
文档

cube_out_and_in...
文档

Audit Results

Preview Seed000 Results



思考了 3m 18s >

ChatGPT 说：

你说：

这样你的TRO思路是什么，为什么现在做这些toy实验

思考了 1m 42s >

ChatGPT 说：

你说：

那我需要对比哪些算法，选择哪个当baseline

思考了 2m 50s >

ChatGPT 说：

你说：

那线怎么开始

思考了 5m 2s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ python -m mani_skill.examples.motionplanning.panda.run \
>     -e tools_usage_compose \
>     -o none \
>     -n 3 \
>     -b cpu
2026-10-26 21:47:39,841 - mani_skill  - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-26 21:47:39,867 - mani_skill  - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:47:39,868 - mani_skill  - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:47:39,868 - mani_skill  - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:47:39,868 - mani_skill  - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
Motion Planning Running on tools_usage_compose
proc_id: 0:   0%|                                                  | 0/3 [00:00<?, ?it/s]tensor([0.2849])
Cannot find valid solution because of an error in motion planning solution: 'RecordEpisode' object has no attribute 'reset_agent'
```

```
/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/site-packages/numpy/core/fromnumeric.py:3504: RuntimeWarning: Mean of empty slice.
  return _methods._mean(a, axis=axis, dtype=dtype,
/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/site-packages/numpy/core/_methods.py:129: RuntimeWarning: invalid value encountered in scalar divide
  ret = ret.dtype.type(ret / rcount)
Traceback (most recent call last):
  File "/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/runpy.py", line 196, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/runpy.py", line 86, in _run_code
    exec(code, run_globals)
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/panda/run.py", line 177, in <module>
    main(parse_args())
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/panda/run.py", line 171, in main
    _main(args)
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/panda/run.py", line 140, in _main
    max_episode_length=np.max(solution_episode_lengths),
  File "/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/site-packages/numpy/core/fromnumeric.py", line 2810, in max
    return _wrapreduction(a, np.maximum, 'max', axis, None, out,
  File "/home/users/lemon/miniconda3/envs/skillcompose/lib/python3.10/site-packages/numpy/core/fromnumeric.py", line 88, in _wrapreduction
    return ufunc.reduce(obj, axis, dtype, out, **passkwargs)
ValueError: zero-size array to reduction operation maximum which has no identity
```

proc_id: 0: 33%|████████████▌ | 1/3 [00:00<00:00, 11.09it/s]

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ python -m mani_skill.examples.motionplanning.panda.run \
>   -e tools_usage_compose \
>   -o pointcloud \
>   -n 1 \
>   -b cpu \
>   --save-video \
>   --record-dir demos/tools_test
2026-10-26 21:51:32,206 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-26 21:51:32,231 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:51:32,232 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:51:32,232 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 21:51:32,232 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/utils/common.py:191: UserWarning: Creating a tensor from a list of numpy.ndarrays is extremely

slow. Please consider converting the list to a single numpy.ndarray with numpy.array() before converting to a tensor. (Triggered internally at ../torch/csrc/utils/tensor_new.cpp:278.)
  return torch.tensor(array).cpu()
Motion Planning Running on tools_usage_compose
proc_id: 0:   0%| | 0/1 [00:00<?, ?it/s]tensor([0.2849])
proc_id: 0: 100%|█| 1/1 [01:03<00:00, 63.91s/it, success_rate=1, failed_motion_pla

ChatGPT 说：

你说：

视频展示的是红色的直角去勾蓝色方块，然后红色的长方体去推绿色方块，这是那个json：{
  "env_info": {
    "env_id": "tools_usage_compose",
    "env_kwargs": {
      "obs_mode": "pointcloud",
      "control_mode": "pd_joint_pos",
      "render_mode": "rgb_array",
      "sensor_configs": {
        "shader_pack": "default"
      },
      "human_render_camera_configs": {
        "shader_pack": "default"
      },
      "viewer_camera_configs": {
        "shader_pack": "default"
      },
      "sim_backend": "cpu"
    },
    "max_episode_steps": 100
  },
  "commit_info": {
    "commit_id":
"66a3c810552648e8f420d094ed03664341c43671",
    "branch": "main"
  },
  "episodes": [
    {
      "episode_id": 0,
      "episode_seed": 0,

```
    "control_mode": "pd_joint_pos",
    "elapsed_steps": 593,
    "reset_kwargs": {
      "options": null,
      "seed": 0
    },
    "success": true
  }
 ],
 "source_type": "motionplanning",
 "source_desc": "official motion planning solution from
ManiSkill contributors"
}
```


思考了 2m 28s >

ChatGPT 说：

你说：

Trajectory length: 593
Truncated steps: 494
First truncated step: 100

思考了 1m 19s >

ChatGPT 说：

你说：

Trajectory length: 593
Truncated count: 0
Terminated count: 361
Expected: truncated count = 0

思考了 2m 44s >

ChatGPT 说：

你说：

2026-10-26 23:03:15,509 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with

other objects at their initial poses.
2026-10-26 23:03:15,766 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:03:16,086 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:03:16,447 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:03:16,777 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
tensor([0.2849])

=== Blue cube / Hook ===
Initial: [0.11459383 -0.30561447 0.02   ]
Final: [-0.19113138 -0.30346248 0.01999989]
Displacement: [-3.05725217e-01 2.15199590e-03 -1.09896064e-07]

=== Green cube / Push ===
Initial: [0.1505845 0.28488934 0.02   ]
Final: [0.29171506 0.28662688 0.01999992]
Displacement: [1.4113055e-01 1.7375350e-03 -8.3819032e-08]

Total steps: 593

思考了 2m 44s >

ChatGPT 说：

tool_skills_stage0.py
代码

你说：

怎么看结果

ChatGPT 说：

你说：

```
========== 四个技能执行结果 ==========
S1_grasp_l_tool       Success=True   Steps=114   Error=
S2_hook_and_pull      Success=True   Steps=219   Error=
S3_grasp_i_tool       Success=True   Steps=83    Error=
S4_push               Success=True   Steps=177   Error=

========== 物体最终位移 ==========
Blue cube:
  Initial: [ 0.1146 -0.3056  0.02  ]
  Final:   [-0.1911 -0.3035  0.02  ]
  Delta:   [-0.3057  0.0022 -0.    ]
Green cube:
  Initial: [0.1506 0.2849 0.02  ]
  Final:   [0.2917 0.2866 0.02  ]
  Delta:   [ 0.1411  0.0017 -0.    ]

Total steps: 593
Error: None
```

ChatGPT 说：

你说：

粘贴的文本 (1)(20...
文档

没有找到你说的东西啊：

思考了 25s >

ChatGPT 说：

你说：

完整代码，并给注释，方便我按照你的建议改

思考了 5m 15s >

ChatGPT 说：

tool_skills_stage1.py
代码

你说：

第一步：总结：
seed=0 case=release steps=593 planned=True chain_proxy=True blue=-0.305725 green=0.141131 error=None

已保存: demos/tool_stage1/summary.csv； 第二步：总结：
seed=0 case=release steps=593 planned=True chain_proxy=True blue=-0.305725 green=0.141131 error=None
seed=0 case=home steps=695 planned=True chain_proxy=True blue=-0.30839 green=0.141271 error=None
seed=0 case=left steps=702 planned=True chain_proxy=False blue=-0.30839 green=0.0 error=None
seed=0 case=near_i steps=677 planned=True chain_proxy=False blue=-0.30839 green=0.0 error=None

已保存: demos/tool_stage1/summary.csv，执行结果：near_i Steps: 677 Planner: True Physical: False Blue: -0.30839 Green: 0.0 Error:

思考了 1m 5s >

ChatGPT 说：

你说：

==========
==========
CASE: release
S1_grasp_l_tool Planner=True Physical=True Steps=114
S2_hook_and_pull Planner=True Physical=True Steps=219

```
S3_grasp_i_tool       Planner=True   Physical=True   Steps=83
S4_push               Planner=True   Physical=True   Steps=177
S2 parking error (m):
S3 I-tool grasped: True
I-tool movement during S2: [-0. -0. -0.]
I-tool movement during S3: [-0.0044  0.0006  0.1947]
Green cube movement during S4: [0.1411 0.0017 0.    ]
L-tool final position: [-0.3205 -0.3642  0.025 ]

========================================
==============
CASE: home
S1_grasp_l_tool       Planner=True   Physical=True   Steps=114
S2_hook_and_pull      Planner=True   Physical=True
Steps=309
S3_grasp_i_tool       Planner=True   Physical=True   Steps=89
S4_push               Planner=True   Physical=True   Steps=183
S2 parking error (m): 0.007265849751064404
S3 I-tool grasped: True
I-tool movement during S2: [-0. -0. -0.]
I-tool movement during S3: [-0.0044  0.0011  0.1947]
Green cube movement during S4: [ 0.1413  0.0019 -0.    ]
L-tool final position: [-0.2064 -0.1197  0.025 ]

========================================
==============
CASE: left
S1_grasp_l_tool       Planner=True   Physical=True   Steps=114
S2_hook_and_pull      Planner=True   Physical=True
Steps=314
S3_grasp_i_tool       Planner=True   Physical=False  Steps=88
S4_push               Planner=True   Physical=False  Steps=186
S2 parking error (m): 0.0073455699382352
S3 I-tool grasped: False
I-tool movement during S2: [-0. -0. -0.]
I-tool movement during S3: [-0.0002 -0.0017 -0.    ]
Green cube movement during S4: [ 0. -0.  0.]
L-tool final position: [-0.2663 -0.1192  0.025 ]

========================================
==============
CASE: near_i
S1_grasp_l_tool       Planner=True   Physical=True   Steps=114
S2_hook_and_pull      Planner=True   Physical=True
Steps=293
```

S3_grasp_i_tool Planner=True Physical=False Steps=87
S4_push Planner=True Physical=False Steps=183
S2 parking error (m): 0.015177612345113105
S3 I-tool grasped: False
I-tool movement during S2: [0.0154 0.0125 -0.]
I-tool movement during S3: [0.012 0.006 0.]
Green cube movement during S4: [0. -0. 0.]
L-tool final position: [-0.1025 -0.0171 0.0236]

思考了 1m 20s >

ChatGPT 说：

你说：

tensor([0.3169])
tensor([0.2849])
seed000_left S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000
L工具当前位置: [-0.3202 -0.3644 0.0253]
目标停放XY: [-0.2593 -0.1212]
seed000_left S2_hook_and_pull: steps=314 planner=True physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769584 0.02791669 0.02499983]
[S3] TCP 初始位置: [-0.00624986 -0.00810818 0.17095667]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499983]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4368693e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.07347381 0.02231084 0.05867423]
[S3] 到达后直杆: [-0.11769588 0.02791611 0.02499985]
[S3] 闭合夹爪后是否抓住: False
[S3] 抬起后直杆: [-0.11785556 0.02619296 0.02499981]
[S3] 直杆位移: [-1.59718096e-04 -1.72373466e-03 -2.60770321e-08]
[S3] 抬起后是否抓住: False
seed000_left S3_grasp_i_tool: steps=88 planner=True physical=False blue_dx=-0.308 green_dx=+0.000
seed000_left S4_push: steps=186 planner=True physical=False blue_dx=-0.308 green_dx=+0.000

==========
==========
总结：

```
  seed=0 case=left     steps=702   planned=True
chain_proxy=False blue=-0.30839 green=0.0 error=None

已保存: demos/tool_stage1/summary.csv
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$ python tool_skills_stage1.py \
>     --seed 0 \
>     --parking-case near_i


==========================================
==================================
seed=0 parking_case=near_i
2026-10-26 23:37:07,648 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-26 23:37:07,914 - mani_skill  - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:37:08,226 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_2, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:37:08,551 - mani_skill  - WARNING - No initial
pose set for actor builder of l_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-26 23:37:08,875 - mani_skill  - WARNING - No initial
pose set for actor builder of i_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_near_i S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644  0.0253]
```

```
  目标停放XY: [-0.1177 -0.0171]
seed000_near_i S2_hook_and_pull: steps=293 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.10226856  0.0403953   0.02499445]
[S3] TCP 初始位置: [-0.00419353 -0.01503149  0.17240778]
[S3] 目标抓取位置: [-0.08226925  0.04022978  0.02499445]
[S3] 目标抓取四元数: [ 0.          0.9999915  -0.00413808  0.        ]
[S3] 到达后 TCP: [-0.05307923  0.03818686  0.06692442]
[S3] 到达后直杆: [-0.09592304  0.03824171  0.02503425]
[S3] 闭合夹爪后是否抓住: False
[S3] 抬起后直杆: [-0.09026755  0.0463926   0.02499603]
[S3] 直杆位移: [1.20010152e-02 5.99730387e-03 1.57393515e-
06]
[S3] 抬起后是否抓住: False
seed000_near_i S3_grasp_i_tool: steps=87 planner=True
physical=False blue_dx=-0.308 green_dx=+0.000
seed000_near_i S4_push: steps=183 planner=True
physical=False blue_dx=-0.308 green_dx=+0.000

==========================================
==================================
总结:
  seed=0 case=near_i   steps=677   planned=True
chain_proxy=False blue=-0.30839 green=0.0 error=None

已保存: demos/tool_stage1/summary.csv
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composit
```

思考了 1m 57s >

已分析

ChatGPT 说:

‹/›

你说:

看视频好像是放那个L到绿色的长方体了:

ChatGPT 说：

你说：

在哪里修改，怎么执行

思考了 1m 33s >

ChatGPT 说：

你说：

```
tensor([0.3169])
tensor([0.2849])
seed000_near_i S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644  0.0253]
  目标停放XY: [-0.1177 -0.0171]

========== S2 干扰诊断 ==========
直杆工具位移: 19.84 mm
绿色方块位移: 0.00 mm
[注意] S2 明显扰动了直杆工具
================================

seed000_near_i S2_hook_and_pull: steps=293 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.10226856  0.0403953   0.02499445]
[S3] TCP 初始位置: [-0.00419353 -0.01503149  0.17240778]
[S3] 目标抓取位置: [-0.08226925  0.04022978  0.02499445]
[S3] 目标抓取四元数: [ 0.          0.9999915  -0.00413808  0.        ]
[S3] 到达后 TCP: [-0.05307923  0.03818686  0.06692442]
[S3] 到达后直杆: [-0.09592304  0.03824171  0.02503425]
[S3] 闭合夹爪后是否抓住: False
[S3] 抬起后直杆: [-0.09026755  0.0463926   0.02499603]
[S3] 直杆位移: [1.20010152e-02 5.99730387e-03 1.57393515e-
06]
[S3] 抬起后是否抓住: False
seed000_near_i S3_grasp_i_tool: steps=87 planner=True
physical=False blue_dx=-0.308 green_dx=+0.000
seed000_near_i S4_push: steps=183 planner=True
physical=False blue_dx=-0.308 green_dx=+0.000
```

```
==========================================
==================================
总结:
  seed=0 case=near_i   steps=677   planned=True
chain_proxy=False blue=-0.30839 green=0.0 error=None

已保存: demos/tool_stage1_diag/summary.csv
```

ChatGPT 说:

你说:

```
修改摆放位置后正常，下一步呢：tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644  0.0253]
  目标停放XY: [-0.2  -0.12]

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585  0.02791669  0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931  0.17212337]
[S3] 目标抓取位置: [-0.09769584  0.02791655  0.02499987]
[S3] 目标抓取四元数: [ 0.0000000e+00  1.0000000e+00
-3.4568034e-06  0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359  0.02779171  0.02592734]
[S3] 到达后直杆: [-0.11769594  0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586  0.02900716  0.21965735]
[S3] 直杆位移: [-0.00442001  0.00109047  0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000
seed000_custom S4_push: steps=183 planner=True
physical=True blue_dx=-0.308 green_dx=+0.141
```

```
==========================================
==================================
总结:
  seed=0 case=custom   steps=695   planned=True
chain_proxy=True blue=-0.30839 green=0.141269 error=None
```

思考了 4m 36s >

ChatGPT 说:

probe_delayed_dependency.py
代码

你说:

```
p00_safe S2: True S3: True S4: True 结果: all_four_passed
p01 S2: False S3: True S4: True 结果: S1_or_S2_failed
p02 S2: False S3: False S4: False 结果: S1_or_S2_failed
p03 S2: False S3: True S4: False 结果: S1_or_S2_failed
p04 S2: False S3: False S4: False 结果: S1_or_S2_failed
p05 S2: False S3: False S4: False 结果: S1_or_S2_failed
```

思考了 2m 32s >

ChatGPT 说:

你说:

```
Traceback (most recent call last):
  File "<stdin>", line 20, in <module>
KeyError: 'parking_target_xy'
```

思考了 43s >

ChatGPT 说:

你说:

```
==========================================
===============
CASE: p01
S2 planner: True
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False
```

目标停放 XY: [0.25 0.2]
实际停放 XY: [0.1447 0.1756]
停放误差: 108.04 mm
>>> 停放误差超出35mm阈值
S3 physical: True
S4 physical: True
绿色方块最终位移: 0.141706 m
S2 error:

==========
==========
CASE: p02
S2 planner: False
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: True
目标停放 XY: [0.27 0.24]
实际停放 XY: [-0.3253 -0.3607]
停放误差: 845.71 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/parking transit

==========
==========
CASE: p03
S2 planner: True
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False
目标停放 XY: [0.31 0.285]
实际停放 XY: [0.1306 0.2289]
停放误差: 187.94 mm
>>> 停放误差超出35mm阈值
S3 physical: True
S4 physical: False
绿色方块最终位移: -0.00809 m
S2 error:

==========
==========
CASE: p04
S2 planner: False
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False

目标停放 XY: [0.35 0.285]
实际停放 XY: [0.134 0.2198]
停放误差: 225.66 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/TCP back home

==========
==========
CASE: p05
S2 planner: False
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: True
目标停放 XY: [0.31 0.34]
实际停放 XY: [-0.3253 -0.3607]
停放误差: 945.82 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/parking transit

思考了 2m 14s >

ChatGPT 说：

你说：

==========
==========
CASE: p01
S2 planner: True
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False
目标停放 XY: [0.25 0.2]
实际停放 XY: [0.1447 0.1756]
停放误差: 108.04 mm
>>> 停放误差超出35mm阈值
S3 physical: True
S4 physical: True
绿色方块最终位移: 0.141706 m
S2 error:

==========
==========
CASE: p02
S2 planner: False

S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: True
目标停放 XY: [0.27 0.24]
实际停放 XY: [-0.3253 -0.3607]
停放误差: 845.71 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/parking transit

==========
==========
CASE: p03
S2 planner: True
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False
目标停放 XY: [0.31 0.285]
实际停放 XY: [0.1306 0.2289]
停放误差: 187.94 mm
>>> 停放误差超出35mm阈值
S3 physical: True
S4 physical: False
绿色方块最终位移: -0.00809 m
S2 error:

==========
==========
CASE: p04
S2 planner: False
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: False
目标停放 XY: [0.35 0.285]
实际停放 XY: [0.134 0.2198]
停放误差: 225.66 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/TCP back home

==========
==========
CASE: p05
S2 planner: False
S2 physical: False
蓝色方块位移: -0.30839 m
L工具仍被抓住: True

目标停放 XY: [0.31 0.34]
实际停放 XY: [-0.3253 -0.3607]
停放误差: 945.82 mm
>>> 停放误差超出35mm阈值
S2 error: PlanningFailure: 运动规划失败: S2/parking transit

思考了 46s >

ChatGPT 说：

你说：

========== p01 ==========
目标 XY: [0.25 0.2]
松手前 XY: [0.1459 0.1754]
S2结束 XY: [0.1447 0.1756]
松手前目标误差: 106.94 mm
S2结束目标误差: 108.04 mm
松手后至S2结束的位移: 1.19 mm

========== p03 ==========
目标 XY: [0.31 0.285]
松手前 XY: [0.1324 0.2289]
S2结束 XY: [0.1306 0.2289]
松手前目标误差: 186.24 mm
S2结束目标误差: 187.94 mm
松手后至S2结束的位移: 1.78 mm

思考了 2m 57s >

ChatGPT 说：

tool_skills_stage1_trace.py
代码

你说：

==========
==========
seed=0 parking_case=custom
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644 0.0253]

目标停放XY: [0.31 0.285]
[PARK TRACE] S2/before parking | step=246 | grasped=True | tcp=[-0.3027, -0.3641, 0.0242] | tool=[-0.3202, -0.3644, 0.0253] | rel_drift=0.0mm | rel_rot=0.0deg | rigid_gap=0.0mm
[PARK TRACE] S2/after lift | step=278 | grasped=True | tcp=[-0.3079, -0.3604, 0.1613] | tool=[-0.3253, -0.3607, 0.1627] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=6.9mm
[PARK TRACE] S2/after transit | step=4035 | grasped=True | tcp=[0.1477, 0.2582, 0.1455] | tool=[0.1303, 0.258, 0.147] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=182.4mm | tool_target_err=182.4mm
[PARK TRACE] S2/after lower | step=8217 | grasped=True | tcp=[0.1499, 0.2291, 0.0353] | tool=[0.1324, 0.2289, 0.0369] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=186.3mm | tool_target_err=186.3mm
[PARK TRACE] S2/after release | step=8223 | grasped=False | tcp=[0.15, 0.2289, 0.0354] | tool=[0.1306, 0.2289, 0.025] | rel_drift=11.7mm | rel_rot=1.0deg | rigid_gap=12.0mm | tcp_target_err=186.2mm | tool_target_err=188.2mm
seed000_custom S2_hook_and_pull: steps=8192 planner=True physical=False blue_dx=-0.308 green_dx=-0.008
seed000_custom S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.308 green_dx=-0.008
screw plan failed
  [ERROR] S4_push: PlanningFailure: 运动规划失败: S4/push green
seed000_custom S4_push: steps=63 planner=False physical=False blue_dx=-0.308 green_dx=-0.008

==========
==========
总结:
  seed=0 case=custom steps=8452 planned=False chain_proxy=False blue=-0.30839 green=-0.00809 error=S4_push: PlanningFailure: 运动规划失败: S4/push green

已保存: demos/stage2_trace_p03/summary.csv
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ grep "PARK TRACE" /tmp/stage2_trace_p03.log
[PARK TRACE] S2/before parking | step=246 | grasped=True | tcp=[-0.3027, -0.3641, 0.0242] | tool=[-0.3202, -0.3644, 0.0253] | rel_drift=0.0mm | rel_rot=0.0deg | rigid_gap=0.0mm
[PARK TRACE] S2/after lift | step=278 | grasped=True | tcp=

[-0.3079, -0.3604, 0.1613] | tool=[-0.3253, -0.3607, 0.1627] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=6.9mm
[PARK TRACE] S2/after transit | step=4035 | grasped=True | tcp=[0.1477, 0.2582, 0.1455] | tool=[0.1303, 0.258, 0.147] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=182.4mm | tool_target_err=182.4mm
[PARK TRACE] S2/after lower | step=8217 | grasped=True | tcp=[0.1499, 0.2291, 0.0353] | tool=[0.1324, 0.2289, 0.0369] | rel_drift=0.1mm | rel_rot=1.0deg | rigid_gap=0.2mm | tcp_target_err=186.3mm | tool_target_err=186.3mm
[PARK TRACE] S2/after release | step=8223 | grasped=False | tcp=[0.15, 0.2289, 0.0354] | tool=[0.1306, 0.2289, 0.025] | rel_drift=11.7mm | rel_rot=1.0deg | rigid_gap=12.0mm | tcp_target_err=186.2mm | tool_target_err=188.2mm
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$

思考了 3m 29s >

ChatGPT 说：

tool_skills_stage1_guarded.py
代码

你说：

不是，你现在到底想确认什么，一直在测试，不能给出完整的测试流程么，省的到时候忘记之前的结果，而且你的目的也不明确

思考了 8m 57s >

ChatGPT 说：

MASTER_PROTOCOL.md
文档

KNOWN_RESULTS.csv
电子表格

archive_results.py
代码

RESULT_SCHEMA.md
文档

你说：

我需要给你保存的东西么。还是?

ChatGPT 说：

你说：

tool_skills_stage...
代码

tools_usage_co...
代码

思考了 10m 45s

ChatGPT 说：

EXPERIMENT_PROTOCOL.md
文档

你说：

保存: experiments/layout_v01_pilot_seed000/paired_results.csv
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ cat experiments/layout_v01_pilot_seed000/paired_results.csv
seed,candidate,parking_x_m,parking_y_m,s1_ok,s2_ok,s3_ok,s4_ok,s2_green_shift_mm,s2_i_shift_mm,l_tool_static_after_s2,s2_parking_error_mm,blue_dx_m,green_dx_m,total_steps,verdict,error
0,safe,-0.2,-0.12,True,True,True,True,0.0,0.01,True,7.31,-0.30839,0.141665,670,ALL_STAGES_PASSED,
0,block,-0.2,0.16,True,False,False,False,0.0,0.25,True,45.25,-0.30839,0.0,418,INVALID_LOCAL_EXECUTION,

思考了 3m 38s

ChatGPT 说：

你说：

好的

思考了 12m 39s

ChatGPT 说：

你说：

issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm
l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000
seed000_custom S4_push: steps=161 planner=True
physical=False blue_dx=-0.308 green_dx=+0.001
[P1 STOP] S4_push 未满足局部成功，停止后续技能
[VERDICT] block

DELAYED_CANDIDATE_NEEDS_VIDEO_REPLAN S2 True S3 True S4 False
[PAIR INIT] 0.0 same_initial_state= True

已保存：
experiments/layout_v02_pilot_seed000/paired_results.csv
[CANDIDATE] 仅是潜在延迟依赖；必须通过录像、接触核查和S4重规划对照。
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ cat experiments/layout_v02_pilot_seed000/paired_results.csv
seed,candidate,parking_mode,parking_xy,S1_executed,S2_executed,S3_executed,S4_executed,S1_ok,S2_ok,S3_ok,S4_ok,s2_green_shift_mm,s2_i_shift_mm,l_tool_static_after_s2,s2_parking_error_mm,s2_actual_l_tool_xy,nominal_s4_geometry_overlap,blue_dx_m,green_dx_m,total_steps,s4_error,verdict,error
0,safe,release,,True,True,True,True,True,True,True,True,0.001,0.004,True,,"[-0.3205384612083435,
-0.3642076849937439]",False,-0.305725,0.142579,576,,ALL_STAGES_PASSED,
0,block,custom,"(-0.2,
-0.12)",True,True,True,True,True,True,True,False,0.001,0.005,True,7.309,"[-0.20714610815048218,
-0.11846321821212769]",True,-0.30839,0.001128,673,,DELAYED_CANDIDATE_NEEDS_VIDEO_REPLAN,

思考了 3m 4s >

ChatGPT 说：

validate_layout_v02_causal.py
代码

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ cat experiments/layout_v02_causal_seed000/causal_results.csv
variant,seed,diagnostic_teleport,S1_executed,S1_planner_ok,S1_physical_ok,S1_error,S2_executed,S2_planner_ok,S2_physical_ok,S2_error,S3_executed,S3_planner_ok,S3_physical_ok,S3_error,S4_executed,S4_planner_ok,S4_physical_ok,S4_error,green_dx_m,steps
block_replay,0,False,True,True,True,,True,True,True,,True,True,True,,True,True,False,,0.001128,673

```
block_remove_l_before_s4,0,True,True,True,True,,True,True,True,
,True,True,True,,True,True,True,,0.142809,673
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark$ cat
experiments/layout_v02_causal_seed000/causal_pair_check.jso
n
{
  "same_pre_s4_max_abs_difference": 0.0,
  "comparable_pre_s4_at_1mm": true,
  "baseline_block_fails": true,
  "removing_l_recovers_s4": true,
  "verdict": "CAUSAL_SUPPORT_L_OCCUPANCY_EFFECT"
```

思考了 43s >

ChatGPT 说：

你说：

```
seed=01 MISSING
seed=02 MISSING
seed=03 MISSING
seed=04 MISSING
seed=05 MISSING
seed=06 MISSING
seed=07 MISSING
seed=08 MISSING
seed=09 MISSING
seed=10 MISSING

====== P1 DEV RESULTS ======
有效配对: 0/10
延迟依赖候选: 0/10
有效配对内的延迟依赖比例: 0/0
```

ChatGPT 说：

你说：

```
seed=1 LOG_ONLY
seed=2 LOG_ONLY
seed=3 LOG_ONLY
```

seed=4 LOG_ONLY
seed=5 LOG_ONLY
seed=6 LOG_ONLY
seed=7 LOG_ONLY
seed=8 LOG_ONLY
seed=9 LOG_ONLY
seed=10 LOG_ONLY

ChatGPT 说：

你说：

```
>     echo "日志大小: $(wc -c < "$log") bytes"
> done
========== Seed 1 ==========
日志大小: 135 bytes
========== Seed 2 ==========
日志大小: 135 bytes
========== Seed 3 ==========
日志大小: 135 bytes
========== Seed 4 ==========
日志大小: 135 bytes
========== Seed 5 ==========
日志大小: 135 bytes
========== Seed 6 ==========
日志大小: 135 bytes
========== Seed 7 ==========
日志大小: 135 bytes
========== Seed 8 ==========
日志大小: 135 bytes
========== Seed 9 ==========
日志大小: 135 bytes
========== Seed 10 ==========
日志大小: 135 bytes
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ tail -n 80 experiments/layout_v02_dev10/seed001/run.log
结果目录已存在，拒绝覆盖：experiments/layout_v02_dev10/seed001
请指定新的 --output-dir；不允许覆盖旧试验。
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark$ ps -ef | grep '[r]un_layout_v02_pair.py'
```

ChatGPT 说：

你说：

seed=01 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=02 SAFE=False BLOCK_LOCAL=False BLOCK_S4=NOT_RUN DELAYED=False
seed=03 SAFE=True BLOCK_LOCAL=False BLOCK_S4=False DELAYED=False
seed=04 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=05 SAFE=False BLOCK_LOCAL=False BLOCK_S4=NOT_RUN DELAYED=False
seed=06 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=07 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=08 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=09 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True
seed=10 SAFE=True BLOCK_LOCAL=True BLOCK_S4=False DELAYED=True

========== P1 批量复验 ==========
成功读取结果：10/10
有效配对：7/10
延迟依赖候选：7/7

ChatGPT 说：

你说：

好的，设计吧

思考了 4m 32s >

ChatGPT 说：

BENCHMARK_SPEC_v2.md
文档

TASK_CONFIG_v2.yaml

代码

IMPLEMENTATION_ORDER.md
文档

你说：

好的，现在生成完成的实验目的，实验方法，对比的baseline选择，还有创新点设计和描述，需要对比的算法，以及最终需要跑的代码

思考了 9m 37s

ChatGPT 说：

BASELINE_MATRIX.csv
电子表格

FVS_Compose_研究与实验总方案_v3.docx
Word

你说：

这是一个专家的意见，你先思考，是否他的合理，你再根据建议修改：这份《FVS-Compose 研究与实验总方案 v3.0》在**科学问题切割、反作弊实验契约、以及止损门控（G1–G5）设计上非常严密，完全符合 IEEE T-RO 对长序列操作机理与算法严谨性的审稿偏好；在单卡 RTX 4090 算力约束下，该路线在工程上是可完成的，但目前存在两个核心机制设计隐患（均值与风险决策同质化、S2 决策时刻对 S4 物理参数的不可观测性）以及较高的仿真环境调优工作量，必须在写代码前先行修正**。

一、 路线合理性评估：为什么这个方案底子很好？

1. **切入点精准且避开了算力泥潭**：方案明确不卷自然语言规划、不端到端训 VLA、不依赖灵巧手或触觉，而是固定底层 Panda 运动规划技能，专注于状态空间（SE(3)、接触标志、速度）上的局部转移学习与多步不确定性传播。这种"冻结底层原子技能 + 仅用单技能短时过渡数据训练轻量级概率效果模型"的设计，完美绕开了长序列像素级强化学习/扩散策略的高昂算力需求，单张 4090 完全可以在数小时内完成单技能数据采集与模型训练迭代。

2. **科学边界与对照组设置极具说服力**：方案非常清醒地将 T1 定

性为“几何负对照”（承认完整几何/TAMP 可轻松解决，仅用于证伪固定规则 B0 和单步交接 B1），而将真正的创新主战场放在具有动态接触不确定性与工具复用的 T2 和未见组合图的 T3。同时明确要求在论文中把 B1–B5 定义为共同技能库下的 adapted baseline，并严格区分局部合法率与无条件完整任务成功率，这种诚实且严密的实验规范极易获得 T-RO 审稿人的信任。

3. **止损机制（G1–G5）避免了“沉没成本陷阱”**：方案制定了明确的 Pilot 预算（T1/T2 各最多 20 条全任务执行验证，允许一次几何修订）和退出条件。这保证了不会在不可行的物理布局上无限耗费时间。

二、 核心隐患与潜在“死穴”（决定能否发 T-RO 的关键瓶颈）

虽然整体框架逻辑自洽，但在从“抽象原型（abstract smoke）”走向“ManiSkill 真实物理实现”的过程中，有 **4 个致命细节**如果不提前调整，极易触发 G3/G4 止损条件（即你提出的 B6 跑不过 B3-TAMP 或 B4-STACK 均值搜索）：

1. 均值与风险的决策同质化陷阱（直接威胁 RQ3 与 H3、G4）

* **问题所在**：方案在 T1 和 T2 中默认将 S2 的停放候选设为二元或少量离散区域（`PAD_A` / `PAD_B`），并在 A7 中测试 2/4/8 个候选终态。如果 `PAD_A` 和 `PAD_B` 只是左右对称的两个独立停放区，当 S4 推动挡板去 `LEFT` 时，一侧停放区会被挡板或机械臂包络面遮挡（未来成功概率低，如 0.1），另一侧完全通畅（未来成功概率高，如 0.85）。此时，**基于均值效果的 B4（STACK-style）和基于风险感知 CVaR 的 B6（Proposed）会选出完全相同的 `PAD`！**

* **后果**：只有当候选集存在“高均值、高方差（高尾部失败风险）” vs. “次优均值、低方差（稳健安全）”的经典权衡（Mean-Variance Trade-off）时，$\arg\max [E_\phi(J_{\text{future}}) - \lambda \text{Risk}_\phi(J_{\text{future}})]$ 才会与纯均值搜索产生分歧。

* **修正方案**：在 T2/T3 中，不能只设完全对称的 A/B 两个安全垫。必须引入**连续参数化或具有风险梯度的至少 3–4 类候选区**

（例如：PAD_Aggressive 离 S3/S5 抓取位最近、运动代价 $\text{Cost}(\theta)$ 最低且名义均值最高，但紧贴 S4 挡板滑动接触不确定性边界，摩擦稍小就会发生擦碰导致 S5 失败；PAD_Conservative 距离稍远或姿态略刁钻、局部及 S5 标称裕度稍低，但完全避开 S4 的动态接触发散锥）。只有这样，B4 才会贪婪地选 PAD_Aggressive 并栽跟头，而 B6 通过 CVaR_α 选出 PAD_Conservative 胜出。

2. S2 决策时刻对 S4 隐藏物理参数的"因果不可观测性"（威胁 Test-P 与 C1）

* **问题所在**：方案 3.1 节规定隐藏摩擦/质量不能直接作为输入，4.1 节提出用观测历史特征 o_k 来辅助推断接触不确定性。但在 T2 序列（GetL → HookBlueAndDock → GetI → PushGate → ReacquireL → HookYellow）中，选择停放参数 θ_2 发生在 **S2（步数 $k=2$）**，而真正受挡板摩擦/质量影响的操作发生在 **S4（PushGate，步数 $k=4$）**。

* **后果**：在执行 S2 时，机器人根本还没有接触过 Gate（挡板）！如果桌面/物体的摩擦系数是独立随机化的，那么在 $k=2$ 时刻，o_2（仅包含抓 L 工具和勾蓝块的历史）在物理上**不包含任何关于 S4 挡板摩擦力的信息**。

* **修正方案**：
1. **桌面全局物理耦合设定**：在 Test-P 中，将摩擦系数设定为**全局桌面材质属性**（即蓝块、L 工具与桌面在 S1/S2 滑动时的阻尼特性，与 S4 挡板在桌面滑动的摩擦系数同源相关）。这样模型在执行 S2 勾蓝块（或 S1 拿 L）时，才能通过 o_2 的接触/速度偏差合法地在线推断当前桌面的摩擦分布。

2. **先验分布传播**：若各物体物理参数独立，则在 $k=2$ 时不依赖 o_2 辨识 S4 参数，而是直接利用局部模型在训练分布上习得的**先验随机多模态粒子**向后传播，并在 S4 执行完毕后的闭环重规划（C3）中利用 o_4 更新 S5/S6 的决策。

3. 六至八步非固定抓取操作（Non-Prehensile）的仿真信噪比

问题

* **问题所在**：T2 包含 6 个技能（含勾取、推动挡板、重抓 L 工具、勾黄块），T3 包含 7–8 个技能。在 ManiSkill（基于 SAPIEN/PhysX）中，L 型工具勾物（`HookBlue`）和直杆推挡板（`PushGate`）属于非固定抓取（non-prehensile）操作，极易产生微小的旋转漂移；而 S5（`ReacquireL`）需要再次抓取此前停放的 L 工具。

* **后果**：如果底层技能库设计不够鲁棒，S5 或 S6 的失败有很大概率是因为 S2 释放 L 工具时的微小弹跳或夹爪碰撞，而不是因为远期空间占位冲突，从而导致实验噪声掩盖核心科学假设（H2）。

* **修正方案**：在实现 `fvsbench/skills/` 时，`ReacquireL`（S5）必须基于 S2 结束后的**实际观测物体 SE(3) 位姿**通过 IK 和 Motion Planner 动态生成抓取轨迹（而非死记硬背相对于 PAD 中心的开环轨迹），确保局部微小位姿偏差被底层规划器吸收，让失败纯粹来源于“挡板推过去后遮挡了机械臂抓取 L 工具的 IK 可达空间/扫掠体”。

4. 工程开发与基线复现的工作量评估（能否按期完成？）

目前 `maniskill_bridge.py`、T1/T2/T3 环境、真实几何碰撞检查 `geometry_real.py` 以及 B1–B5 真实物理版均处于“尚未实现”状态。我们可以通过下表评估工作量与完成可行性：

| 模块阶段 | 方案内具体任务

难度与耗时预估（单人）	可完成性评估与建议
Phase 1：环境与契约	`maniskill_bridge.py`、`t1_lane.py`、`t2_gate.py`、S1–S6 技能封装与 Pilot G1/G2 验证

| ⭐⭐⭐⭐（约 2–3 周） | **全案最关键瓶颈**。T2 挡板铰接/滑动阻尼与 L 工具重抓的物理稳定性调优最耗时间。 |
| **Phase 2：几何与规划基线** | `geometry_real.py`、B2（扫掠体碰撞）、B3（Symbolic TAMP / Skill-RRT）、Oracle

| ⭐⭐⭐（约 1.5–2 周） | **完全可控**。可复用 ManiSkill/Pinocchio 自带的碰撞检测接口与采样器，无需从零写网格求交。 |

| **Phase 3：数据与核心模型** | 采集 `D_local_train`（每技能 300–800 条）、实现 C1–C3（`real_effect.py`）与 B4/B5

| ⭐⭐⭐（约 2 周） | **极易完成**。状态空间维度低（SE(3)+接触标志），MLP/GNN 集成模型在 4090 上训练仅需分钟级。

|
| **Phase 4：T3 与消融表格** | T3 组合 OOD 环境、A1–A9 消融实験、100 初态正式评测

| ⭐⭐⭐（约 2 周） | **可完成，但需控制仿真评测耗时**。多步粒子展开 + Motion Planner 在线调用需做好并行或缓存优化。 |
| **Phase 5：真机最小验证** | Fig. 7 实机平行夹爪最小对照实验

| ⭐⭐⭐⭐（约 1–2 周） | **T-RO 录用关键加分项**。只需在真机复现 T1 或简化版 T2 的 10–20 次配对试验即可闭环。 |

三、 落地执行的优先级与优化建议

为了确保该项目不仅"能完成"，而且能顺利产出达到 T-RO 水准的显著性实验结果，建议按以下顺序推进并微调方案细节：

1. **第一步（本周立即执行）：打通 T1/T2 物理环境并优先攻克 G1/G2**
* 严格按照方案第 10 节的指示，先写 `fvsbench/maniskill_bridge.py`、`t1_lane.py` 和 `t2_gate.py`，绝对不要现在去碰神经网络代码。

* 在设计 `t2_gate.py` 时，直接将 S2 的停放区参数 $\theta_2 \in \Theta_2$ 设计为包含"近距高风险区（靠近 Gate 运动包络边缘）"与"远距安全区"的多候选集合（例如 4–8 个连续采样点或离散锚点），并在 Pilot 阶段直接验证：**当给 Gate 施加微小的滑动角度/摩擦扰动时，近距高风险区会有 40%–60% 的概率在 S5 发生机械臂碰撞/IK 无解，而远距安全区有 >85% 的概率成功**。只要这个物理现象在 Pilot 的 20 条轨迹里跑出来，你这篇 T-RO 的核心故事（RQ3/H3）就已经成功了 80%！

2. **第二步：先实现 Oracle 与 B2/B3，验证 G3 门控**
* 在采集大规模训练数据前，先用 ManiSkill 的 `get_state()` / `set_state()` 实现特权模拟器上界 `real_oracle.py`（即在 S2 处

分叉真机模拟候选参数，看哪个能走通全链），再用 Pinocchio/SAPIEN 碰撞检测实现 B2/B3。

* 确认在 T2 的接触不确定性下，静态几何 B2 和确定性 TAMP B3 因为无法预判 Gate 推动后的真实滑移分布而频繁误选高风险停放位，从而确立学习局部随机效果模型的必要性。

3. **第三步：简化 T3 的工程实现成本**
* 方案 3.5 节要求 T3 测试"未见技能组合（七至八技能）"。为了不重新开发一套全新的底层动作，建议 **T3 完全复用 T1/T2 已经写好的 6 个原子技能模块**，仅通过**技能图拓扑重排或串联**来构造（例如：`GetL → HookBlueAndDock → GetI → PushGate → PushTarget(Green) → ReacquireL → HookYellow`），这样既满足"训练期只见过单技能独立过渡、未见过该 7 步任务图后缀标签"的严格组合 OOD 定义，又能将 T3 的环境代码开发量压缩到 2 天以内。

4. **第四步：控制 A1–A9 消融实验的组合爆炸**
* 方案第 6 节列出了 A1–A9 共 9 组消融实验。如果每组消融都在 T1/T2/T3 的 IID/C/P 全部子集上跑 3 个训练种子 × 100 条初态，总仿真次数将高达数万条 episode（且每条包含多次 Motion Planning 调用）。

* 建议将 **Table II（主对比 B0–B6, Oracle）** 在 T1/T2/T3 全量运行，而将 **Table IV（A1–A9 核心消融）** 聚焦在最具代表性的主战场 **T2（Test-IID 与 Test-P）和 T3（Test-C）** 上运行，从而将实验计算时间控制在 1 周以内。

要不要针对当前最紧迫的第一步（**T2 动态挡板环境的几何布局参数与"均值-风险分歧"机制设计**），一起细化一下具体的坐标系布局和 `execute_skill` 接口代码结构？

思考了 11m 9s

ChatGPT 说：

T2_LAYOUT_AND_INTERFACE_v31.md
文档

V31_CHANGELOG.md
文档

RUNBOOK_v31.md
文档

FVS_Compose_研究与实验总方案_v3.1.docx
Word

你说：

这是一个专家的意见：**v3.1 版本在数学严密性、因果信息边界和抗审稿人质疑（Reviewer-proof）的设计上比 v3.0 上了一个大台阶，已经是一份逻辑完全闭环、可直接冻结并转入工程落地的 T-RO 级主规格说明**。

一、 v3.1 最关键的四大升级（为什么这一版彻底立住了？）

1. **从数学根源上排除了"伯努利 CVaR 排序同质化"陷阱（Section 1.2 & 4.1）**
* v3.1 极其敏锐地指出了一个核心数学事实：如果损失函数 L 仅采用二值指标 $\mathbb{I}[\text{全任务失败}]$，由于伯努利随机变量的 $\text{CVaR}_\alpha(L)$ 是期望失败率 $\mathbb{E}[L]$ 的严格单调递增函数，无论把候选停放位增加到多少个，均值最小化与 CVaR 最小化在数学上都会给出**完全相同的候选排序**。

* 为此，v3.1 引入了五项连续非负物理损失组合：

$$L(\tau) = w_f \cdot \mathbb{I}[\text{全任务失败}] + w_c \cdot \text{collision_severity} + w_m \cdot \text{clearance_violation} + w_t \cdot \text{normalized_recovery_time} + w_e \cdot \text{terminal_pose_error}$$

并增设了共享同一模型、同一损失 L、同一粒子与算力预算但令 $\lambda=0$ 的对照基线 **B4M（same probabilistic effect model, mean-only）**。这一改动让“尾部风险感知”相对于“期望损失最小化”的独立贡献具备了严格的数学可证伪性。

2. **彻底厘清了 S2 决策时刻的因果可观测性边界（Section 3.4 & 4.2 & A8）**
* v3.1 明确禁止在未接触挡板（gate）的 S2 阶段假装从历史观测推断独立的隐藏物理参数 ξ，并将实验显式拆分为 `independent_gate_physics`（S2 仅传播离线训练习得的先验分布，S4 接触后才更新信念用于 S5/S6 闭环）与 `shared_surface_physics`（共享桌面物理属性，但必须实测验证 S1/S2 接触信号对该共因子的可辨识性），还在消融实验 **A8** 中将两者正面对照。这完全堵死了因果推理层面的审稿漏洞。

3. **确立了“反结果导向（Anti-p-hacking）”的两阶段 T2 验证与预注册机制（Section 3.4 & 3.6）**
* 将 T2 拆解为 **T2-D（动态依赖，20 条预算验证六步通路与延迟状态依赖）** 与 **T2-R（风险选择，按工作空间几何覆盖预注册 4–8 个连续 SE(3) 候选 $\theta_2 = (\text{pad_pose}, \text{yaw}, \text{approach_side}, \text{release_height})$）**。明确禁止“看着算法结果去凑 40%–60% vs >85% 的失败率数字”，这种预注册（pre-registration）实验规范非常契合 T-RO 对实验客观性的高要求。

4. **补齐了最强规划基线 B3R 与物理引擎状态分叉规范（Section 5 & Section 6）**
* 新增 **B3R（robust TAMP，在相同离线物理先验与在线预算下进行物理扰动情景采样）**，直接回应了规划领域审稿人最常问的“为什么不用带扰动采样的鲁棒 TAMP”。

* 明确指出了 ManiSkill/SAPIEN 中 `get_state/set_state` 存在接触求解器缓存、控制器积分状态和 RNG 等隐状态丢失风险，规定 Oracle 优先采用“同初态 reset + 确定性重放到分支点”（或经严格等价测试后才用快照）。

* 将 A1–A9 消融聚焦于核心分组（`T2-IID`、`T2-P`、`T3-C`），避免了 $9 \times \text{task} \times \text{split}$ 的算力爆炸。

二、 进入代码实现阶段前，仅剩的 3 个工程落地细节提醒

在规范层面 v3.1 已经无需再改；但接下来在编写 `fvsbench/models/real_effect.py`、`robust_tamp.py` 和调节 `L(τ)` 权重时，有 **3 个实操细节**将直接决定你能否顺利通过 **G3/G4 门控**：

1. 连续损失 $L(\tau)$ 的权重标定：防止退化回伯努利排序，并打通“连续裕度 $\to$ 二值成功率”的断崖机制

* **潜在风险**：v3.1 的 G4 门控明确指出，如果风险方法只降低连续损伤却不提高完整任务成功率（主指标 `N_full_success / N_all_attempted`），就必须收缩论文贡献。同时，在公式 $L(\tau)$ 中包含二值项 $w_f \cdot \mathbb{I}[\text{全任务失败}]$ 和四个连续项（碰撞冲量、间隙违规、恢复时间、末端位姿误差）。

* **实操准则**：
* **切勿让 w_f 支配 $L(\tau)$**：在 Validation 集上锁定权重 w 时，如果把 w_f 设得过大（例如 $w_f=10$，而连续项归一化后只有 $0.05 \sim 0.2$），那么 $L(\tau)$ 的分布就会退化为近似二值分布，`B4M`（均值）和 `B6`（CVaR）的排序又会高度重合。应当让连续安全违规项（特别是 `clearance_violation` 和

`terminal_pose_error`）与 w_f 处于同一数量级。

* **利用物理"断崖效应（Cliff Effect）"赢下二值成功率**：为什么惩罚连续尾部损失的 `B6` 能在最终的**二值完整成功率**上击败 `B4M`？因为真实物理执行中，当最小安全间隙 `min_signed_clearance` 跌破 0（发生擦碰）或位姿误差超出 S5 夹爪闭合捕获域时，会触发非线性断崖式的二值失败。某个激进候选位在 80% 的摩擦样本下间隙裕度极好（均值损失低），但在 20% 的低摩擦尾部样本下间隙跌破 0；由于局部效果模型 p_ϕ 存在有限样本认知不确定性（epistemic）与接触随机性（aleatoric），`B4M` 对预测的连续损失求均值会低估跨越断崖的概率，而 `B6` 对预测的 `clearance_violation` 上尾求 CVaR_α 会主动避开靠近断崖边缘的候选位，从而在真实 rollout 中转化为更高的 `N_full_success`。

2. 严格界定 `B3R`（Robust TAMP）的物理预测模型，避免将其写成"在线模拟器 Oracle"

* **潜在风险**：v3.1 规定 `B3R` 使用与 `B6` 相同的离线物理先验和在线预算进行物理扰动情景采样。

* **实操准则**：在实现 `fvsbench/baselines/robust_tamp.py` 时，必须明确 `B3R` 作为"非学习规划基线"，在推演 S4（`PushGate`）的扰动情景时，使用的是**解析准静态接触近似模型（Analytical/Quasi-static Pushing Model）或刚体几何扫掠包络**，而**不能在在线决策时直接克隆 ManiSkill 真实物理引擎去 step 未来轨迹**（否则 `B3R` 就变成了带采样预算的在线特权 Simulator Oracle `O`）。这样才能客观体现出：当真实非固定抓取（Non-prehensile）接触存在多模态滑移/卡滞时，从局部真实交互数据中学出的概率效果模型 p_ϕ（`B6`）比传统解析近似的 `B3R` 更准、且比在线调用几何/IK 规划器的多情景搜索计算开销更低（对应 Table III）。

3. 多步粒子传播中连续损失项的可预测性设计

* **潜在风险**：在 S2 决策时刻，模型需要通过多步后缀粒子 $\tau \sim p_\phi(\tau \mid h_2, G, \theta_2)$ 预测未来 S3–S6 的 $L(\tau)$。但此时 S3–S6 尚未真实执行，没有真实的物理日志可供读取。

* **实操准则**：在设计 `fvsbench/models/real_effect.py` 的输出头（C1 组件）时，除了输出下一技能后的物体/工具/gate 的 $\text{SE}(3)$ 与速度增量外，**必须直接将单步的 `(collision_impulse, min_signed_clearance, step_duration, local_fail_prob)` 作为局部效果网络 p_ϕ 的显式回归/分类输出头**（训练标签由采集 `D_local_train` 时的真实物理日志提供）。这样在多步粒子传播（C2）时，就能直接累加/取极值算出每条虚拟粒子轨迹的 $L(\tau)$，而无需在神经网络前向传播循环内部去实时跑高开销的 3D 网格碰撞检测。

三、 总结与下一步行动

这份 v3.1 规范已经非常成熟，不需要再做文档层面的大改。按照规范第 10 节与第 13 节的路线，当前的**唯一关键路径**就是将已有的类型化接口（`skill_api_v31.py`）和布局草案（`T2_LAYOUT_AND_INTERFACE_v31.md`）接入 ManiSkill：

1. 编写 `fvsbench/maniskill_bridge.py` 与 `fvsbench/envs/t1_lane.py`，跑通 **G1/T1 的 20 条全链执行预算**，验证双目标下的合法 PAD 翻转；

2. 紧接着实现 `fvsbench/envs/t2_gate.py` 与基于真实观测位姿重抓的 S5，跑通 **G2a/T2-D 的 20 条动态依赖验证**。

如果你准备现在开始写 `maniskill_bridge.py` 或审视 `T2_LAYOUT_AND_INTERFACE_v31.md` 中的具体坐标与门轨阻尼

参数，随时把该 Markdown 或现有 `layout_v02_env.py` 发出来，我们可以直接对接代码实现。这是第二个是专家的意见：我仔细阅读了《FVS-Compose：面向未见技能组合的风险感知未来可执行性推理》v3.1。我的总体判断是：
这条研究路线在科学问题、实验逻辑和研究方法上是合理的，具备继续推进的价值；核心仿真研究也有完成的可能。但按照当前完整方案直接推进，工作量偏大，且距离形成一篇有竞争力的 IEEE T-RO 论文，还有几个必须通过实验解决的不确定性。
我尤其认可这份方案的一点：它没有把已经完成的物理现象验证、离线软件测试和真正的算法效果混为一谈。文件明确指出，T0 已有初步证据，而 T1、T2、T3 的真实 ManiSkill 物理交互环境尚未实现或验证。这种区分是正确的研究态度。
FVS_Compose_v3.1.docx

一、整体评价：路线合理，但核心贡献尚未成立

研究问题：合理，值得研究
前序技能的终态即使合法、稳定，也不代表它对更远期的技能有利。将这种远期可执行性作为研究对象，具有明确的机器人操作意义。
研究逻辑：比较完整
T0 解释问题存在，T1 验证终态选择翻转，T2 验证真实物理动态依赖，T3 检验未见技能组合泛化，整体有递进关系。
工程准备度：仍处早期
当前缺少实际的 T1/T2 环境桥接、完整几何检查、真实局部效果模型和可运行的完整基线体系。研究规范比较成熟，不等于实验系统已经建成。
T-RO 投稿前景：有条件，不能保证
最大问题不是能否把算法写出来，而是最终能否证明：相较于强规划方法和同模型的均值优化方法，你提出的机制确实带来了不可替代的收益。
这里需要区分三个不同的问题：

- 能否完成软件和仿真实验？ 可以规划出可执行的路线。
- 能否验证研究假设？ 可以，但部分假设可能被实验否定。
- 能否达到 T-RO 的发表要求？ 取决于新颖性、实验结果和强基线对比，不能在实验前承诺。

二、为什么我认为这条路线在科学上是成立的?

1. 核心问题不是人为制造出来的

假设机器人依次执行四个技能：

S1

抓取工具 L
完成局部操作

S2

将 L 放在 PAD_A 或 PAD_B
两种终态均合法，但会改变后续可用空间

S3

抓取另一个工具 I
两种 PAD 都允许当前技能成功

S4

执行远期推送或接触操作
S2 的终态可能使该步骤变得困难甚至失败
这类问题的关键在于：

- 只检查 S2 是否成功，不足以选择最合适的终态。
- 只检查 S3 是否能继续，也可能忽略 S4 之后的困难。
- 如果提前知道全部几何信息，传统规划器可能直接解决问题。
- 如果物理后果存在随机性或不确定性，单纯几何规划也未必能准确估计后续风险。

因此，研究局部技能效果如何传播到远期可执行性，是有明确逻辑的。
但要注意：存在远期依赖，不等于一定需要学习算法；存在不确定性，也不等于一定需要 CVaR。 这两点必须由后续对照实验分别证明。

2. T0 到 T3 的实验递进是合理的

阶段	需要回答的问题	我的评价
T0	远期失败是否真实存在?	已有初步机制证据，但不能单独支撑算法贡献
T1	是否存在依任务变化而翻转的最优终态?	必须做，且容易由强几何规划解决
T2-D	真实接触和动态物理效果是否改变后续技能?	全文最重要的机制验证
T2-R	连续物理损失是否存在可利用的尾部风险差异?	决定 CVaR 是否有实际意义
T3		

方法能否泛化到训练中未见的技能组合?

决定能否主张组合泛化
这套设计的优点是，每一步都设有准入或止损条件，而不是默认假设所有实验都会支持自己的方法。
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特别是 T1，我赞同把它定位为几何负对照，而不是论文的核心创新。它的作用是先证明固定规则和只看直接后继的策略存在不足，并检验强规划基线能解决到什么程度。
真正值得投入精力的是 T2。

三、目前最值得警惕的五个问题

问题一：T2 的物理因果链必须真正成立

这是我认为整个研究路线的第一道关键关卡。
当前 T2 的技能序列是：
GetL → HookBlueAndDock → GetI → PushGate → ReacquireL → HookYellow。
方案要求 S4 通过 I 工具与 gate 发生真实接触，S5 根据 S4 之后 L 工具的实际位姿重新生成抓取路径，而不是直接前往名义停放点。
这是正确的设计方向。
FVS_Compose_v3.1.docx

但仍有一个必须回答的问题：
S2 选择不同的 L 工具停放位，究竟通过什么物理机制改变 S4、S5 或 S6 的执行结果?
例如，假设 L 工具被停放在 gate 的运动包络附近：

- 候选 A：gate 推动过程中与 L 工具发生接触，改变 gate 的运动轨迹，甚至导致 L 工具被带动。
- 候选 B：L 工具位于运动包络之外，gate 可以正常运动。
- 两种情况下，后续重新抓取 L、执行 HookYellow 的可行性可能不同。

这是一条合理的因果链，但目前还只是需要验证的设计。不能仅凭场景几何示意图就认定它成立。
我建议专门检查以下三个条件：

1. S2 的不同终态均能稳定实现，且 S3 都能正常成功。

2. S4 的真实接触日志、gate 的位姿和速度变化能够解释不同终态为何产生不同后果。
3. 在控制其他条件后，移除 L 工具与 gate 的相关接触，或改变其停放位，能够产生符合物理机理的结果变化。

如果候选终态根本不影响 gate 的物理运动，那么 T2-R 就不能靠这条因果链成立。需要重新设计场景，而不是继续训练网络。

问题二：即使 T2 成功，也不代表你的算法一定有优势

这是比工程实现更深层的问题。
假设 T2 最终证明了远期物理依赖确实存在。仍然可能出现以下结果：

- 完整几何规划器 B2 已经能找到合适的终态。
- 允许在线重规划的 TAMP 基线 B3 可以绕开障碍。
- 稳健 TAMP B3R 通过物理扰动情景采样，获得了与所提方法接近的鲁棒性。
- 你的概率效果模型 B4M 即使不使用 CVaR，仅通过最小化期望损失，也获得了同样好的结果。

如果出现这些结果，说明问题本身成立，但当前算法的必要性尚未得到证明。
方案已经考虑了这些风险，并要求在训练大型效果模型之前先运行 B0–B3 和稳健 TAMP。我完全赞同。
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因此，不要把“我的方法优于固定终态选择”当成最终目标。真正要回答的是：我的方法为什么比一个认真实现的、允许重规划的强基线更好，或者为什么在达到相近成功率时成本更低？
这是论文能否进入高水平机器人期刊竞争的关键。

问题三：C1–C3 目前是研究框架，还不是完整的算法定义

文件提出了三个组件：

- C1：局部概率效果与物理关系图。
- C2：多步信念传播。
- C3：尾部损失选择与观测闭环。

三者组合在逻辑上合理，但现在还缺少足够具体的算法细节。

需要明确的内容

当前需要补充什么

局部效果模型

输入特征、输出分布、网络结构、训练目标

物理状态表达

物体与工具的 SE(3)、速度、接触模式如何联合表示

随机效果传播

多模态接触结果如何采样，粒子如何更新和筛选

不确定性估计

如何区分接触随机性与模型不确定性，如何验证预测校准

候选终态选择

可行域约束、风险目标、计算预算和优化算法

闭环重规划

哪些观测触发更新，哪些信息允许进入决策
例如，如果一个技能有两种不同的接触结果，仅预测平均位姿可能得到一个实际中根本不会出现的中间状态。因此，C1 不能只做普通的均值状态回归，还需要验证模型能否表达对后续决策真正重要的多模态物理结果。
我的建议是先采用一个简单、可解释、便于验证的概率效果模型，建立可靠的状态转移和不确定性传播基线，再决定是否需要更复杂的关系图网络。

另外，算法新颖性不能仅仅来自“局部效果模型 + 粒子传播 + CVaR + 重规划”的组合。论文必须明确这些组件之间存在什么不可替代的机制，以及为什么已有方法不能通过简单组合相同工具达到相同效果。

问题四：CVaR 方向正确，但实验规模不能低估

这份方案对 CVaR 的数学问题处理得比较严谨。
如果损失只有“成功或失败”两个取值，最小化期望失败率与最小化对应的 CVaR 可能产生相同的候选排序。此时，仅仅引入 CVaR 并不能证明方法更好。
所以方案将损失扩展为碰撞严重程度、最小安全间隙违规、恢复时间和终态误差等连续指标。这是合理的。
FVS_Compose_v3.1.docx

不过还有一个实际困难：连续损失的 CVaR 本身也需要足够的数据才能可靠估计。
例如，若采用 \(\alpha=0.9\)，关注的是最差 10% 的损失：

- 每个候选只有 20 条独立轨迹时，尾部大约只有 2 条样本，估计非常不稳定。
- 每个候选有 100 条轨迹时，尾部大约有 10 条样本，仍需报告不确定性。
- 如果比较的是多个候选、多个任务和物理 OOD 条件，就必须预先计算实际需要多少轨迹，而不是把所有轨迹合在一起估计一个 CVaR。

因此，不能只规定“至少 100 条测试初态”，还应明确这是每个测试组、每个候选，还是多个候选的总量。最终数量应根据配对成功率差异、连续损失方差和尾部估计精度进行功效分析。

问题五：T3 的组合泛化不能只靠改变场景或增加一个技能

方案已经明确，测试集必须包含训练期间未见的技能顺序或目标约束关系，不能把换随机种子或调整 gate 位置称为未见组合泛化。
FVS_Compose_v3.1.docx

这一点非常重要。
我建议至少准备两到三个结构不同的留出任务图，并严格检查训练集和测试集之间是否存在完整后缀标签泄漏。测试时还要区分：

- 未见目标布局；
- 未见技能组合；

- 未见物理参数。

三者对应的泛化能力并不相同。
如果只做一个未见任务图，实验仍然有价值，但论文结论应该限定为该任务图上的组合泛化证据，而不宜过度推广到一般性的技能组合能力。

四、真正的工程难度有多大?

当前最大的工程工作并不是训练神经网络，而是建立一个可信、可复现的物理实验系统。
文件已经列出了尚待实现的模块，包括 ManiSkill 环境桥接、T1/T2 场景、统一技能执行接口、完整几何检查、局部效果模型、真实基线和评估脚本。当前的抽象软件冒烟测试及风险单元测试不能替代这些物理验证。
FVS_Compose_v3.1.docx

我会把工程工作分成三个层次：
第一层：场景和技能能够可靠执行
建立统一的执行接口，确保抓取、放置、推送、重新抓取和碰撞检测能在 ManiSkill 中正常工作；准确记录技能是否执行、局部成功、真实终态及失败原因。
这是所有后续实验的基础。
第二层：基线和物理因果可靠
实现 T1、T2 和强规划基线；确认不同终态确实造成不同的后续执行结果；排除 IK、碰撞检测、重抓控制器自身不稳定造成的假失败。
这是最可能消耗大量调试时间的部分。
第三层：学习、风险推理和泛化评估
在物理环境和基线可信后，才收集局部数据、训练效果模型、执行多步传播，并完成风险消融、OOD 和组合泛化实验。
此时模型的收益才具有可解释性。
从计算资源看，单张 RTX 4090 足以启动这种规模的研究，模型训练也不一定是瓶颈。真正的限制更可能是仿真吞吐、规划器调用、物理场景调试和成批运行实验所需的时间。

实验量是否过大?

是的，完整版相当重。
方案建议每个测试组至少 100 个独立初态，并要求多个方法在相同任务实例上进行配对比较。如果把 T1/T2/T3、IID/C/P 和约 8 个方法全部组合起来，粗略就可能达到数千次完整任务执行；还没有计入局部数据采集、模型训练种子、消融和失败案例复验。

因此，不能一开始就同时开发所有模块。必须先用少量实验判断场景和研究假设是否成立，再逐步扩大样本规模。

五、我建议把原方案调整成一条更稳妥的执行路线

不建议推翻 v3.1，而是保留核心研究问题，压缩前期任务，把每一步的准入条件变得更明确。

01

先完成 T1 和强规划基线
阶段目标：验证问题，而不是训练模型。
实现双合法停放位、双目标通道及完整任务判据。比较固定终态、随机策略、完整几何规划和允许重规划的 TAMP。先证明不同任务目标下确实存在终态选择翻转。
通过条件： A/B 均局部合法且可执行，不同目标下最优选择确实发生翻转。

02

验证 T2-D 的真实动态依赖
阶段目标：先证明物理机制，不急着引入风险算法。
实现 gate 的真实接触、运动和工具复用，确保 S5 根据实际工具位姿重新规划。检查接触日志、gate 运动、重抓结果和共同控制器的可靠性。
通过条件： 局部成功与远期执行结果存在可复现的差异，并有物理因果证据解释。

03

只在 T2-D 通过后开发概率效果模型和 CVaR
阶段目标：证明算法的独立贡献。
先用小规模数据训练局部效果模型，完成状态转移预测与概率校准。随后比较同模型的 mean-only、mean+CVaR、mean+variance，以及稳健 TAMP。
通过条件： 在预先定义的连续损失、任务成功率或计算成本指标上获得有统计依据的收益。

04

完成 T3、OOD 和核心消融

阶段目标：证明泛化，而不是仅在单个场景中有效。
冻结训练与验证配置，评估未见任务图和物理参数，报告所有尝试的完整成功率、置信区间、连续风险指标及资源开销。
通过条件： 至少有明确的未见组合测试证据，且主结果可以独立复现。

05

最后考虑实机验证和论文定稿
先确定仿真中的主要结论，再用真实 Panda 平行夹爪做一个范围有限、控制条件明确的验证实验。实机实验不应成为前期开发的阻塞因素。
这一顺序基本保留了原方案的科学目标，但能避免在 T2 的物理机制还没有站稳时，就投入大量时间开发复杂模型。文件本身也明确要求从 T1 环境和统一执行契约开始，而不是先训练新网络。
FVS_Compose_v3.1.docx

六、这条路线需要多长时间?

下面是我的工程估计，不是保证工期。假设现有 ManiSkill 环境可复用、基础 Panda 技能能够正常执行，并且主要由一名熟悉 Python 和机器人仿真的研究者推进。

阶段	粗略时间
环境桥接、统一执行接口、T1 和基线	2–4 周
T2 动态场景、接触验证、可靠重抓	3–6 周
局部数据采集、概率效果模型、风险方法	

4–8 周
T3、OOD、核心对比与统计分析
4–8 周
论文整合、复核及可选实机实验
4–8 周或更长

这些阶段可以部分并行，实际时间也可能因场景调试而明显增加。
我的判断是：

- 核心仿真研究： 如果已有代码和技能可以复用，约 3–6 个月是一个可考虑的规划区间。
- 完整版研究： 如果要完成全部任务图、强基线、系统消融、实机验证和高质量论文复核，按 6–12 个月甚至更长来规划更稳妥。
- T-RO 投稿： 即使实验按计划完成，也不能保证达到录用要求；决定性因素仍是方法贡献和实验证据，而不只是工作量。

七、最终结论：应该继续，但要明确止损条件

我认为这份研究路线值得继续推进，不需要推倒重来。它最大的优点是研究问题明确，实验递进合理，而且已经考虑到隐藏物理参数、CVaR 排序同质化、真实重抓和强规划基线等关键问题。
不过，建议你把下面三条作为不可妥协的原则：

1. T1 不成立，不要急着训练模型。 先解决任务构造问题。
2. T2 没有真实物理因果证据，不要宣称远期风险推理有效。 先排除技能执行器和规划器的错误。
3. B6 没有超过 B4M、B3R 等关键对照，就不要为了保住原定故事而反复调整实验。 如果优势仅体现在计算成本、后缀标签成本或连续物理损失上，就围绕真实成立的优势重新定位贡献；如果没有可靠优势，就缩小或更换方法主张。

这也意味着，即使 CVaR 最终没有带来额外收益，前面的实验仍可能形成有价值的研究结果，只是不能再把尾部风险优化作为核心创新。
我最推荐的下一步是：先审计已有 T0 代码和数据，然后只实现 T1 的真实 ManiSkill 环境、统一执行接口，以及 B0/B2/B3 的第一轮比较。 在这一步完成之前，不要大规模采集训练数据，也不要投入时间实现全部 B0–B6。
只要把研究目标从“一次性完成所有实验”改成“每过一道关卡，再决定是否投入下一阶段”，这条路线就有较现实的完成机会。

思考了 3m 46s >

ChatGPT 说：

FVS_Compose_v31_双专家评审实施决议.md
文档

你说：

好的，现在需要完整代码，实施路径，创新等东西

思考了 12m 54s >

ChatGPT 说：

RESEARCH_AND_INNOVATION.md
文档

EXECUTION_RUNBOOK.md
文档

BASELINE_AND_EXPERIMENT_MATRIX.md
文档

IMPLEMENTATION_MATRIX.md
文档

你说：

我怎么做实验啊，不知道看哪个

思考了 31s >

ChatGPT 说：

你说：

第二步结果：(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat experiments/t1_smoke_lower/episodes.csv
goal,seed,pad,env_id,s2_legal,successor_valid,full_success,run_status,initial_hash,initial_blue_xyz,green_dx_m,blue_dx_m,s4_executed,s4_local_success
lower,0,natural,tools_usage_t1_lower,True,True,True,COMPLETED,39fa4120fc74d51dc3e5843ad4afe6c191da7e219d9ea868ac7db88065ae3b0c,"[0.11459382623434067, -0.3056144714355469, 0.019999999552965164]",0.142579,-0.305725,True,True
lower,0,dock,tools_usage_t1_lower,True,True,False,COMPLETED,39fa4120fc74d51dc3e5843ad4afe6c191da7e219d9ea868ac7db88065ae3b0c,"[0.11459382623434067, -0.3056144714355469, 0.019999999552965164]",0.001128,-0.30839,True,False
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat experiments/t1_smoke_lower/audit_summary.json

```
{
  "num_physical_episodes": 2,
  "num_complete_pairs": 1,
  "goals": {
    "lower": {
      "legal_both": 1,
      "natural_only_wins": 1,
      "dock_only_wins": 0
    },
    "upper": {
      "legal_both": 0,
      "natural_only_wins": 0,
      "dock_only_wins": 0
    }
  },
  "g1_provisional_flip": false,
  "g1_formal_pass": false,
  "formal_reason": "natural is not yet a fixed validated SE3 docking region; full B2/B3 not implemented",
  "limitations": [
    "20-episode pilot is not statistical significance",
    "geometry_proxy is 2D only",
    "no complete TAMP or IK reachability checks",
```

"the upper lane remains unvalidated"
]
第三部结果：(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat experiments/t1_smoke_upper/episodes.csv
goal,seed,pad,env_id,s2_legal,successor_valid,full_success,run_status,initial_hash,initial_blue_xyz,green_dx_m,blue_dx_m,s4_executed,s4_local_success
upper,0,natural,tools_usage_t1_upper,False,False,False,COMPLETED,0bceac8363804bc9443f6a0bdf77a2bea1db6dbac752cf45563b4abc3b8cf3f5,"[0.11459382623434067, -0.3056144714355469, 0.019999999552965164]",0.0115,-0.208163,False,
upper,0,dock,tools_usage_t1_upper,False,False,False,COMPLETED,0bceac8363804bc9443f6a0bdf77a2bea1db6dbac752cf45563b4abc3b8cf3f5,"[0.11459382623434067, -0.3056144714355469, 0.019999999552965164]",0.0115,-0.211988,False,

思考了 2m 12s >

ChatGPT 说：

你说：

========== UPPER natural ==========
S1_grasp_l_tool planner= True physical= True blue_dx= -0.0 green_dx= 0.0 error=
S2_hook_and_pull planner= True physical= True blue_dx= -0.208163 green_dx= 0.0115 error=

绿色方块各阶段水平位移：
initial 0.00 mm
S1_grasp_l_tool 0.00 mm
S2_hook_and_pull 11.75 mm

========== UPPER dock ==========
S1_grasp_l_tool planner= True physical= True blue_dx= -0.0 green_dx= 0.0 error=
S2_hook_and_pull planner= True physical= True blue_dx= -0.211988 green_dx= 0.0115 error=

绿色方块各阶段水平位移：
initial 0.00 mm

S1_grasp_l_tool 0.00 mm
S2_hook_and_pull 11.75 mm
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$

思考了 4m 40s >

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat experiments/t1_v2_seed000/episodes.csv
goal,seed,pad,x,y,env_id,run_status,s1_executed,s2_executed,s3_executed,s4_executed,s1_ok,s2_local_ok,s2_legal,s3_ok,s4_ok,full_success,s2_green_shift_mm,s2_i_shift_mm,park_error_mm,park_static,green_dx_m,blue_dx_m,initial_hash,total_steps,error
lower,0,south,-0.2,-0.12,tools_usage_t1v2_lower,COMPLETED,True,True,True,True,True,True,True,True,False,False,0.001,0.005,7.309,True,0.001128,-0.30839,39fa4120fc74d51dc3e5843ad4afe6c191da7e219d9ea868ac7db88065ae3b0c,673,
lower,0,north,-0.2,0.16,tools_usage_t1v2_lower,COMPLETED,True,True,False,False,True,False,False,False,False,False,0.001,0.249,45.253,True,0.0,-0.30839,39fa4120fc74d51dc3e5843ad4afe6c191da7e219d9ea868ac7db88065ae3b0c,418,
upper,0,south,-0.2,-0.12,tools_usage_t1v2_upper,COMPLETED,True,True,True,True,True,True,True,True,True,True,0.001,0.005,7.309,True,0.140011,-0.30839,ebff5ea0b1effd2d3767a5249b347963cc1f635d430ce24318fe24ba2d3b5e03,681,
upper,0,north,-0.2,0.16,tools_usage_t1v2_upper,COMPLETED,True,True,False,False,True,False,False,False,False,False,0.001,0.249,45.253,True,0.0,-0.30839,ebff5ea0b1effd2d3767a5249b347963cc1f635d430ce24318fe24ba2d3b5e03,418,
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat experiments/t1_v2_seed000/audit_summary.json

```
{
  "num_physical_episodes": 4,
  "num_complete_pairs": 2,
  "per_goal": {
    "lower": {
      "valid_pairs": 0,
      "hypothesis_supported_pairs": 0
    },
```

```
    "upper": {
      "valid_pairs": 0,
      "hypothesis_supported_pairs": 0
    }
  },
  "g1_provisional_flip": false,
  "g1_formal_pass": false,
  "formal_reason": "S2 two-pad T1-v2 is new and unvalidated;
B2 full 3D and B3 TAMP not implemented",
  "limitations": [
    "same 4-skill controller used but goals have different green
initial positions",
    "S2 3mm collateral, S2 35mm parking precision frozen",
    "2D geometry is not formal collision planning",
    "development seed(s) are not held-out test",
    "do not treat non-executed S3/S4 as physical failures"
  ]
}(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/real_t1$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/real_t1$ clear
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/real_t1$ cat
experiments/t1_v2_seed000/episodes.csv
goal,seed,pad,x,y,env_id,run_status,s1_executed,s2_executed,s3
_executed,s4_executed,s1_ok,s2_local_ok,s2_legal,s3_ok,s4_ok,
full_success,s2_green_shift_mm,s2_i_shift_mm,park_error_mm,
park_static,green_dx_m,blue_dx_m,initial_hash,total_steps,error
lower,0,south,-0.2,-0.12,tools_usage_t1v2_lower,COMPLETED,Tr
ue,True,True,True,True,True,True,True,True,False,False,0.001,0.005,7.3
09,True,0.001128,-0.30839,39fa4120fc74d51dc3e5843ad4afe6c
191da7e219d9ea868ac7db88065ae3b0c,673,
lower,0,north,-0.2,0.16,tools_usage_t1v2_lower,COMPLETED,Tru
e,True,True,False,False,True,False,False,False,False,False,0.001,0.249,4
5.253,True,0.0,-0.30839,39fa4120fc74d51dc3e5843ad4afe6c19
1da7e219d9ea868ac7db88065ae3b0c,418,
upper,0,south,-0.2,-0.12,tools_usage_t1v2_upper,COMPLETED,T
rue,True,True,True,True,True,True,True,True,True,True,0.001,0.005,7.3
09,True,0.140011,-0.30839,ebff5ea0b1effd2d3767a5249b3479
63cc1f635d430ce24318fe24ba2d3b5e03,681,
upper,0,north,-0.2,0.16,tools_usage_t1v2_upper,COMPLETED,Tr
ue,True,True,False,False,True,False,False,False,False,False,0.001,0.249,
45.253,True,0.0,-0.30839,ebff5ea0b1effd2d3767a5249b347963
cc1f635d430ce24318fe24ba2d3b5e03,418,
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
```

composition-benchmark/real_t1$ cat experiments/t1_v2_seed000/audit_summary.json

```
{
  "num_physical_episodes": 4,
  "num_complete_pairs": 2,
  "per_goal": {
    "lower": {
      "valid_pairs": 0,
      "hypothesis_supported_pairs": 0
    },
    "upper": {
      "valid_pairs": 0,
      "hypothesis_supported_pairs": 0
    }
  },
  "g1_provisional_flip": false,
  "g1_formal_pass": false,
  "formal_reason": "S2 two-pad T1-v2 is new and unvalidated; B2 full 3D and B3 TAMP not implemented",
  "limitations": [
    "same 4-skill controller used but goals have different green initial positions",
    "S2 3mm collateral, S2 35mm parking precision frozen",
    "2D geometry is not formal collision planning",
    "development seed(s) are not held-out test",
    "do not treat non-executed S3/S4 as physical failures"
  ]
}
```

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$

ChatGPT 说：

你说：

可以的

思考了 7m 36s

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$ cat

```
experiments/terminal_harvest_v1/candidates.csv
candidate_id,mode,target_xy,historical_observations,valid_s2_o
bservations,valid_s2_s3_observations,eligible_for_replay,termin
al_kind,source_goals
dock_x-0.200_y-0.120,custom,"[-0.2,
-0.12]",3,3,3,True,COMMANDED_FIXED_XY_NOT_SE3,"lower,up
per"
release_observed_family,release,,1,1,1,True,STATE_DEPENDENT
_RELEASE_FAMILY,lower
dock_x-0.200_y+0.160,custom,"[-0.2,
0.16]",2,0,0,False,COMMANDED_FIXED_XY_NOT_SE3,"lower,up
per"
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/real_t1$ python -u
terminal_state_validator.py replay \
>   --catalog experiments/terminal_harvest_v1/catalog.json \
>   --env-id tools_usage_t1v2_lower \
>   --seeds 1 \
>   --max-candidates 2 \
>   --out experiments/terminal_replay_seed001
```

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/real_t1$ cat
experiments/terminal_replay_seed001/replays.csv
candidate_id,mode,terminal_kind,seed,target_xy,run_status,s2_
valid,s3_executed,s3_ok,s4_executed,s4_ok,blue_dx_m,green_s
hift_mm,i_shift_mm,parking_error_mm,l_static,ground_obb_bo
ttom_z_m,reason,error,s1_ok,s2_local_ok,l_released,geometry_
check,total_steps,execution_error
release_observed_family,release,STATE_DEPENDENT_RELEASE_
FAMILY,1,,COMPLETED,True,True,True,True,True,-0.309635,0.000
8,0.0009,,True,-2.2508762256845394e-
08,S2_VALID_AND_S3_PASSED,,True,True,True,SOURCE_BOX_GE
OMETRY_BOTTOM_ONLY,589,
dock_x-0.200_y-
0.120,custom,COMMANDED_FIXED_XY_NOT_SE3,1,"[-0.2,
-0.12]",COMPLETED,True,True,True,True,False,-0.311854,0.0008,
0.0009,7.0715,True,-2.109374933403907e-
08,S2_VALID_AND_S3_PASSED,,True,True,True,SOURCE_BOX_GE
OMETRY_BOTTOM_ONLY,693,
```

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$

ChatGPT 说：

你说：

好的，继续

思考了 8m 3s >

ChatGPT 说：

你说：

```
===== T1 OFFLINE FLIP AUDIT =====
Observed L Release: {'p': [-0.34624868631362915,
-0.3536216914653778, 0.02499997802078724], 'q_wxyz':
[0.9999862909317017, 1.0691517005056994e-08,
-1.9129080541802068e-08, 0.005247443914413452]}
Observed L South:   {'p': [-0.20686572790145874,
-0.11830634623765945, 0.024999985471367836], 'q_wxyz':
[0.9999946355819702, -4.0516145105584656e-08,
-5.2713113873892326e-09, 0.0032793134450912476]}
Known Lower 2.5D check: Release overlap= False South
overlap= True
Physical S4 labels from replays.csv: {'release': {'s2_valid': True,
's3_ok': True, 's4_executed': True, 's4_ok': True}, 'south':
{'s2_valid': True, 's3_ok': True, 's4_executed': True, 's4_ok':
False}}
Proxy agrees with physical outcomes: True
Reverse candidates (initial-contact filtered): 75
S2 swept-motion uncertainty among these: 63
Decision:
REVERSE_GEOMETRIC_HYPOTHESES_ONLY_REQUIRE_S2_AND_
S4_PHYSICAL_VALIDATION
Saved report:
experiments/t1_reverse_geometry_seed001/report.json
Saved candidate table:
experiments/t1_reverse_geometry_seed001/reverse_flip_candi
dates.csv
Saved preview:
experiments/t1_reverse_geometry_seed001/geometry_map.sv
g
```

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/real_t1$

思考了 2m 28s >

ChatGPT 说：

你说：

反向几何候选总数: 75
未触发S2近似扫掠警告: 12

编号 | Green X | Green Y | Release间隙(mm) | South间隙(mm)
01 | +0.100 | -0.390 | -2.32 | 222.25
02 | +0.080 | -0.390 | -11.93 | 222.12
03 | +0.060 | -0.390 | -12.14 | 221.99
04 | +0.040 | -0.390 | -12.35 | 221.86
05 | +0.020 | -0.390 | -12.56 | 221.75
06 | +0.000 | -0.390 | -12.77 | 221.63
07 | -0.020 | -0.390 | -12.98 | 221.52
08 | -0.040 | -0.390 | -13.19 | 221.41
09 | -0.060 | -0.390 | -13.4 | 221.3
10 | -0.080 | -0.390 | -13.61 | 221.19
11 | -0.100 | -0.390 | -13.8 | 221.07
12 | -0.120 | -0.390 | -13.86 | 220.96

思考了 2m 36s >

ChatGPT 说：

你说：

好的，建议一次给出详细完善的下一步行动，要不然一直在尝试一直测试，不知道该干什么

思考了 5m 30s >

ChatGPT 说：

FVS_Compose_T2_D_冻结实施协议_20261029.md
文档

你说：

好的，你直接弄

思考了 9m 28s >

ChatGPT 说：

ROADMAP_AND_INNOVATION.md
文档

README.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W0_W1_Executable_20261029$ EXP_ROOT="../real_t1/experiments" bash RUN_W0_W1.sh
test_absent_cannot_claim_failure (test_contract.ContractTest) ... ok
test_absent_stages_have_null_values (test_contract.ContractTest) ... ok
test_s4_failure_preserves_s5_unexecuted (test_contract.ContractTest) ... ok
test_stop_at_s2 (test_contract.ContractTest) ... ok
test_success (test_contract.ContractTest) ... ok
test_no_tool_teleport_during_skills (test_source_guard.SourceGuard) ... ok
test_w1_only_initialization_teleports (test_source_guard.SourceGuard) ... ok
test_audit_rejects_inside_old_dir (test_w0.AuditTest) ... ok
test_w0_read_only_and_i_release (test_w0.AuditTest) ... ok

Ran 9 tests in 0.049s

OK
W0: SOURCE_EVIDENCE_INDEXED files= 75 S4 I release samples= 19 I still grasped= 4
Saved: w0_audit_20261029_035810/audit_report.json
2026-10-29 03:58:39,780 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation

issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 03:58:40,047 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:40,367 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:40,699 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:41,039 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:41,363 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[seed000_release] S1_GetL: steps=114 plan=True phys=True valid=True
[seed000_release] S2_HookBlueAndDock: steps=219 plan=True phys=True valid=True

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300772 -0.00158694 0.172769]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.1000126   0.02803093 0.02557869]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1221   0.02853507 0.21965088]
[S3] 直杆位移: [-0.00440425 0.00061716 0.19465101]

[S3] 抬起后是否抓住: True
[seed000_release] S3_GetI: steps=83 plan=True phys=True valid=True
[seed000_release] S4_PushGreenAndReleaseI: steps=177 plan=True phys=True valid=True
[seed000_release] S5_ReacquireL: steps=112 plan=True phys=True valid=True
[seed000_release] S6_HookYellow: steps=138 plan=True phys=True valid=True
[W1] seed000_release: FULL_SIX_SKILL_SUCCESS failure=None
2026-10-29 03:58:43,076 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 03:58:43,395 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:43,735 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:44,069 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:44,399 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 03:58:44,733 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[seed000_south] S1_GetL: steps=114 plan=True phys=True valid=True
L工具当前位置: [-0.3203 -0.3641 0.0253]

```
  目标停放XY: [-0.2  -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm
l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.2mm
l_grasped=True
[seed000_south] S2_HookBlueAndDock: steps=309 plan=True
phys=True valid=True

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585  0.02791669  0.02499987]
[S3] TCP 初始位置: [-0.00491277 -0.00901996  0.17214051]
[S3] 目标抓取位置: [-0.09769584  0.02791655  0.02499987]
[S3] 目标抓取四元数: [ 0.0000000e+00  1.0000000e+00
-3.4568034e-06  0.0000000e+00]
[S3] 到达后 TCP: [-0.10004333  0.02779158  0.02592658]
[S3] 到达后直杆: [-0.11769594  0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211583  0.0290028   0.21965724]
[S3] 直杆位移: [-0.00441998  0.00108611  0.19465737]
[S3] 抬起后是否抓住: True
[seed000_south] S3_GetI: steps=89 plan=True phys=True
valid=True
[seed000_south] S4_PushGreenAndReleaseI: steps=183
plan=True phys=True valid=True
[seed000_south] S5_ReacquireL: steps=94 plan=True
phys=True valid=True
[seed000_south] S6_HookYellow: steps=141 plan=True
phys=True valid=True
[W1] seed000_south: FULL_SIX_SKILL_SUCCESS failure=None

==== W1 SUMMARY ====
{
  "phase": "W1",
  "attempted": 2,
  "full_six_success": 2,
  "by_policy": {
    "release": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
```

```
    },
    "south": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
    }
  },
  "positive_capability_observed": true,
  "gate":
"DO_NOT_START_GATE_UNTIL_POSITIVE_SIX_SKILL_TRACE",
  "warning": "Success here does not validate T2 Gate physical
dependence."
}
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T2D_W0_W1_Executable_20261029$
```


思考了 8m 56s >

ChatGPT 说：

你说：


```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$ bash
RUN_W2.sh w2_gate_seed000
test_absent_cannot_claim_failure (test_contract.ContractTest) ...
ok
test_absent_stages_have_null_values
(test_contract.ContractTest) ... ok
test_s4_failure_preserves_s5_unexecuted
(test_contract.ContractTest) ... ok
test_stop_at_s2 (test_contract.ContractTest) ... ok
test_success (test_contract.ContractTest) ... ok
test_no_tool_teleport_during_skills
(test_source_guard.SourceGuard) ... ok
test_w1_only_initialization_teleports
(test_source_guard.SourceGuard) ... ok
test_audit_rejects_inside_old_dir (test_w0.AuditTest) ... ok
test_w0_read_only_and_i_release (test_w0.AuditTest) ... ok
test_contact_impulse_norm
(test_w2_contact.W2ContactUnitTest) ... ok
test_gate_scene_fixed_from_historical_upper
(test_w2_contact.W2ContactUnitTest) ... ok
test_monitor_rejects_unobservable_as_positive
```

(test_w2_contact.W2ContactUnitTest) ... ok
test_zero_impulse_not_contact
(test_w2_contact.W2ContactUnitTest) ... ok

-
Ran 13 tests in 0.102s

OK
OUTPUT_EXISTS: w2_gate_seed000
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat
w2_gate_seed000/episodes.csv
cat: w2_gate_seed000/episodes.csv: 没有那个文件或目录
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat
w2_gate_seed000/w2_contact_summary.json
cat: w2_gate_seed000/w2_contact_summary.json: 没有那个文件
或目录

思考了 1m 55s >

ChatGPT 说：

你说：

[W2] 当前新实验输出目录：w2_gate_contact_20261029_044519
Traceback (most recent call last):
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029/scripts/01_r
un_w2.py", line 326, in <module>
    if __name__=='__main__':main()
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029/scripts/01_r
un_w2.py", line 308, in main
    one=one_episode(seed,policy,opt,opt.out)
  File "/home/users/lemon/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029/scripts/01_r
un_w2.py", line 109, in one_episode
    import tool_skills_t1 as core

File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029/tool_skills_t1.py", line 26, in <module>
import t1_lane # noqa: F401; registers two T1 physical task variants
ModuleNotFoundError: No module named 't1_lane'

思考了 2m 8s >

ChatGPT 说：

你说：

怎么查看，在哪里

ChatGPT 说：

你说：

seed,policy,env_id,episode,initial_hash,status,full_success,first_failure_stage,**s4_executed,**gate_displacement_x_m,gate_lateral_drift_mm,i_gate_contact_observed,i_gate_contact_steps,num_steps,wall_seconds,error
0,release,tools_usage_t2d_w2_sliding_gate,seed000_release,661da0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b534782b5c7,FULL_SIX_SKILL_SUCCESS,True,,**True,**0.08986757695674896,1.5339255332946777,True,29,832,4.318,
0,south,tools_usage_t2d_w2_sliding_gate,seed000_south,661da0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b534782b5c7,PLANNER_OR_EXECUTION_FAILURE,False,S4_PushGreenAndReleaseI,**True,**-3.5762786865234375e-07,0.0006556510925292969,False,0,512,3.392,PlanningFailure: 运动规划失败: S4/approach green

{
"phase": "W2_CONTACT_PILOT",
"attempted": 2,
"full_six_success": 1,
"by_policy": {
"release": {
"attempted": 1,
"full_success": 1,

```
      "failure_stages": []
    },
    "south": {
      "attempted": 1,
      "full_success": 0,
      "failure_stages": [
        "S4_PushGreenAndReleaseI"
      ]
    }
  },
  "positive_capability_observed": true,
  "gate": "W2_CONTACT_PILOT_NOT_W3_CAUSALITY",
  "warning": "A guide-constrained cube carriage is a movable gate CONTACT PILOT, not an articulated prismatic Gate. It is far from both L tool terminals. No delayed-effect claim."
}
```


思考了 2m 5s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ python -u scripts/01_run_w2.py \
>     --seed-start 0 \
>     --num-seeds 1 \
>     --policy both \
>     --rrt-fallback \
>     --out "$OUT" \
>     --traceback

2026-10-29 05:08:17,203 - mani_skill  - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 05:08:17,471 - mani_skill  - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
```

2026-10-29 05:08:17,792 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:18,116 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:18,438 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:18,766 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:19,106 - mani_skill - WARNING - No initial pose set for actor builder of gate_rail_positive, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:19,436 - mani_skill - WARNING - No initial pose set for actor builder of gate_rail_negative, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[seed000_release] S1_GetL: steps=114 plan=True phys=True valid=True
[seed000_release] S2_HookBlueAndDock: steps=219 plan=True phys=True valid=True

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300772 -0.00158694 0.172769]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.1000126   0.02803093 0.02557869]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]

[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1221 0.02853507 0.21965088]
[S3] 直杆位移: [-0.00440425 0.00061716 0.19465101]
[S3] 抬起后是否抓住: True
[seed000_release] S3_GetI: steps=83 plan=True phys=True valid=True
[seed000_release] S4_PushGreenAndReleaseI: steps=169 plan=True phys=True valid=True
[seed000_release] S5_ReacquireL: steps=107 plan=True phys=True valid=True
[seed000_release] S6_HookYellow: steps=140 plan=True phys=True valid=True
[W2] seed000_release: FULL_SIX_SKILL_SUCCESS failure=None contact={'s4_control_steps_logged': 169, 'contact_observable': True, 'contact_steps_gt_1e_7_ns': 29, 'peak_impulse_ns': 0.006224642318978204, 'integrated_observed_impulse_ns': 0.03638056236435851, 'monitor_errors': []}
2026-10-29 05:08:21,189 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 05:08:21,508 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:21,838 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:22,170 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:22,513 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:22,841 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial

poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:23,170 - mani_skill - WARNING - No initial pose set for actor builder of gate_rail_positive, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 05:08:23,515 - mani_skill - WARNING - No initial pose set for actor builder of gate_rail_negative, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[seed000_south] S1_GetL: steps=114 plan=True phys=True valid=True
  L工具当前位置: [-0.3203 -0.3641 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.2mm l_grasped=True
[seed000_south] S2_HookBlueAndDock: steps=309 plan=True phys=True valid=True

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00491277 -0.00901996 0.17214051]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004333 0.02779158 0.02592658]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211598 0.02868463 0.21966594]
[S3] 直杆位移: [-0.00442014 0.00076794 0.19466607]
[S3] 抬起后是否抓住: True
[seed000_south] S3_GetI: steps=89 plan=True phys=True valid=True
screw plan failed

[seed000_south] S4_PushGreenAndReleaseI: steps=168 plan=True phys=True valid=True
[seed000_south] S5_ReacquireL: steps=97 plan=True phys=True valid=True
[seed000_south] S6_HookYellow: steps=143 plan=True phys=True valid=True
[W2] seed000_south: FULL_SIX_SKILL_SUCCESS failure=None contact={'s4_control_steps_logged': 168, 'contact_observable': True, 'contact_steps_gt_1e_7_ns': 20, 'peak_impulse_ns': 0.0031859059449567332, 'integrated_observed_impulse_ns': 0.02767820742258398, 'monitor_errors': []}

==== W2 SUMMARY ====

```
{
  "phase": "W2_CONTACT_PILOT",
  "attempted": 2,
  "full_six_success": 2,
  "by_policy": {
    "release": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
    },
    "south": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
    }
  },
  "positive_capability_observed": true,
  "gate": "W2_CONTACT_PILOT_NOT_W3_CAUSALITY",
  "warning": "A guide-constrained cube carriage is a movable gate CONTACT PILOT, not an articulated prismatic Gate. It is far from both L tool terminals. No delayed-effect claim."
}
```

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat "$OUT/episodes.csv"

```
seed,policy,env_id,episode,initial_hash,status,full_success,first_failure_stage,s4_executed,gate_displacement_x_m,gate_lateral_drift_mm,i_gate_contact_observed,i_gate_contact_steps,num_s
```

teps,wall_seconds,error
0,release,tools_usage_t2d_w2_sliding_gate,seed000_release,66
1da0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b
534782b5c7,FULL_SIX_SKILL_SUCCESS,True,,True,0.0898675769
5674896,1.5339255332946777,True,29,832,4.338,
0,south,tools_usage_t2d_w2_sliding_gate,seed000_south,661d
a0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b53
4782b5c7,FULL_SIX_SKILL_SUCCESS,True,,True,0.089476719498
63434,2.611011266708374,True,20,920,4.213,
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat
"$OUT/w2_contact_summary.json"

```
{
  "phase": "W2_CONTACT_PILOT",
  "attempted": 2,
  "full_six_success": 2,
  "by_policy": {
    "release": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
    },
    "south": {
      "attempted": 1,
      "full_success": 1,
      "failure_stages": []
    }
  },
  "positive_capability_observed": true,
  "gate": "W2_CONTACT_PILOT_NOT_W3_CAUSALITY",
  "warning": "A guide-constrained cube carriage is a movable
gate CONTACT PILOT, not an articulated prismatic Gate. It is
far from both L tool terminals. No delayed-effect claim."
}(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$
```


思考了 4m 49s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ W2_DIR=$(find . -maxdepth 2 -name w2_contact_summary.json \
>   -printf '%T@ %h\n' | sort -nr | head -1 | cut -d' ' -f2-)
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ echo "分析实验：$W2_DIR"
分析实验：./w2_rrt_seed000_20261029_050748
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ python W3_Offline_Geometry_Decision_v1/w3_audit.py \
>   --w2 "$W2_DIR" \
>   --out w3_offline_seed000
===== W3 OFFLINE PREFLIGHT =====
release: six=True contact_steps=29 L_shift_during_S4=0.00mm L-Gate_xy_min=584.37mm L-Rail_xy_min=519.24mm
south: six=True contact_steps=20 L_shift_during_S4=0.00mm L-Gate_xy_min=313.88mm L-Rail_xy_min=252.58mm
Decision: CURRENT_GATE_HAS_NO_OBSERVED_L_OCCUPANCY_MECHANISM
Read-only. NO physical experiment run. NO W3 causal claim.
Saved: w3_offline_seed000/w3_preflight.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat w3_offline_seed000/w3_policy_distances.csv
policy,all_six_physical_success,observed_contact_steps,l_move_s2_to_s4_mm,gate_dx_mm,gate_dy_mm,min_l_to_gate_xy_clearance_mm_conditional,l_to_static_rails_xy_clearance_mm_at_s2,next_stage_s5_physical_success
release,True,29,0.0,89.86793458461761,-1.5332698822021484,584.367,519.241,True
south,True,20,0.00010746994418465835,89.47707712650299,-2.6103556156158447,313.876,252.58,True
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
```

composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ cat w3_offline_seed000/w3_preflight.json


```
{
  "phase": "W3_OFFLINE_PREFLIGHT_ONLY",
  "source_w2": "/home/users/lemon/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029/w2_rrt_seed000_20261029_050748",
  "source_w2_episodes_sha256": "595ed65e3fdda587eae6677f14a70e1ac7273df54cf287c143f396f5c8b109a4",
  "policies": [
    {
      "policy": "release",
      "all_six_physical_success": true,
      "s2_l_pose": {
        "p": [
          -0.320588082075119,
          -0.36431723833084106,
          0.024999991059303284
        ],
        "q_wxyz": [
          0.9999908208847046,
          -3.846864871093203e-08,
          -1.4398917613789308e-08,
          0.004299938678741455
        ],
        "is_grasped": false,
        "is_static": true,
        "linear_velocity": [
          0.0,
          0.0,
          0.0
        ],
        "angular_velocity": [
          0.0,
          0.0,
          0.0
        ]
      },
      "s4_l_pose": {
        "p": [
          -0.320588082075119,
          -0.36431723833084106,
```

```
        0.024999991059303284
       ],
       "q_wxyz": [
        0.9999908208847046,
        -3.846864871093203e-08,
        -1.4398917613789308e-08,
        0.004299938678741455
       ],
       "is_grasped": false,
       "is_static": true,
       "linear_velocity": [
        0.0,
        0.0,
        0.0
       ],
       "angular_velocity": [
        0.0,
        0.0,
        0.0
       ]
      },
      "l_move_s2_to_s4_mm": 0.0,
      "s4_gate_initial_xyz": [
       0.14999964833259583,
       0.28489935398101807,
       0.019999882206320763
      ],
      "s4_gate_final_xyz": [
       0.23986758291721344,
       0.2833660840988159,
       0.01999991200864315
      ],
      "gate_dx_mm": 89.86793458461761,
      "gate_dy_mm": -1.5332698822021484,
      "logged_s4_control_steps": 169,
      "contact_observable": true,
      "observed_contact_steps": 29,
      "min_l_to_gate_xy_clearance_mm_conditional": 584.367,
      "max_l_to_gate_xy_clearance_mm_conditional": 627.011,
      "l_to_static_rails_xy_clearance_mm_at_s2": 519.241,
      "dynamic_clearance_trace_assumption": "L tool treated
static at S2 pose during S4; only supported if S2->S4 L
displacement is negligible; XY projection only",
      "s4_i_released": true,
      "next_stage_s5_physical_success": true,
```

```
  "full_robot_sweep_available": false,
  "min_3d_robot_swept_clearance_mm": null,
  "s5_ik_margin_available": false,
  "s5_ik_margin": null
},
{
  "policy": "south",
  "all_six_physical_success": true,
  "s2_l_pose": {
    "p": [
      -0.20696550607681274,
      -0.11873704195022583,
      0.024999991059303284
    ],
    "q_wxyz": [
      0.9999918341636658,
      5.894432675290773e-08,
      -7.302724469582245e-08,
      0.004052102565765381
    ],
    "is_grasped": false,
    "is_static": true,
    "linear_velocity": [
      0.0,
      0.0,
      0.0
    ],
    "angular_velocity": [
      0.0,
      0.0,
      0.0
    ]
  },
  "s4_l_pose": {
    "p": [
      -0.20696541666984558,
      -0.11873698234558105,
      0.024999989196658134
    ],
    "q_wxyz": [
      0.999991774559021,
      -7.174060812076277e-08,
      -1.059025809627832e-07,
      0.004051834344863892
    ],
```

```
        "is_grasped": false,
        "is_static": true,
        "linear_velocity": [
          0.0,
          0.0,
          0.0
        ],
        "angular_velocity": [
          0.0,
          0.0,
          0.0
        ]
      },
      "l_move_s2_to_s4_mm": 0.00010746994418465835,
      "s4_gate_initial_xyz": [
        0.14999964833259583,
        0.28489935398101807,
        0.019999882206320763
      ],
      "s4_gate_final_xyz": [
        0.23947672545909882,
        0.2822889983654022,
        0.019999910145998
      ],
      "gate_dx_mm": 89.47707712650299,
      "gate_dy_mm": -2.6103556156158447,
      "logged_s4_control_steps": 168,
      "contact_observable": true,
      "observed_contact_steps": 20,
      "min_l_to_gate_xy_clearance_mm_conditional": 313.876,
      "max_l_to_gate_xy_clearance_mm_conditional": 360.131,
      "l_to_static_rails_xy_clearance_mm_at_s2": 252.58,
      "dynamic_clearance_trace_assumption": "L tool treated
static at S2 pose during S4; only supported if S2->S4 L
displacement is negligible; XY projection only",
      "s4_i_released": true,
      "next_stage_s5_physical_success": true,
      "full_robot_sweep_available": false,
      "min_3d_robot_swept_clearance_mm": null,
      "s5_ik_margin_available": false,
      "s5_ik_margin": null
    }
  ],
  "same_initial_hash": true,
  "both_six_skills_passed": true,
```

```
  "both_gate_physical_contact": true,
  "l_pose_stable_across_s4_at_3mm": true,
  "min_clearance_both_above_50mm_xy": true,
  "decision":
"CURRENT_GATE_HAS_NO_OBSERVED_L_OCCUPANCY_MECH
ANISM",
  "next_action": "Freeze W2; collect actual full robot S5
sweep/IK feasibility and design a causally interacting W3
mechanism from reachability, not by relocating W2 carriage at
random.",
  "limits": [
    "2D L/gate footprint, not continuous 3D robot collision",
    "L position throughout S4 approximated by S2 endpoint;
check stability field",
    "W2 contains no true robot arm swept trajectory for S5",
    "cannot prove all alternative S5 paths infeasible",
    "no causal effect of S2 terminal on W3 has been tested"
  ]
}(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$
```


思考了 7m 19s >

ChatGPT 说：

你说：


```
W3B: TRACE_COMPLETE_FOR_GEOMETRY_FEASIBILITY_ONLY
seed000_release S4/S5 traced= 169 107 links= 15 min link-
origin to current gate 3D m= 0.387401
seed000_south S4/S5 traced= 168 97 links= 15 min link-origin
to current gate 3D m= 0.389016
===== COMPLETE RESULT =====
seed,policy,env_id,episode,initial_hash,status,full_success,first_f
ailure_stage,s4_executed,gate_displacement_x_m,gate_lateral_
drift_mm,i_gate_contact_observed,i_gate_contact_steps,num_s
teps,wall_seconds,error
0,release,tools_usage_t2d_w2_sliding_gate,seed000_release,66
1da0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b
534782b5c7,FULL_SIX_SKILL_SUCCESS,True,,True,0.0898675769
5674896,1.5339255332946777,True,29,832,5.595,
0,south,tools_usage_t2d_w2_sliding_gate,seed000_south,661d
a0db91b95587ccf8ed352846c2905c8316e61f69e40a87675b53
4782b5c7,FULL_SIX_SKILL_SUCCESS,True,,True,0.089476719498
```

63434,2.611011266708374,True,20,920,5.629,
===== W3B DECISION (only this block needs to be shared) =====
DECISION:
TRACE_COMPLETE_FOR_GEOMETRY_FEASIBILITY_ONLY
SAME_INITIAL_STATE: True
seed000_release full_success= True S1-S4_logged= 585 S5_logged= 107 link_count= 15
min_link_origin_to_W2_gate_m= 0.387401 errors= 0
seed000_south full_success= True S1-S4_logged= 680 S5_logged= 97 link_count= 15
min_link_origin_to_W2_gate_m= 0.389016 errors= 0
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$

思考了 6m 41s >

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ bash W3C_CollisionGeometry_Audit_20261029/RUN_W3C.sh
test_bounds_positive_separation (test_w3c.GeometryTests) ... ok
test_missing_mesh_fails_closed (test_w3c.GeometryTests) ... ok
test_quat_normalization (test_w3c.GeometryTests) ... ok
test_synthetic_complete_audit (test_w3c.GeometryTests) ... ok
test_tool_dimensions (test_w3c.GeometryTests) ... ok
test_ascii_stl (test_w3c.MeshParsingTests) ... ok
test_binary_stl (test_w3c.MeshParsingTests) ... ok

Ran 7 tests in 0.016s

OK
===== W3C FINAL GEOMETRY DECISION =====
DECISION:
CURRENT_GATE_FAR_FROM_OBSERVED_S5_COLLISION_BODIES_STOP_W3_CURRENT_LAYOUT
COLLISION_LINKS: 11 MESHES: 9

release: min_S5_robot_to_gate_AABB=308.3mm;
min_S5_L_tool_to_gate_AABB=624.3mm;
nearest_link=panda_hand
south: min_S5_robot_to_gate_AABB=309.1mm;
min_S5_L_tool_to_gate_AABB=359.2mm;
nearest_link=panda_hand
READ_ONLY: True; no new physical episode.
REPORT:
w3c_collision_audit_20261029_053813/w3c_report.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$

ChatGPT 说：

你说：

可以

思考了 12m 39s >

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T2D_W2_Gate_ContactPilot_20261029$ bash W3D_GeometrySynthesis_20261029/RUN_W3D.sh
test_fail_closed_or_candidate_is_frozen (test_e2e.SyntheticEndToEnd) ... ok
test_early_interference_rejected (test_w3d.GatePreflight) ... ok
test_missing_blue_checkpoint_fails_closed (test_w3d.GatePreflight) ... ok
test_south_only_overlap_is_geometric_screen (test_w3d.GatePreflight) ... ok
test_unexecuted_s5_not_distal_success (test_w3d.PhysicalVerdict) ... ok
test_bounded_box_geometry (test_w3d.PhysicsShapes) ... ok
test_segment_fails_if_diagonal (test_w3d.PhysicsShapes) ... ok
test_seven_built_parts_share_one_actor (test_w3d.PhysicsShapes) ... ok
test_world_aabb_orientation (test_w3d.PhysicsShapes) ... ok

```
----------------------------------------------------------------------
-
Ran 9 tests in 0.059s

OK
W3B source: ./w3b_s5_trace_20261029_052717
Geometry output: w3d_geometry_20261029_055625
===== W3D GEOMETRY DECISION =====
DECISION:
STOP_NO_SAMPLED_RIGID_GATE_GEOMETRIC_CANDIDATE
CANDIDATES: 1080 PASSED: 0
LAYOUT: NONE
NO PHYSICAL EPISODE RUN IN THIS PHASE
W3D STOP: No geometrically screened single-rigid-body
mechanism. Do NOT move coordinates or rerun with changed
parameters.
REPORT: w3d_geometry_20261029_055625/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T2D_W2_Gate_ContactPilot_20261029$
```

思考了 1m 14s >

ChatGPT 说：

你说：

好的

思考了 8m 15s >

ChatGPT 说：

README.md
文档

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_StrongPlanner_Audit_20261029$
bash RUN_T0.sh
test_early_failure_not_s4 (test_contract.TestContracts) ... ok
test_good_backend (test_contract.TestContracts) ... ok
test_no_teleport (test_contract.TestContracts) ... ok
test_none_state (test_contract.TestContracts) ... ok
```

```
test_requires_flags (test_contract.TestContracts) ... ok
test_s4_ok (test_contract.TestContracts) ... ok
test_s4_physical (test_contract.TestContracts) ... ok
test_90deg_transform (test_math.MathTests) ... ok
test_box_surface (test_math.MathTests) ... ok
test_i_attached_translation (test_math.MathTests) ... ok
test_invalid_quat (test_math.MathTests) ... ok
test_l_two_parts_and_blue (test_math.MathTests) ... ok
test_quat_roundtrip (test_math.MathTests) ... ok

----------------------------------------------------------------------
-
Ran 13 tests in 0.003s

OK
===== release screw_only =====
2026-10-29 06:12:55,197 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-29 06:12:55,454 - mani_skill  - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:55,771 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_2, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:56,088 - mani_skill  - WARNING - No initial
pose set for actor builder of l_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:56,417 - mani_skill  - WARNING - No initial
pose set for actor builder of i_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
```

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300789 -0.0015866   0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342   0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True
seed000_release S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.306 green_dx=+0.000
seed000_release S4_push: steps=160 planner=True physical=True blue_dx=-0.306 green_dx=+0.143
STATUS FULL_S1_S4_PHYSICAL_SUCCESS green_dx 0.142579
===== south screw_only =====
2026-10-29 06:12:57,697 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 06:12:58,016 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:58,344 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:58,667 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial

poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:12:58,997 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True physical=True blue_dx=-0.308 green_dx=+0.000

seed000_custom S4_push: steps=161 planner=True physical=False blue_dx=-0.308 green_dx=+0.001
[P1 STOP] S4_push 未满足局部成功，停止后续技能
STATUS S4_PHYSICAL_GOAL_FAILURE green_dx 0.001128
===== release screw_rrt_no_scene =====
2026-10-29 06:13:00,480 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 06:13:00,800 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:01,127 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:01,455 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:01,776 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]

[S3] TCP 初始位置: [-0.01300789 -0.0015866 0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178 0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06
0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342 0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True
seed000_release S3_grasp_i_tool: steps=83 planner=True
physical=True blue_dx=-0.306 green_dx=+0.000
seed000_release S4_push: steps=160 planner=True
physical=True blue_dx=-0.306 green_dx=+0.143
STATUS FULL_S1_S4_PHYSICAL_SUCCESS green_dx 0.142579
===== south screw_rrt_no_scene =====
2026-10-29 06:13:03,003 - mani_skill - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-29 06:13:03,319 - mani_skill - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:03,645 - mani_skill - WARNING - No initial
pose set for actor builder of cube_2, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:03,971 - mani_skill - WARNING - No initial
pose set for actor builder of l_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:04,305 - mani_skill - WARNING - No initial
pose set for actor builder of i_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000

```
  L工具当前位置: [-0.3202 -0.3644  0.0253]
  目标停放XY: [-0.2  -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm
l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
================================

seed000_custom S2_hook_and_pull: steps=309 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585  0.02791669  0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931  0.17212337]
[S3] 目标抓取位置: [-0.09769584  0.02791655  0.02499987]
[S3] 目标抓取四元数: [ 0.0000000e+00  1.0000000e+00
-3.4568034e-06  0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359  0.02779171  0.02592734]
[S3] 到达后直杆: [-0.11769594  0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586  0.02900716  0.21965735]
[S3] 直杆位移: [-0.00442001  0.00109047  0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000
seed000_custom S4_push: steps=161 planner=True
physical=False blue_dx=-0.308 green_dx=+0.001
[P1 STOP] S4_push 未满足局部成功，停止后续技能
STATUS S4_PHYSICAL_GOAL_FAILURE green_dx 0.001128
===== release scene_attached_rrt =====
2026-10-29 06:13:05,772 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-29 06:13:06,092 - mani_skill  - WARNING - No initial
```

pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:06,424 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:06,751 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:07,080 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300789 -0.0015866   0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342   0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True
seed000_release S3_grasp_i_tool: steps=83 planner=True

physical=True blue_dx=-0.306 green_dx=+0.000
  [ERROR] S4_push: CapabilityUnavailable: missing:
detach_object
seed000_release S4_push: steps=0 planner=False
physical=False blue_dx=-0.306 green_dx=+0.000
[P1 STOP] S4_push 未满足局部成功，停止后续技能
STATUS UNSUPPORTED_MPLIB_NOT_EVALUATED green_dx 0.0
===== south scene_attached_rrt =====
2026-10-29 06:13:08,035 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 06:13:08,352 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:08,679 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:09,018 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:13:09,351 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True
physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm
l_grasped=True

[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000
  [ERROR] S4_push: CapabilityUnavailable: missing:
detach_object
seed000_custom S4_push: steps=0 planner=False
physical=False blue_dx=-0.308 green_dx=+0.000
[P1 STOP] S4_push 未满足局部成功，停止后续技能
STATUS UNSUPPORTED_MPLIB_NOT_EVALUATED green_dx 0.0
===== T0 REPLANNING DECISION =====
DECISION:
INCONCLUSIVE_ATTACHED_COLLISION_API_UNSUPPORTED
release screw_only FULL_S1_S4_PHYSICAL_SUCCESS
green_dx= 0.142579
south screw_only S4_PHYSICAL_GOAL_FAILURE green_dx=
0.001128
release screw_rrt_no_scene FULL_S1_S4_PHYSICAL_SUCCESS
green_dx= 0.142579
south screw_rrt_no_scene S4_PHYSICAL_GOAL_FAILURE
green_dx= 0.001128
release scene_attached_rrt
UNSUPPORTED_MPLIB_NOT_EVALUATED green_dx= 0.0

south scene_attached_rrt
UNSUPPORTED_MPLIB_NOT_EVALUATED green_dx= 0.0
OUTPUT: t0_strong_planner_20261029_061226_00/results.csv
t0_strong_planner_20261029_061226_00/pre_s4_comparability
.json
===== VIEW =====

```
{
  "decision":
"INCONCLUSIVE_ATTACHED_COLLISION_API_UNSUPPORTED",
  "rows": [
    {
      "seed": 0,
      "policy": "release",
      "variant": "screw_only",
      "status": "FULL_S1_S4_PHYSICAL_SUCCESS",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "True",
      "s4_physical_ok": "True",
      "green_dx_m": 0.142579,
      "wall_seconds": 2.861,
      "planner_calls": 0,
      "collision_cloud_points": null,
      "first_error": null,
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/release_sc
rew_only"
    },
    {
      "seed": 0,
      "policy": "south",
      "variant": "screw_only",
      "status": "S4_PHYSICAL_GOAL_FAILURE",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "True",
      "s4_physical_ok": "False",
      "green_dx_m": 0.001128,
      "wall_seconds": 2.783,
      "planner_calls": 0,
      "collision_cloud_points": null,
```

```
      "first_error": null,
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/south_scre
w_only"
    },
    {
      "seed": 0,
      "policy": "release",
      "variant": "screw_rrt_no_scene",
      "status": "FULL_S1_S4_PHYSICAL_SUCCESS",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "True",
      "s4_physical_ok": "True",
      "green_dx_m": 0.142579,
      "wall_seconds": 2.524,
      "planner_calls": 0,
      "collision_cloud_points": null,
      "first_error": null,
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/release_sc
rew_rrt_no_scene"
    },
    {
      "seed": 0,
      "policy": "south",
      "variant": "screw_rrt_no_scene",
      "status": "S4_PHYSICAL_GOAL_FAILURE",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "True",
      "s4_physical_ok": "False",
      "green_dx_m": 0.001128,
      "wall_seconds": 2.766,
      "planner_calls": 0,
      "collision_cloud_points": null,
      "first_error": null,
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/south_scre
w_rrt_no_scene"
    },
```

```
    {
      "seed": 0,
      "policy": "release",
      "variant": "scene_attached_rrt",
      "status": "UNSUPPORTED_MPLIB_NOT_EVALUATED",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "False",
      "s4_physical_ok": "False",
      "green_dx_m": 0.0,
      "wall_seconds": 2.264,
      "planner_calls": 0,
      "collision_cloud_points": null,
      "first_error": "CapabilityUnavailable: missing:
detach_object",
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/release_sc
ene_attached_rrt"
    },
    {
      "seed": 0,
      "policy": "south",
      "variant": "scene_attached_rrt",
      "status": "UNSUPPORTED_MPLIB_NOT_EVALUATED",
      "s1_valid": true,
      "s2_valid": true,
      "s3_valid": true,
      "s4_executed": true,
      "s4_planner_ok": "False",
      "s4_physical_ok": "False",
      "green_dx_m": 0.0,
      "wall_seconds": 2.496,
      "planner_calls": 0,
      "collision_cloud_points": null,
      "first_error": "CapabilityUnavailable: missing:
detach_object",
      "data_dir":
"t0_strong_planner_20261029_061226_00/episodes/south_sce
ne_attached_rrt"
    }
  ],
  "same_pre_s4": {
    "release": {
```

```
      "screw_only": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      },
      "screw_rrt_no_scene": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      },
      "scene_attached_rrt": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      }
    },
    "south": {
      "screw_only": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      },
      "screw_rrt_no_scene": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      },
      "scene_attached_rrt": {
        "max_absolute_difference": 0.0,
        "same_pre_s4_at_1mm": true
      }
    }
  }
}(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_StrongPlanner_Audit_20261029$
```


思考了 2m 54s >

ChatGPT 说：

你说：

```
===== MPLIB API =====
VERSION: 0.1.1
plan_screw (self, target_pose, qpos, qpos_step=0.1,
time_step=0.1, use_point_cloud=False, use_attach=False,
wrt_world=True, verbose=False)
plan_qpos_to_pose (self, goal_pose, current_qpos,
mask=None, time_step=0.1, rrt_range=0.1, planning_time=1,
fix_joint_limits=True, use_point_cloud=False, use_attach=False,
```

```
wrt_world=True, planner_name='RRTConnect',
no_simplification=False, constraint_function=None,
constraint_jacobian=None, constraint_tolerance=0.001,
verbose=False)
update_point_cloud (self, pc, radius=0.001)
update_attached_box (self, size, pose, link_id=-1)
detach_object MISSING
===== PLANNING WORLD =====
PlanningWorld: <class 'mplib.pymp.PlanningWorld'>
detach_object False
attach_object False
remove_object False
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_StrongPlanner_Audit_20261029$
```

思考了 6m 19s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_StrongPlanner_Audit_20261029$
bash RUN_B2_ONLY.sh
t0_strong_planner_20261029_061226_00
test_early_failure_not_s4 (test_contract.TestContracts) ... ok
test_good_backend (test_contract.TestContracts) ... ok
test_no_teleport (test_contract.TestContracts) ... ok
test_none_state (test_contract.TestContracts) ... ok
test_requires_flags (test_contract.TestContracts) ... ok
test_s4_ok (test_contract.TestContracts) ... ok
test_s4_physical (test_contract.TestContracts) ... ok
test_90deg_transform (test_math.MathTests) ... ok
test_box_surface (test_math.MathTests) ... ok
test_i_attached_translation (test_math.MathTests) ... ok
test_invalid_quat (test_math.MathTests) ... ok
test_l_two_parts_and_blue (test_math.MathTests) ... ok
test_quat_roundtrip (test_math.MathTests) ... ok
test_b2_unobserved_not_failure
(test_mplib011_patch.LegacyPatchTests) ... ok
test_historical_root_fails_before_physics
(test_mplib011_patch.LegacyPatchTests) ... ok
test_legacy_api_does_not_require_detach_object
(test_mplib011_patch.LegacyPatchTests) ... ok
test_missing_release_api_fails_closed
```

(test_mplib011_patch.LegacyPatchTests) ... ok
test_prepare_uses_numeric_seven_pose_and_scene
(test_mplib011_patch.LegacyPatchTests) ... ok
test_release_removes_attached_once
(test_mplib011_patch.LegacyPatchTests) ... ok
test_runtime_planning_flags_change_after_release
(test_mplib011_patch.LegacyPatchTests) ... ok

-
Ran 20 tests in 0.009s

OK
REFERENCE: t0_strong_planner_20261029_061226_00
B2 OUTPUT: t0_b2_mplib011_20261029_062657_00
===== B2 ONLY release legacy mplib 0.1.1 =====
2026-10-29 06:27:26,702 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 06:27:26,950 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:27,269 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:27,587 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:27,912 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300789 -0.0015866   0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342   0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True
seed000_release S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.306 green_dx=+0.000
seed000_release S4_push: steps=160 planner=True physical=True blue_dx=-0.306 green_dx=+0.143
B2 RESULT: release FULL_S1_S4_PHYSICAL_SUCCESS green_dx 0.142579 planning_calls 5 same_pre_s4 True
===== B2 ONLY south legacy mplib 0.1.1 =====
2026-10-29 06:27:29,162 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 06:27:29,465 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:29,786 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:30,105 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default

pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 06:27:30,423 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000
  L工具当前位置: [-0.3202 -0.3644 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True

physical=True blue_dx=-0.308 green_dx=+0.000
[ERROR] S4_push: PlannerResultFailure: No attached-aware path for S4/lower I
seed000_custom S4_push: steps=29 planner=False physical=False blue_dx=-0.308 green_dx=+0.000
[P1 STOP] S4_push 未满足局部成功，停止后续技能
B2 RESULT: south S4_PLAN_OR_EXECUTION_FAILURE green_dx 0.0 planning_calls 2 same_pre_s4 True
===== B2 ONLY REPLANNING DECISION =====
DECISION:
B2_SOUTH_BOUNDED_PLANNER_OR_EXECUTION_FAILURE
release FULL_S1_S4_PHYSICAL_SUCCESS green_dx= 0.142579 planner_calls= 5
south S4_PLAN_OR_EXECUTION_FAILURE green_dx= 0.0 planner_calls= 2
OUTPUT: t0_b2_mplib011_20261029_062657_00/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_StrongPlanner_Audit_20261029$

思考了 2m 33s >

ChatGPT 说：

你说：

可以的

思考了 10m 15s >

ChatGPT 说：

README.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_ContactMode_Replanning_20261029$ bash RUN_CONTACT_MODE.sh \
>
../T0_StrongPlanner_Audit_20261029/t0_strong_planner_20261029_061226_00
test_bounded_candidates_and_yaw_set (test_geometry.Geometry) ... ok

```
test_invalid_cube_position_fails (test_geometry.Geometry) ...
ok
test_release_nominal_is_not_l_overlap
(test_geometry.Geometry) ... ok
test_south_nominal_l_overlap_and_side_option
(test_geometry.Geometry) ... ok
test_yaw_candidate_uses_actual_tool_tcp_transform
(test_geometry.Geometry) ... ok
test_collateral_not_valid (test_guard.Guard) ... ok
test_failed_south_not_tamp_impossibility (test_guard.Guard) ...
ok
test_missing_contact_fail_closed (test_guard.Guard) ... ok
test_physical_contact_success (test_guard.Guard) ... ok
test_prestate_mismatch_invalid (test_guard.Guard) ... ok
test_bounded_candidate_rejects_nominal_collision
(test_planning.Planning) ... ok
test_budget_preflight (test_planning.Planning) ... ok
test_fail_all_not_claim_infeasible (test_planning.Planning) ... ok
test_planning_flags_never_disabled (test_planning.Planning) ...
ok
test_qpos_saves_gripper_joints (test_planning.Planning) ... ok
test_changed_source_detected (test_reference.HistoricalGuard)
... ok
test_complete_ref_accepted (test_reference.HistoricalGuard) ...
ok
test_same_pre_s4_delta (test_reference.HistoricalGuard) ... ok

----------------------------------------------------------------------
-
Ran 18 tests in 0.032s

OK
REFERENCE:
../T0_StrongPlanner_Audit_20261029/t0_strong_planner_20261
029_061226_00
NEW OUTPUT: t0_contact_mode_20261029_212056_2092497
===== CONTACT MODE release =====
2026-10-29 21:21:31,309 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-29 21:21:31,588 - mani_skill  - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
```

slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:31,929 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:32,273 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:32,626 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300789 -0.0015866   0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342   0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True
seed000_release S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.306 green_dx=+0.000
seed000_release S4_push: steps=157 planner=True

physical=True blue_dx=-0.306 green_dx=+0.159
CONTACT RESULT release: S4_PHYSICAL_GOAL_SUCCESS
yaw=0 green_dx=0.159072 push_contacts=34
same_pre_s4=True
===== CONTACT MODE south =====
2026-10-29 21:21:33,937 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 21:21:34,266 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:34,607 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:34,947 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 21:21:35,293 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000
L工具当前位置: [-0.3202 -0.3644 0.0253]
目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True physical=True blue_dx=-0.308 green_dx=+0.000
seed000_custom S4_push: steps=207 planner=True physical=True blue_dx=-0.308 green_dx=+0.122
CONTACT RESULT south: S4_PHYSICAL_GOAL_SUCCESS yaw=35 green_dx=0.121936 push_contacts=44 same_pre_s4=True
===== T0 CONTACT-MODE REPLANNING DECISION =====
DECISION:
BOTH_CONTACT_MODES_PHYSICALLY_SUCCEEDED_IN_ONE_SEED
release status= S4_PHYSICAL_GOAL_SUCCESS yaw= 0 green_dx= 0.159072 push_contact_steps= 34 pre_s4_match= True
south status= S4_PHYSICAL_GOAL_SUCCESS yaw= 35 green_dx= 0.121936 push_contact_steps= 44 pre_s4_match= True
OUTPUT:
t0_contact_mode_20261029_212056_2092497/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_ContactMode_Replanning_20261029$

ChatGPT 说：

你说：

可以

思考了 2m 38s >

ChatGPT 说：

你说：

不是，你现在在干啥，接着弄啊，都跑了一天算法了

思考了 6m 52s >

ChatGPT 说：

README_B4.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B4_ModeGraph_20261030$ bash RUN_B4.sh \
> ../T0_StrongPlanner_Audit_20261029/t0_strong_planner_20261029_061226_00 \
> ../T0_ContactMode_Replanning_20261029/t0_contact_mode_20261029_212056_2092497
test_bad_prefix_cannot_be_scored (test_b4_audit.B4AuditTests) ... ok
test_no_import_time_physics (test_b4_audit.B4AuditTests) ... ok
test_prefix_cost_from_successful_physical_stages (test_b4_audit.B4AuditTests) ... ok
test_branch_choice_never_uses_physical_outcomes (test_b4_mode_graph.B4Tests) ... ok
test_chooses_shorter_feasible_contact_mode (test_b4_mode_graph.B4Tests) ... ok
test_enumerates_all_instead_of_first_feasible (test_b4_mode_graph.B4Tests) ... ok
test_graph_reports_recourse_not_physical_success

```
(test_b4_mode_graph.B4Tests) ... ok
test_missing_data_fails_closed (test_b4_mode_graph.B4Tests)
... ok
test_zero_candidates_does_not_claim_global_failure
(test_b4_mode_graph.B4Tests) ... ok
test_bounded_candidates_and_yaw_set
(test_geometry.Geometry) ... ok
test_invalid_cube_position_fails (test_geometry.Geometry) ...
ok
test_release_nominal_is_not_l_overlap
(test_geometry.Geometry) ... ok
test_south_nominal_l_overlap_and_side_option
(test_geometry.Geometry) ... ok
test_yaw_candidate_uses_actual_tool_tcp_transform
(test_geometry.Geometry) ... ok
test_collateral_not_valid (test_guard.Guard) ... ok
test_failed_south_not_tamp_impossibility (test_guard.Guard) ...
ok
test_missing_contact_fail_closed (test_guard.Guard) ... ok
test_physical_contact_success (test_guard.Guard) ... ok
test_prestate_mismatch_invalid (test_guard.Guard) ... ok
test_bounded_candidate_rejects_nominal_collision
(test_planning.Planning) ... ok
test_budget_preflight (test_planning.Planning) ... ok
test_fail_all_not_claim_infeasible (test_planning.Planning) ... ok
test_planning_flags_never_disabled (test_planning.Planning) ...
ok
test_qpos_saves_gripper_joints (test_planning.Planning) ... ok
test_changed_source_detected (test_reference.HistoricalGuard)
... ok
test_complete_ref_accepted (test_reference.HistoricalGuard) ...
ok
test_same_pre_s4_delta (test_reference.HistoricalGuard) ... ok

----------------------------------------------------------------------
-
Ran 27 tests in 0.012s

OK
===== B4 EXHAUSTIVE CONTACT release =====
2026-10-29 23:09:46,754 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
```

2026-10-29 23:09:47,038 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:47,382 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:47,734 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:48,088 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_release S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_release S2_hook_and_pull: steps=219 planner=True physical=True blue_dx=-0.306 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300789 -0.0015866   0.17276858]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.10001301 0.02803092 0.02557832]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12209986 0.0285342   0.21965037]
[S3] 直杆位移: [-0.00440411 0.00061629 0.1946505]
[S3] 抬起后是否抓住: True

seed000_release S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.306 green_dx=+0.000
seed000_release S4_push: steps=157 planner=True physical=True blue_dx=-0.306 green_dx=+0.155
B4 MODE RESULT: release S4= S4_PHYSICAL_GOAL_SUCCESS yaw= -20 feasible= 6 backup= 4 estimated_s4_steps= 91 green_dx= 0.154779
===== B4 EXHAUSTIVE CONTACT south =====
2026-10-29 23:09:49,671 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:09:50,000 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:50,345 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:50,696 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:09:51,048 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
seed000_custom S1_grasp_l_tool: steps=114 planner=True physical=True blue_dx=-0.000 green_dx=+0.000
L工具当前位置: [-0.3202 -0.3644 0.0253]
目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True

[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.3mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed000_custom S2_hook_and_pull: steps=309 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00492197 -0.00903931 0.17212337]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004359 0.02779171 0.02592734]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211586 0.02900716 0.21965735]
[S3] 直杆位移: [-0.00442001 0.00109047 0.19465747]
[S3] 抬起后是否抓住: True
seed000_custom S3_grasp_i_tool: steps=89 planner=True
physical=True blue_dx=-0.308 green_dx=+0.000
seed000_custom S4_push: steps=207 planner=True
physical=True blue_dx=-0.308 green_dx=+0.122
B4 MODE RESULT: south S4= S4_PHYSICAL_GOAL_SUCCESS
yaw= 35 feasible= 2 backup= 1 estimated_s4_steps= 120
green_dx= 0.121936
===== B4 MODE GRAPH DECISION =====
PHYSICAL_VERDICT:
BOTH_CONTACT_MODES_PHYSICALLY_SUCCEEDED_IN_ONE_S
EED
release feasible= 6 backup= 4 yaw= -20 green_dx= 0.154779
push_contacts= 33 prestate_match= True
south feasible= 2 backup= 1 yaw= 35 green_dx= 0.121936
push_contacts= 44 prestate_match= True
COUNTERFACTUAL_TERMINAL_CHOICE: release (retrospective
oracle, NOT deployable online)
OUTPUT: /home/users/lemon/skill_future_viability/skill-
composition-
benchmark/T0_B4_ModeGraph_20261030/b4_mode_graph_20
261029_230912_185574/decision.json

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B4_ModeGraph_20261030$

思考了 7m 9s >

ChatGPT 说：

README_B5.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B5_OnlineTerminal_20261030$ bash RUN_B5.sh \
>
../T0_B4_ModeGraph_20261030/b4_mode_graph_20261029_230912_185574
test_bad_prefix_cannot_be_scored (test_b4_audit.B4AuditTests) ... ok
test_no_import_time_physics (test_b4_audit.B4AuditTests) ... ok
test_prefix_cost_from_successful_physical_stages (test_b4_audit.B4AuditTests) ... ok
test_branch_choice_never_uses_physical_outcomes (test_b4_mode_graph.B4Tests) ... ok
test_chooses_shorter_feasible_contact_mode (test_b4_mode_graph.B4Tests) ... ok
test_enumerates_all_instead_of_first_feasible (test_b4_mode_graph.B4Tests) ... ok
test_graph_reports_recourse_not_physical_success (test_b4_mode_graph.B4Tests) ... ok
test_missing_data_fails_closed (test_b4_mode_graph.B4Tests) ... ok
test_zero_candidates_does_not_claim_global_failure (test_b4_mode_graph.B4Tests) ... ok
test_compile_from_completed_history_snapshot (test_b5_terminal.B5SyntheticB4Integration) ... ok
test_can_choose_south_in_other_task_no_hardcoding (test_b5_terminal.B5Tests) ... ok
test_dominance_release (test_b5_terminal.B5Tests) ... ok
test_incorrect_model_rejected (test_b5_terminal.B5Tests) ... ok
test_no_future_fields_in_model_action (test_b5_terminal.B5Tests) ... ok
test_ood_fails_closed_to_declared_baseline (test_b5_terminal.B5Tests) ... ok

```
test_readonly_calibration_hash_check
(test_b5_terminal.B5Tests) ... ok
test_run_wrapper_decides_before_s2_in_context
(test_b5_terminal.B5Tests) ... ok
test_seed0_and_duplicate_rejected (test_b5_terminal.B5Tests)
... ok
test_bounded_candidates_and_yaw_set
(test_geometry.Geometry) ... ok
test_invalid_cube_position_fails (test_geometry.Geometry) ...
ok
test_release_nominal_is_not_l_overlap
(test_geometry.Geometry) ... ok
test_south_nominal_l_overlap_and_side_option
(test_geometry.Geometry) ... ok
test_yaw_candidate_uses_actual_tool_tcp_transform
(test_geometry.Geometry) ... ok
test_collateral_not_valid (test_guard.Guard) ... ok
test_failed_south_not_tamp_impossibility (test_guard.Guard) ...
ok
test_missing_contact_fail_closed (test_guard.Guard) ... ok
test_physical_contact_success (test_guard.Guard) ... ok
test_prestate_mismatch_invalid (test_guard.Guard) ... ok
test_bounded_candidate_rejects_nominal_collision
(test_planning.Planning) ... ok
test_budget_preflight (test_planning.Planning) ... ok
test_fail_all_not_claim_infeasible (test_planning.Planning) ... ok
test_planning_flags_never_disabled (test_planning.Planning) ...
ok
test_qpos_saves_gripper_joints (test_planning.Planning) ... ok
test_changed_source_detected (test_reference.HistoricalGuard)
... ok
test_complete_ref_accepted (test_reference.HistoricalGuard) ...
ok
test_same_pre_s4_delta (test_reference.HistoricalGuard) ... ok

----------------------------------------------------------------------
-
Ran 36 tests in 0.035s

OK
B5 FROZEN CALIBRATION: seed0, eval_seeds= [1, 2, 3]
B5 calibrated scores only: {'release': {'prefix_s1_s3_steps': 416,
'feasible_modes': 6, 'backup_modes': 4, 'predicted_s4_cost':
91.4, 'mode_yaws': [-50, -35, -20, 0, 20, 35]}, 'south':
{'prefix_s1_s3_steps': 512, 'feasible_modes': 2, 'backup_modes':
```

1, 'predicted_s4_cost': 120.7, 'mode_yaws': [-50, 35]}}
===== B5 ONLINE DECISION SEED 1 =====
2026-10-29 23:49:27,826 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:28,173 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:28,547 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:28,932 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:29,322 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3139])
seed001_custom S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed001_custom S2_hook_and_pull: steps=216 planner=True physical=True blue_dx=-0.310 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.18061393 0.06408017 0.02499982]
[S3] TCP 初始位置: [-0.01434325 -0.00231081 0.17239553]
[S3] 目标抓取位置: [-0.16061394 0.06408016 0.02499982]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00

-4.4777065e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.16260044 0.06399948 0.0236516]
[S3] 到达后直杆: [-0.18061414 0.06407838 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1861673   0.06493839 0.22259043]
[S3] 直杆位移: [-0.00555336 0.00085822 0.19759061]
[S3] 抬起后是否抓住: True
seed001_custom S3_grasp_i_tool: steps=101 planner=True physical=True blue_dx=-0.310 green_dx=-0.000
seed001_custom S4_push: steps=174 planner=True physical=True blue_dx=-0.310 green_dx=+0.159
B5 SEED RESULT: 1 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6 green_dx= 0.158684
===== B5 ONLINE DECISION SEED 2 =====
2026-10-29 23:49:30,994 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:31,360 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:31,741 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:32,117 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:32,503 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3066])
seed002_custom S1_grasp_l_tool: steps=111 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
screw plan failed

[ERROR] S2_hook_and_pull: PlanningFailure: 运动规划失败: S2/pull blue

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed002_custom S2_hook_and_pull: steps=87 planner=False physical=False blue_dx=-0.000 green_dx=-0.000
[P1 STOP] S2_hook_and_pull 未满足局部成功，停止后续技能
B5 SEED RESULT: 2 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= INVALID_EARLY_SKILL modes= None green_dx= -1e-06
===== B5 ONLINE DECISION SEED 3 =====
2026-10-29 23:49:33,105 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:33,464 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:33,847 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:34,226 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:34,616 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2951])
seed003_custom S1_grasp_l_tool: steps=93 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed003_custom S2_hook_and_pull: steps=212 planner=True physical=True blue_dx=-0.305 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.15716857 0.02161637 0.02499986]
[S3] TCP 初始位置: [-0.0150999 -0.00312714 0.17281465]
[S3] 目标抓取位置: [-0.13716857 0.02161635 0.02499986]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.9628551e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.13902062 0.02138357 0.02319042]
[S3] 到达后直杆: [-0.15716876 0.02161407 0.02499982]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.16285971 0.02230435 0.22409341]
[S3] 直杆位移: [-0.00569114 0.00068798 0.19909355]
[S3] 抬起后是否抓住: True
seed003_custom S3_grasp_i_tool: steps=101 planner=True physical=True blue_dx=-0.305 green_dx=-0.000
seed003_custom S4_push: steps=165 planner=True physical=True blue_dx=-0.305 green_dx=+0.159
B5 SEED RESULT: 3 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6 green_dx= 0.158957
===== B5 ONLINE TERMINAL DECISION =====
DECISION: OOD_OR_EARLY_FAILURE_NO_GENERALIZATION_CLAIM
HELDOUT_SEEDS: [1, 2, 3] VALID: 3 OOD: 3
PHYSICALLY_VALID_S4: 2
seed 1 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6
seed 2 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= INVALID_EARLY_SKILL modes= None
seed 3 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6
ALWAYS_RELEASE_COMPARISON: not demonstrated superior; do not claim advantage
OUTPUT: /home/users/lemon/skill_future_viability/skill-composition-

benchmark/T0_B5_OnlineTerminal_20261030/b5_online_terminal_20261029_234851_365594/decision.json

思考了 5m 42s

ChatGPT 说：

README.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B5_OnlineTerminal_20261030$ bash RUN_B5.sh \
>
../T0_B4_ModeGraph_20261030/b4_mode_graph_20261029_230912_185574
test_bad_prefix_cannot_be_scored (test_b4_audit.B4AuditTests) ... ok
test_no_import_time_physics (test_b4_audit.B4AuditTests) ... ok
test_prefix_cost_from_successful_physical_stages (test_b4_audit.B4AuditTests) ... ok
test_branch_choice_never_uses_physical_outcomes (test_b4_mode_graph.B4Tests) ... ok
test_chooses_shorter_feasible_contact_mode (test_b4_mode_graph.B4Tests) ... ok
test_enumerates_all_instead_of_first_feasible (test_b4_mode_graph.B4Tests) ... ok
test_graph_reports_recourse_not_physical_success (test_b4_mode_graph.B4Tests) ... ok
test_missing_data_fails_closed (test_b4_mode_graph.B4Tests) ... ok
test_zero_candidates_does_not_claim_global_failure (test_b4_mode_graph.B4Tests) ... ok
test_compile_from_completed_history_snapshot (test_b5_terminal.B5SyntheticB4Integration) ... ok
test_can_choose_south_in_other_task_no_hardcoding (test_b5_terminal.B5Tests) ... ok
test_dominance_release (test_b5_terminal.B5Tests) ... ok
test_incorrect_model_rejected (test_b5_terminal.B5Tests) ... ok
test_no_future_fields_in_model_action (test_b5_terminal.B5Tests) ... ok
test_ood_fails_closed_to_declared_baseline (test_b5_terminal.B5Tests) ... ok

```
test_readonly_calibration_hash_check
(test_b5_terminal.B5Tests) ... ok
test_run_wrapper_decides_before_s2_in_context
(test_b5_terminal.B5Tests) ... ok
test_seed0_and_duplicate_rejected (test_b5_terminal.B5Tests)
... ok
test_bounded_candidates_and_yaw_set
(test_geometry.Geometry) ... ok
test_invalid_cube_position_fails (test_geometry.Geometry) ...
ok
test_release_nominal_is_not_l_overlap
(test_geometry.Geometry) ... ok
test_south_nominal_l_overlap_and_side_option
(test_geometry.Geometry) ... ok
test_yaw_candidate_uses_actual_tool_tcp_transform
(test_geometry.Geometry) ... ok
test_collateral_not_valid (test_guard.Guard) ... ok
test_failed_south_not_tamp_impossibility (test_guard.Guard) ...
ok
test_missing_contact_fail_closed (test_guard.Guard) ... ok
test_physical_contact_success (test_guard.Guard) ... ok
test_prestate_mismatch_invalid (test_guard.Guard) ... ok
test_bounded_candidate_rejects_nominal_collision
(test_planning.Planning) ... ok
test_budget_preflight (test_planning.Planning) ... ok
test_fail_all_not_claim_infeasible (test_planning.Planning) ... ok
test_planning_flags_never_disabled (test_planning.Planning) ...
ok
test_qpos_saves_gripper_joints (test_planning.Planning) ... ok
test_changed_source_detected (test_reference.HistoricalGuard)
... ok
test_complete_ref_accepted (test_reference.HistoricalGuard) ...
ok
test_same_pre_s4_delta (test_reference.HistoricalGuard) ... ok

----------------------------------------------------------------------
-
Ran 36 tests in 0.035s

OK
B5 FROZEN CALIBRATION: seed0, eval_seeds= [1, 2, 3]
B5 calibrated scores only: {'release': {'prefix_s1_s3_steps': 416,
'feasible_modes': 6, 'backup_modes': 4, 'predicted_s4_cost':
91.4, 'mode_yaws': [-50, -35, -20, 0, 20, 35]}, 'south':
{'prefix_s1_s3_steps': 512, 'feasible_modes': 2, 'backup_modes':
```

1, 'predicted_s4_cost': 120.7, 'mode_yaws': [-50, 35]}}
===== B5 ONLINE DECISION SEED 1 =====
2026-10-29 23:49:27,826 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:28,173 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:28,547 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:28,932 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:29,322 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3139])
seed001_custom S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed001_custom S2_hook_and_pull: steps=216 planner=True physical=True blue_dx=-0.310 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.18061393 0.06408017 0.02499982]
[S3] TCP 初始位置: [-0.01434325 -0.00231081 0.17239553]
[S3] 目标抓取位置: [-0.16061394 0.06408016 0.02499982]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00

-4.4777065e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.16260044 0.06399948 0.0236516]
[S3] 到达后直杆: [-0.18061414 0.06407838 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1861673   0.06493839 0.22259043]
[S3] 直杆位移: [-0.00555336 0.00085822 0.19759061]
[S3] 抬起后是否抓住: True
seed001_custom S3_grasp_i_tool: steps=101 planner=True physical=True blue_dx=-0.310 green_dx=-0.000
seed001_custom S4_push: steps=174 planner=True physical=True blue_dx=-0.310 green_dx=+0.159
B5 SEED RESULT: 1 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6 green_dx= 0.158684
===== B5 ONLINE DECISION SEED 2 =====
2026-10-29 23:49:30,994 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:31,360 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:31,741 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:32,117 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:32,503 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3066])
seed002_custom S1_grasp_l_tool: steps=111 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
screw plan failed

[ERROR] S2_hook_and_pull: PlanningFailure: 运动规划失败: S2/pull blue

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed002_custom S2_hook_and_pull: steps=87 planner=False physical=False blue_dx=-0.000 green_dx=-0.000
[P1 STOP] S2_hook_and_pull 未满足局部成功，停止后续技能
B5 SEED RESULT: 2 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= INVALID_EARLY_SKILL modes= None green_dx= -1e-06
===== B5 ONLINE DECISION SEED 3 =====
2026-10-29 23:49:33,105 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-29 23:49:33,464 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:33,847 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:34,226 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-29 23:49:34,616 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2951])
seed003_custom S1_grasp_l_tool: steps=93 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed003_custom S2_hook_and_pull: steps=212 planner=True physical=True blue_dx=-0.305 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.15716857 0.02161637 0.02499986]
[S3] TCP 初始位置: [-0.0150999 -0.00312714 0.17281465]
[S3] 目标抓取位置: [-0.13716857 0.02161635 0.02499986]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.9628551e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.13902062 0.02138357 0.02319042]
[S3] 到达后直杆: [-0.15716876 0.02161407 0.02499982]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.16285971 0.02230435 0.22409341]
[S3] 直杆位移: [-0.00569114 0.00068798 0.19909355]
[S3] 抬起后是否抓住: True
seed003_custom S3_grasp_i_tool: steps=101 planner=True physical=True blue_dx=-0.305 green_dx=-0.000
seed003_custom S4_push: steps=165 planner=True physical=True blue_dx=-0.305 green_dx=+0.159
B5 SEED RESULT: 3 decision= OOD_ABSTAIN_FALLBACK_RELEASE terminal= release S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6 green_dx= 0.158957
===== B5 ONLINE TERMINAL DECISION =====
DECISION: OOD_OR_EARLY_FAILURE_NO_GENERALIZATION_CLAIM
HELDOUT_SEEDS: [1, 2, 3] VALID: 3 OOD: 3
PHYSICALLY_VALID_S4: 2
seed 1 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6
seed 2 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= INVALID_EARLY_SKILL modes= None
seed 3 policy= release gate= OOD_ABSTAIN_FALLBACK_RELEASE S4= S4_PHYSICAL_GOAL_SUCCESS modes= 6
ALWAYS_RELEASE_COMPARISON: not demonstrated superior; do not claim advantage
OUTPUT: /home/users/lemon/skill_future_viability/skill-composition-

benchmark/T0_B5_OnlineTerminal_20261030/b5_online_terminal_20261029_234851_365594/decision.json

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B6_PairedDecision_20261030$ bash RUN_B6.sh \
>
../T0_B4_ModeGraph_20261030/b4_mode_graph_20261029_230912_185574
test_below_three_is_insufficient (test_b6.B6AuditTests) ... ok
test_early_failure_is_censored_not_zero_modes
(test_b6.B6AuditTests) ... ok
test_initial_xyz_delta (test_b6.B6AuditTests) ... ok
test_mismatched_initial_not_comparable
(test_b6.B6AuditTests) ... ok
test_missing_initial_fails_closed (test_b6.B6AuditTests) ... ok
test_no_heldout_seed_zero (test_b6.B6AuditTests) ... ok
test_no_teleport_in_new_b6 (test_b6.B6AuditTests) ... ok
test_read_prefix_stages (test_b6.B6AuditTests) ... ok
test_real_tradeoff_only_suggests_further_validation
(test_b6.B6AuditTests) ... ok
test_release_dominant_pair (test_b6.B6AuditTests) ... ok
test_south_can_win_when_proxy_truly_better
(test_b6.B6AuditTests) ... ok
test_symmetric_upstream_rrt (test_b6.B6AuditTests) ... ok
test_three_comparable_release_dominant_stop
(test_b6.B6AuditTests) ... ok
test_verified_requires_true_contacts_no_collateral
(test_b6.B6AuditTests) ... ok

----------------------------------------------------------------------
Ran 14 tests in 0.001s

OK
===== B6 SEED 1 release =====
2026-10-30 06:55:59,115 - mani_skill  - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
```

2026-10-30 06:55:59,371 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:55:59,699 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:00,038 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:00,380 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3139])
seed001_release S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed001_release S2_hook_and_pull: steps=216 planner=True physical=True blue_dx=-0.310 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.18061393 0.06408017 0.02499982]
[S3] TCP 初始位置: [-0.01434325 -0.00231081 0.17239553]
[S3] 目标抓取位置: [-0.16061394 0.06408016 0.02499982]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -4.4777065e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.16260044 0.06399948 0.0236516]
[S3] 到达后直杆: [-0.18061414 0.06407838 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1861673   0.06493839 0.22259043]
[S3] 直杆位移: [-0.00555336 0.00085822 0.19759061]
[S3] 抬起后是否抓住: True

seed001_release S3_grasp_i_tool: steps=101 planner=True physical=True blue_dx=-0.310 green_dx=-0.000
seed001_release S4_push: steps=174 planner=True physical=True blue_dx=-0.310 green_dx=+0.159
B6 BRANCH: seed=1 policy=release upstream=True S4=S4_PHYSICAL_GOAL_SUCCESS modes=6 green_dx=0.158684
===== B6 SEED 1 south =====
2026-10-30 06:56:01,995 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:02,323 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:02,663 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:03,003 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:03,342 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3139])
seed001_custom S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
  L工具当前位置: [-0.3463 -0.3536 0.0251]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=30
[PARK GUARD] S2/parking lift: tcp_error=7.7mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=42
[PARK GUARD] S2/parking transit: tcp_error=13.6mm l_grasped=True

[PARK GUARD] S2/parking lower: planned_steps=34
[PARK GUARD] S2/parking lower: tcp_error=7.0mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed001_custom S2_hook_and_pull: steps=317 planner=True
physical=True blue_dx=-0.312 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.18061393 0.06408017 0.02499982]
[S3] TCP 初始位置: [-0.00513008 -0.00947008 0.17211622]
[S3] 目标抓取位置: [-0.16061394 0.06408016 0.02499982]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-4.4777065e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.16282254 0.06420478 0.02472669]
[S3] 到达后直杆: [-0.18061417 0.06407782 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.18593064 0.06570549 0.22308098]
[S3] 直杆位移: [-0.0053167   0.00162531 0.19808116]
[S3] 抬起后是否抓住: True
seed001_custom S3_grasp_i_tool: steps=100 planner=True
physical=True blue_dx=-0.312 green_dx=-0.000
  [ERROR] S4_push: ExecutionGuard:
PREMATURE_GREEN_DISPLACEMENT_BEFORE_PUSH
seed001_custom S4_push: steps=81 planner=False
physical=False blue_dx=-0.312 green_dx=+0.002
[P1 STOP] S4_push 未满足局部成功，停止后续技能
B6 BRANCH: seed=1 policy=south upstream=True
S4=S4_CONTACT_PLAN_OR_EXECUTION_FAILURE modes=2
green_dx=0.002319
B6 PAIR seed=1: same_initial=True comparable=True
proxy_oracle=release
===== B6 SEED 2 release =====
2026-10-30 06:56:04,974 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:05,303 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may

slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:05,635 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:05,970 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:06,298 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3066])
seed002_release S1_grasp_l_tool: steps=111 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
screw plan failed

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed002_release S2_hook_and_pull: steps=223 planner=True physical=True blue_dx=-0.301 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.22742298 0.04846676 0.02499985]
[S3] TCP 初始位置: [-0.01322646 -0.00316675 0.17300743]
[S3] 目标抓取位置: [-0.20742299 0.04846674 0.02499985]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -4.1324006e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.20953368 0.04855344 0.02536289]
[S3] 到达后直杆: [-0.2274231   0.04846557 0.02499985]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.23249796 0.04866115 0.22096884]
[S3] 直杆位移: [-5.07497787e-03 1.94389373e-04 1.95968989e-01]
[S3] 抬起后是否抓住: True
seed002_release S3_grasp_i_tool: steps=102 planner=True

physical=True blue_dx=-0.301 green_dx=-0.000
seed002_release S4_push: steps=172 planner=True
physical=True blue_dx=-0.301 green_dx=+0.159
B6 BRANCH: seed=2 policy=release upstream=True
S4=S4_PHYSICAL_GOAL_SUCCESS modes=6
green_dx=0.158789
===== B6 SEED 2 south =====
2026-10-30 06:56:07,936 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:08,254 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:08,590 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:08,943 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:09,280 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3066])
seed002_custom S1_grasp_l_tool: steps=111 planner=True
physical=True blue_dx=-0.000 green_dx=-0.000
screw plan failed
  L工具当前位置: [-0.2838 -0.3776 0.0251]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=29
[PARK GUARD] S2/parking lift: tcp_error=7.3mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=40
[PARK GUARD] S2/parking transit: tcp_error=12.0mm
l_grasped=True

[PARK GUARD] S2/parking lower: planned_steps=17
[PARK GUARD] S2/parking lower: tcp_error=4.2mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed002_custom S2_hook_and_pull: steps=295 planner=True physical=False blue_dx=-0.333 green_dx=-0.000
[P1 STOP] S2_hook_and_pull 未满足局部成功，停止后续技能
B6 BRANCH: seed=2 policy=south upstream=False S4=INVALID_EARLY_SKILL modes=None green_dx=-1e-06
B6 PAIR seed=2: same_initial=True comparable=False proxy_oracle=None
===== B6 SEED 3 release =====
2026-10-30 06:56:10,256 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:10,573 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:10,909 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:11,243 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:11,580 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2951])
seed003_release S1_grasp_l_tool: steps=93 planner=True

physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed003_release S2_hook_and_pull: steps=212 planner=True
physical=True blue_dx=-0.305 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.15716857 0.02161637 0.02499986]
[S3] TCP 初始位置: [-0.0150999 -0.00312714 0.17281465]
[S3] 目标抓取位置: [-0.13716857 0.02161635 0.02499986]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-3.9628551e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.13902062 0.02138357 0.02319042]
[S3] 到达后直杆: [-0.15716876 0.02161407 0.02499982]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.16285971 0.02230435 0.22409341]
[S3] 直杆位移: [-0.00569114 0.00068798 0.19909355]
[S3] 抬起后是否抓住: True
seed003_release S3_grasp_i_tool: steps=101 planner=True
physical=True blue_dx=-0.305 green_dx=-0.000
seed003_release S4_push: steps=165 planner=True
physical=True blue_dx=-0.305 green_dx=+0.159
B6 BRANCH: seed=3 policy=release upstream=True
S4=S4_PHYSICAL_GOAL_SUCCESS modes=6
green_dx=0.158957
===== B6 SEED 3 south =====
2026-10-30 06:56:13,155 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:13,473 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:13,807 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.

2026-10-30 06:56:14,159 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:14,489 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2951])
seed003_custom S1_grasp_l_tool: steps=93 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
  L工具当前位置: [-0.3063 -0.4043 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=30
[PARK GUARD] S2/parking lift: tcp_error=6.8mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=43
[PARK GUARD] S2/parking transit: tcp_error=13.5mm
l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=15
[PARK GUARD] S2/parking lower: tcp_error=4.1mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed003_custom S2_hook_and_pull: steps=286 planner=True physical=False blue_dx=-0.335 green_dx=-0.000
[P1 STOP] S2_hook_and_pull 未满足局部成功，停止后续技能
B6 BRANCH: seed=3 policy=south upstream=False S4=INVALID_EARLY_SKILL modes=None green_dx=-1e-06
B6 PAIR seed=3: same_initial=True comparable=False proxy_oracle=None
===== B6 SEED 4 release =====
2026-10-30 06:56:15,424 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:15,736 - mani_skill - WARNING - No initial

pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:16,060 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:16,387 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:16,707 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2819])
seed004_release S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed004_release S2_hook_and_pull: steps=224 planner=True physical=True blue_dx=-0.305 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11829142 0.03259928 0.02499987]
[S3] TCP 初始位置: [-0.01588458 -0.0021978   0.17241585]
[S3] 目标抓取位置: [-0.09829143 0.03259926 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -4.1256808e-07 0.0000000e+00]
[S3] 到达后 TCP: [-0.10061185 0.03274807 0.02557359]
[S3] 到达后直杆: [-0.11829151 0.03259809 0.02499988]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12269211 0.03273845 0.21968739]
[S3] 直杆位移: [-4.40068543e-03 1.39173120e-04 1.94687517e-01]
[S3] 抬起后是否抓住: True

seed004_release S3_grasp_i_tool: steps=83 planner=True physical=True blue_dx=-0.305 green_dx=-0.000
seed004_release S4_push: steps=157 planner=True physical=True blue_dx=-0.305 green_dx=+0.156
B6 BRANCH: seed=4 policy=release upstream=True S4=S4_PHYSICAL_GOAL_SUCCESS modes=6 green_dx=0.15619
===== B6 SEED 4 south =====
2026-10-30 06:56:18,305 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 06:56:18,621 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:18,953 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:19,277 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 06:56:19,601 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2819])
seed004_custom S1_grasp_l_tool: steps=116 planner=True physical=True blue_dx=-0.000 green_dx=-0.000
L工具当前位置: [-0.3432 -0.4059 0.0253]
目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=28
[PARK GUARD] S2/parking lift: tcp_error=8.1mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=40
[PARK GUARD] S2/parking transit: tcp_error=14.5mm l_grasped=True

[PARK GUARD] S2/parking lower: planned_steps=34
[PARK GUARD] S2/parking lower: tcp_error=7.0mm
l_grasped=True

========== S2 干扰诊断 ==========
直杆工具位移: 0.00 mm
绿色方块位移: 0.00 mm
==================================

seed004_custom S2_hook_and_pull: steps=314 planner=True
physical=True blue_dx=-0.307 green_dx=-0.000

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11829151 0.03259807 0.02499986]
[S3] TCP 初始位置: [-0.00502153 -0.00904485 0.17212167]
[S3] 目标抓取位置: [-0.09829152 0.03259802 0.02499986]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-1.1922286e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10063811 0.03253637 0.02584274]
[S3] 到达后直杆: [-0.11829155 0.03259746 0.02499983]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12272234 0.03372491 0.21965107]
[S3] 直杆位移: [-0.00443083 0.00112684 0.19465122]
[S3] 抬起后是否抓住: True
seed004_custom S3_grasp_i_tool: steps=88 planner=True
physical=True blue_dx=-0.307 green_dx=-0.000
seed004_custom S4_push: steps=211 planner=True
physical=True blue_dx=-0.307 green_dx=+0.153
B6 BRANCH: seed=4 policy=south upstream=True
S4=S4_PHYSICAL_GOAL_SUCCESS modes=2
green_dx=0.153011
B6 PAIR seed=4: same_initial=True comparable=True
proxy_oracle=release
===== B6 PAIRED TERMINAL DECISION =====
DECISION: INSUFFICIENT_VALID_PAIRS_NO_TRAINING
PAIRS: 4 COMPARABLE: 2 BOTH_GRAPHS: 2
SOUTH_BETTER_GE_10STEPS: 0
PHYSICAL_VERIFIED: release 4 south 1
seed 1 comparable= True modes= 6 2 total_proxy_steps=
537.0 661.7 retrospective_oracle= release
seed 2 comparable= False modes= 6 None total_proxy_steps=
540.0 None retrospective_oracle= None
seed 3 comparable= False modes= 6 None total_proxy_steps=
502.0 None retrospective_oracle= None
seed 4 comparable= True modes= 6 2 total_proxy_steps=

514.4 638.7 retrospective_oracle= release
OUTPUT: /home/users/lemon/skill_future_viability/skill-composition-benchmark/T0_B6_PairedDecision_20261030/b6_paired_20261030_065529_651025/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B6_PairedDecision_20261030$

思考了 4m 31s >

ChatGPT 说：

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B6_PairedDecision_20261030$ python T0_B6_FinalOfflineAudit_20261030/audit_b6.py \
> b6_paired_20261030_065529_651025
===== B6 FINAL READ-ONLY AUDIT =====
DECISION:
STOP_T0_TERMINAL_POLICY_TRAINING__NO_VALIDATED_STATE_CONDITIONAL_ADVANTAGE
PAIRS: 4 COMPARABLE: 2 BOTH_S4_VERIFIED: 1
S4_VERIFIED: release 4 south 1
SOUTH_PROXY_WINS_GE_10: 0
seed 1 south S2 PHYSICAL_PASS blue_dx= -0.311854 parking_err= 0.007071540333163572 reasons= none
seed 2 south S2 PHYSICAL_FAIL blue_dx= -0.33332 parking_err= 0.054625932615026816 reasons= PARKING_XY_ERROR_GT_35MM
seed 3 south S2 PHYSICAL_FAIL blue_dx= -0.334686 parking_err= 0.04839695178292884 reasons= PARKING_XY_ERROR_GT_35MM
seed 4 south S2 PHYSICAL_PASS blue_dx= -0.307187 parking_err= 0.007063622134031235 reasons= none
NO_PHYSICS: True
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B6_PairedDecision_20261030$

思考了 7m 51s >

ChatGPT 说：

README.md
文档

你说：

(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B7_GoalConditioned_20261030$ bash RUN_B7.sh
test_failed_later_skill_does_not_invalidate_short_goal (test_b7_goal_graph.GoalGraphTests) ... ok
test_no_fabricated_score_for_failure (test_b7_goal_graph.GoalGraphTests) ... ok
test_prefix_is_four_skills (test_b7_goal_graph.GoalGraphTests) ... ok
test_retrospective_conditioning_can_flip (test_b7_goal_graph.GoalGraphTests) ... ok
test_steps_must_be_int (test_b7_goal_graph.GoalGraphTests) ... ok
test_unexecuted_is_not_free (test_b7_goal_graph.GoalGraphTests) ... ok
test_unknown_goal_rejected (test_b7_goal_graph.GoalGraphTests) ... ok
test_9mode_exhaustive (test_source_guard.SourceGuard) ... ok
test_core_s1_s4_unmodified_since_w1 (test_source_guard.SourceGuard) ... ok
test_no_hidden_physical_retries (test_source_guard.SourceGuard) ... ok
test_no_historical_test_in_runner (test_source_guard.SourceGuard) ... ok
test_single_reset_no_set_pose_in_runner (test_source_guard.SourceGuard) ... ok

----
Ran 12 tests in 0.066s

OK
MPLIB: 0.1.1
2026-10-30 07:49:39,522 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 07:49:39,809 - mani_skill - WARNING - No initial

pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:40,148 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:40,491 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:40,831 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:41,179 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_release] S1_GetL: plan=True phys=True valid=True steps=114
[B7 seed000_release] S2_HookBlueAndDock: plan=True phys=False valid=False steps=219 AttributeError("'Namespace' object has no attribute 'policy'")
2026-10-30 07:49:41,993 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 07:49:42,338 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:42,676 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see

https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:43,022 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:43,366 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 07:49:43,709 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_south] S1_GetL: plan=True phys=True valid=True steps=114
  L工具当前位置: [-0.3203 -0.3641 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm
l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm
l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.2mm
l_grasped=True
[B7 seed000_south] S2_HookBlueAndDock: plan=True phys=False valid=False steps=309 AttributeError("'Namespace' object has no attribute 'policy'")
===== B7 GOAL-CONDITIONED SIX-SKILL DECISION =====
DECISION: B7_INTEGRATION_NOT_FULLY_VALIDATED
release status= FAIL_S2_HookBlueAndDock S4_modes= None yaw= None S4_contacts= None physical_steps= 333 first_failure= S2_HookBlueAndDock
south status= FAIL_S2_HookBlueAndDock S4_modes= None yaw= None S4_contacts= None physical_steps= 423 first_failure= S2_HookBlueAndDock
GOAL COSTS: {"release": {"green_only": {"valid": false, "measured_steps": null, "failed_or_missing_stage": 2, "not_online_prediction": true}, "green_then_yellow": {"valid":

```
false, "measured_steps": null, "failed_or_missing_stage": 2,
"not_online_prediction": true}}, "south": {"green_only": {"valid":
false, "measured_steps": null, "failed_or_missing_stage": 2,
"not_online_prediction": true}, "green_then_yellow": {"valid":
false, "measured_steps": null, "failed_or_missing_stage": 2,
"not_online_prediction": true}}}
RETROSPECTIVE GOAL CHOICE: {'green_only': None,
'green_then_yellow': None} (NOT online)
OUTPUT: b7_goal_mode_20261030_074910/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_B7_GoalConditioned_20261030$
```

思考了 2m 6s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_B7_GoalConditioned_20261030$
bash RUN_B7.sh
test_failed_later_skill_does_not_invalidate_short_goal
(test_b7_goal_graph.GoalGraphTests) ... ok
test_no_fabricated_score_for_failure
(test_b7_goal_graph.GoalGraphTests) ... ok
test_prefix_is_four_skills (test_b7_goal_graph.GoalGraphTests)
... ok
test_retrospective_conditioning_can_flip
(test_b7_goal_graph.GoalGraphTests) ... ok
test_steps_must_be_int (test_b7_goal_graph.GoalGraphTests) ...
ok
test_unexecuted_is_not_free
(test_b7_goal_graph.GoalGraphTests) ... ok
test_unknown_goal_rejected
(test_b7_goal_graph.GoalGraphTests) ... ok
test_policy_field_matches_w1_evaluation_contract
(test_s2_options_regression.S2IntegrationRegression) ... ok
test_w1_s2_evaluator_both_real_b7_option_objects
(test_s2_options_regression.S2IntegrationRegression) ... ok
test_9mode_exhaustive (test_source_guard.SourceGuard) ... ok
test_core_s1_s4_unmodified_since_w1
(test_source_guard.SourceGuard) ... ok
test_no_hidden_physical_retries
(test_source_guard.SourceGuard) ... ok
test_no_historical_test_in_runner
```

```
(test_source_guard.SourceGuard) ... ok
test_single_reset_no_set_pose_in_runner
(test_source_guard.SourceGuard) ... ok

----------------------------------------------------------------------
-
Ran 14 tests in 0.001s

OK
MPLIB: 0.1.1
2026-10-30 22:41:15,609 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-30 22:41:15,872 - mani_skill  - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:16,189 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_2, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:16,520 - mani_skill  - WARNING - No initial
pose set for actor builder of l_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:16,854 - mani_skill  - WARNING - No initial
pose set for actor builder of i_shape_tool, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:17,182 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_yellow, setting to default
pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial
poses may slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_release] S1_GetL: plan=True phys=True
valid=True steps=114
[B7 seed000_release] S2_HookBlueAndDock: plan=True
```

phys=True valid=True steps=219

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300772 -0.00158694 0.172769]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.1000126   0.02803093 0.02557869]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1221   0.02853507 0.21965088]
[S3] 直杆位移: [-0.00440425 0.00061716 0.19465101]
[S3] 抬起后是否抓住: True
[B7 seed000_release] S3_GetI: plan=True phys=True valid=True steps=83
[B7 seed000_release] S4_PushGreenAndReleaseI: plan=True phys=True valid=True steps=175
[B7 seed000_release] S5_ReacquireL: plan=True phys=True valid=True steps=106
[B7 seed000_release] S6_HookYellow: plan=True phys=True valid=True steps=140
2026-10-30 22:41:19,618 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 22:41:19,943 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:20,272 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:20,604 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:20,934 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial

poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 22:41:21,268 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_south] S1_GetL: plan=True phys=True valid=True steps=114
  L工具当前位置: [-0.3203 -0.3641 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.2mm l_grasped=True
[B7 seed000_south] S2_HookBlueAndDock: plan=True phys=True valid=True steps=309

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00491277 -0.00901996 0.17214051]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00 -3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004333 0.02779158 0.02592658]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211583 0.0290028   0.21965724]
[S3] 直杆位移: [-0.00441998 0.00108611 0.19465737]
[S3] 抬起后是否抓住: True
[B7 seed000_south] S3_GetI: plan=True phys=True valid=True steps=89
[B7 seed000_south] S4_PushGreenAndReleaseI: plan=True phys=True valid=True steps=172
[B7 seed000_south] S5_ReacquireL: plan=True phys=True valid=True steps=96
[B7 seed000_south] S6_HookYellow: plan=True phys=True valid=True steps=143

```
===== B7 GOAL-CONDITIONED SIX-SKILL DECISION =====
DECISION: B7_TWO_PHYSICAL_SIX_SKILL_TRACES_SUCCEEDED
release status= FULL_SIX_SUCCESS S4_modes= 5 yaw= 20
S4_contacts= 39 physical_steps= 837 first_failure= None
south status= FULL_SIX_SUCCESS S4_modes= 5 yaw= 20
S4_contacts= 39 physical_steps= 923 first_failure= None
GOAL COSTS: {"release": {"green_only": {"valid": true,
"measured_steps": 591, "failed_or_missing_stage": null,
"not_online_prediction": true}, "green_then_yellow": {"valid":
true, "measured_steps": 837, "failed_or_missing_stage": null,
"not_online_prediction": true}}, "south": {"green_only": {"valid":
true, "measured_steps": 684, "failed_or_missing_stage": null,
"not_online_prediction": true}, "green_then_yellow": {"valid":
true, "measured_steps": 923, "failed_or_missing_stage": null,
"not_online_prediction": true}}}
RETROSPECTIVE GOAL CHOICE: {'green_only': 'release',
'green_then_yellow': 'release'} (NOT online)
OUTPUT: b7_goal_mode_20261030_224045/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/T0_B7_GoalConditioned_20261030$
```

思考了 4m 56s >

ChatGPT 说：

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你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-
benchmark/T0_B8_TransitionPlanning_20261031$ bash
RUN_B8.sh \
>
../T0_B7_GoalConditioned_20261030/b7_goal_mode_2026103
0_224045
test_failed_later_skill_does_not_invalidate_short_goal
(test_b7_goal_graph.GoalGraphTests) ... ok
test_no_fabricated_score_for_failure
(test_b7_goal_graph.GoalGraphTests) ... ok
test_prefix_is_four_skills (test_b7_goal_graph.GoalGraphTests)
... ok
test_retrospective_conditioning_can_flip
(test_b7_goal_graph.GoalGraphTests) ... ok
```

```
test_steps_must_be_int (test_b7_goal_graph.GoalGraphTests) ...
ok
test_unexecuted_is_not_free
(test_b7_goal_graph.GoalGraphTests) ... ok
test_unknown_goal_rejected
(test_b7_goal_graph.GoalGraphTests) ... ok
test_both_fail_fallback_home (test_b8_transition.Tests) ... ok
test_bounded_candidate_count (test_b8_transition.Tests) ... ok
test_candidate_with_bad_pose_falls_back
(test_b8_transition.Tests) ... ok
test_flags_no_attach_no_cloud (test_b8_transition.Tests) ... ok
test_goal_ends_at_four_no_planning (test_b8_transition.Tests)
... ok
test_no_qpos_no_planning (test_b8_transition.Tests) ... ok
test_not_enough_budget (test_b8_transition.Tests) ... ok
test_safety_preconditions (test_b8_transition.Tests) ... ok
test_screw_failure_rrt_success (test_b8_transition.Tests) ... ok
test_script_refs_two_goals_w1_stages (test_b8_transition.Tests)
... ok
test_succeeds_bounded_direct (test_b8_transition.Tests) ... ok
test_policy_field_matches_w1_evaluation_contract
(test_s2_options_regression.S2IntegrationRegression) ... ok
test_w1_s2_evaluator_both_real_b7_option_objects
(test_s2_options_regression.S2IntegrationRegression) ... ok

----------------------------------------------------------------------
-
Ran 20 tests in 0.002s

OK
2026-10-30 23:03:48,443 - mani_skill  - WARNING - No initial
pose set for articulation builder of panda, setting to default
pose q=[1,0,0,0], p=[0,0,0]. There may be simulation
issues/bugs if this articulation at it's initial pose collides with
other objects at their initial poses.
2026-10-30 23:03:48,720 - mani_skill  - WARNING - No initial
pose set for actor builder of cube, setting to default pose q=
[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:49,065 - mani_skill  - WARNING - No initial
pose set for actor builder of cube_2, setting to default pose
q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may
slow down simulation, see
https://github.com/haosulab/ManiSkill/issues/421.
```

2026-10-30 23:03:49,411 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:49,769 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:50,108 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_release] S1_GetL: plan=True phys=True valid=True steps=114
[B7 seed000_release] S2_HookBlueAndDock: plan=True phys=True valid=True steps=219

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769575 0.02791791 0.02499987]
[S3] TCP 初始位置: [-0.01300772 -0.00158694 0.172769]
[S3] 目标抓取位置: [-0.09769575 0.0279178   0.02499987]
[S3] 目标抓取四元数: [0.0000e+00 1.0000e+00 -2.6579e-06 0.0000e+00]
[S3] 到达后 TCP: [-0.1000126   0.02803093 0.02557869]
[S3] 到达后直杆: [-0.11769579 0.02791731 0.02499984]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.1221   0.02853507 0.21965088]
[S3] 直杆位移: [-0.00440425 0.00061716 0.19465101]
[S3] 抬起后是否抓住: True
[B7 seed000_release] S3_GetI: plan=True phys=True valid=True steps=83
[B7 seed000_release] S4_PushGreenAndReleaseI: plan=True phys=True valid=True steps=129
[B7 seed000_release] S5_ReacquireL: plan=True phys=True valid=True steps=133
[B7 seed000_release] S6_HookYellow: plan=True phys=True valid=True steps=138
2026-10-30 23:03:52,458 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation

issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 23:03:52,791 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:53,128 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:53,459 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:53,796 - mani_skill - WARNING - No initial pose set for actor builder of i_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:03:54,128 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.2849])
[B7 seed000_south] S1_GetL: plan=True phys=True valid=True steps=114
  L工具当前位置: [-0.3203 -0.3641 0.0253]
  目标停放XY: [-0.2 -0.12]
[PARK GUARD] S2/parking lift: planned_steps=32
[PARK GUARD] S2/parking lift: tcp_error=6.9mm l_grasped=True
[PARK GUARD] S2/parking transit: planned_steps=36
[PARK GUARD] S2/parking transit: tcp_error=13.3mm l_grasped=True
[PARK GUARD] S2/parking lower: planned_steps=33
[PARK GUARD] S2/parking lower: tcp_error=7.2mm l_grasped=True
[B7 seed000_south] S2_HookBlueAndDock: plan=True phys=True valid=True steps=309

[S3] ===== 抓取诊断 =====
[S3] 直杆初始位置: [-0.11769585 0.02791669 0.02499987]
[S3] TCP 初始位置: [-0.00491277 -0.00901996 0.17214051]
[S3] 目标抓取位置: [-0.09769584 0.02791655 0.02499987]
[S3] 目标抓取四元数: [0.0000000e+00 1.0000000e+00
-3.4568034e-06 0.0000000e+00]
[S3] 到达后 TCP: [-0.10004333 0.02779158 0.02592658]
[S3] 到达后直杆: [-0.11769594 0.0279155   0.02499986]
[S3] 闭合夹爪后是否抓住: True
[S3] 抬起后直杆: [-0.12211583 0.0290028   0.21965724]
[S3] 直杆位移: [-0.00441998 0.00108611 0.19465737]
[S3] 抬起后是否抓住: True
[B7 seed000_south] S3_GetI: plan=True phys=True valid=True
steps=89
[B7 seed000_south] S4_PushGreenAndReleaseI: plan=True
phys=True valid=True steps=127
[B7 seed000_south] S5_ReacquireL: plan=True phys=True
valid=True steps=120
[B7 seed000_south] S6_HookYellow: plan=True phys=True
valid=True steps=142
===== B8 DIRECT TRANSITION DECISION =====
DECISION: B8_TWO_PHYSICAL_SIX_SKILL_TRACES_SUCCEEDED
release status= FULL_SIX_SUCCESS S4_modes= 5 yaw= 20
S4_contacts= 39 transition= direct_to_s5 physical_steps= 816
first_failure= None
B7 COMPARISON: release {'b7_steps': 837, 'b8_steps': 816,
'comparable': True, 'same_prefix_s1_s3_steps': True,
'physical_both_success': True, 'saved_steps': 21}
south status= FULL_SIX_SUCCESS S4_modes= 5 yaw= 20
S4_contacts= 39 transition= direct_to_s5 physical_steps= 901
first_failure= None
B7 COMPARISON: south {'b7_steps': 923, 'b8_steps': 901,
'comparable': True, 'same_prefix_s1_s3_steps': True,
'physical_both_success': True, 'saved_steps': 22}
GOAL COSTS: {"release": {"green_only": {"valid": true,
"measured_steps": 545, "failed_or_missing_stage": null,
"not_online_prediction": true}, "green_then_yellow": {"valid":
true, "measured_steps": 816, "failed_or_missing_stage": null,
"not_online_prediction": true}}, "south": {"green_only": {"valid":
true, "measured_steps": 639, "failed_or_missing_stage": null,
"not_online_prediction": true}, "green_then_yellow": {"valid":
true, "measured_steps": 901, "failed_or_missing_stage": null,
"not_online_prediction": true}}}
RETROSPECTIVE GOAL CHOICE: {'green_only': 'release',
'green_then_yellow': 'release'} (NOT online)

OUTPUT: b8_transition_20261030_230319/decision.json
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/T0_B8_TransitionPlanning_20261031$

思考了 2m 52s >

ChatGPT 说：

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你说：

不是，你的创新是什么呢，是Learning方法，感觉速度也很快也不像，另外我就算都按照你的要求做完，最后得到什么，怎么整合，怎么和其他算法对比，你目前是在做什么

思考了 1m 19s >

ChatGPT 说：

你说：

好的

思考了 6m 31s >

ChatGPT 说：

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你说：

ble initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:26:47,431 - mani_skill - WARNING - No initial pose set for actor builder of cube_yellow, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
tensor([0.3169])
tensor([0.3005])

pybind11::handle::inc_ref() is being called while the GIL is either not held or invalid. Please see https://pybind11.readthedocs.io/en/stable/advanced/misc.html#common-sources-of-global-interpreter-lock-errors for debugging advice.
If you are convinced there is no bug in your code, you can #define PYBIND11_NO_ASSERT_GIL_HELD_INCREF_DECREFto disable this check. In that case you have to ensure this #define is consistently used for all translation units linked into a given pybind11 extension, otherwise there will be ODR violations.The failing pybind11::handle::inc_ref() call was triggered on a dict object.
Traceback (most recent call last):
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/SkillLearning_Stage1_20261031/scripts/collect_s4.py", line 109, in collect_one

planner=PandaArmMotionPlanningSolver(env,debug=False,vis=False,
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/panda/motionplanner.py", line 25, in __init__
    super().__init__(env, debug, vis, base_pose, visualize_target_grasp_pose, print_env_info, joint_vel_limits, joint_acc_limits)
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/two_finger_gripper/motionplanner.py", line 26, in __init__
    super().__init__(env, debug, vis, base_pose, print_env_info, joint_vel_limits, joint_acc_limits)
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/base_motionplanner/motionplanner.py", line 32, in __init__
    self.planner = self.setup_planner()
  File "/home/users/lemon/skill_future_viability/skill-composition-benchmark/mani_skill/examples/motionplanning/base_motionplanner/motionplanner.py", line 59, in setup_planner
    planner = mplib.Planner(
  File "/home/users/lemon/miniconda3/envs/skillcompose/lib/pyth

on3.10/site-packages/mplib/planner.py", line 65, in __init__
    self.robot = ArticulatedModel(
RuntimeError: pybind11::handle::inc_ref() PyGILState_Check() failure.
S4_TEACHER seed=35 policy=release split=train accepted=False yaw=None samples=0 reason=ERROR:RuntimeError:pybind11::handle::inc_ref() PyGILState_Check() failure.
2026-10-30 23:26:47,644 - mani_skill - WARNING - No initial pose set for articulation builder of panda, setting to default pose q=[1,0,0,0], p=[0,0,0]. There may be simulation issues/bugs if this articulation at it's initial pose collides with other objects at their initial poses.
2026-10-30 23:26:47,930 - mani_skill - WARNING - No initial pose set for actor builder of cube, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:26:48,235 - mani_skill - WARNING - No initial pose set for actor builder of cube_2, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
2026-10-30 23:26:48,535 - mani_skill - WARNING - No initial pose set for actor builder of l_shape_tool, setting to default pose q=[1,0,0,0], p=[0,0,0]. Not setting reasonable initial poses may slow down simulation, see https://github.com/haosulab/ManiSkill/issues/421.
RUN_STAGE1.sh：行 13: 3221536 段错误   （核心已转储） python -u scripts/collect_s4.py --out "$TAG/dataset" --seed-start 0 --num-seeds "$NUM_SEEDS" --policy release --mode-policy cycle

思考了 3m 20s >

ChatGPT 说：

README_FIX.md
文档

你说：

你的创新在哪里呢

思考了 41s >

ChatGPT 说：

你说：

```
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-
composition-benchmark/SkillLearning_Stage1_20261031$
bash RUN_STAGE1_ISOLATED.sh s4_stage1_20261030_232415
test_bad_phase (test_stage1.TestDataset) ... ok
test_partitions (test_stage1.TestDataset) ... ok
test_training_end_to_end_cpu (test_stage1.TestDataset) ... ok
test_goal_conditioned (test_stage1.TestFeature) ... ok
test_invalid (test_stage1.TestFeature) ... ok
test_options_contract (test_stage1.TestFeature) ... ok
test_seed_grouped (test_stage1.TestFeature) ... ok
test_valid (test_stage1.TestFeature) ... ok
test_heldout_requires_seed4
(test_stage1.TestPhysicalEvalContracts) ... ok
test_model_inference_bounds
(test_stage1.TestPhysicalEvalContracts) ... ok
test_completed_dataset_summary_does_not_fabricate
(test_stage1_isolation.TestIsolationContract) ... ok
test_crash_is_not_silent_success
(test_stage1_isolation.TestIsolationContract) ... ok
test_created_manifest_matches_original_contract
(test_stage1_isolation.TestIsolationContract) ... ok
test_one_worker_is_one_seed_not_loop
(test_stage1_isolation.TestIsolationContract) ... ok
test_original_collector_unmodified
(test_stage1_isolation.TestIsolationContract) ... ok
test_orphan_preserved_not_overwritten
(test_stage1_isolation.TestIsolationContract) ...
QUARANTINED_ORPHAN:
/tmp/tmpe5wlctpg/dataset/episodes/seed0000_release.npz -
>
/tmp/tmpe5wlctpg/dataset/quarantined_orphans/seed0000_r
elease_1793417592098767542.npz
ok
test_resume_completed_skips_and_retries_previous_error
(test_stage1_isolation.TestIsolationContract) ... ok
test_resume_same_source_only
(test_stage1_isolation.TestIsolationContract) ... ok
test_worker_response_and_one_invocation
(test_stage1_isolation.TestIsolationContract) ... ok
test_resumes_seed35_instead_of_repeating_all_40
```

(test_stage1_isolation.TestResumeWorkflow) ...
SKIP_RECORDED seed=0 policy=release accepted=False
SKIP_RECORDED seed=1 policy=release accepted=False
SKIP_RECORDED seed=2 policy=release accepted=False
SKIP_RECORDED seed=3 policy=release accepted=False
SKIP_RECORDED seed=4 policy=release accepted=False
SKIP_RECORDED seed=5 policy=release accepted=False
SKIP_RECORDED seed=6 policy=release accepted=False
SKIP_RECORDED seed=7 policy=release accepted=False
SKIP_RECORDED seed=8 policy=release accepted=False
SKIP_RECORDED seed=9 policy=release accepted=False
SKIP_RECORDED seed=10 policy=release accepted=False
SKIP_RECORDED seed=11 policy=release accepted=False
SKIP_RECORDED seed=12 policy=release accepted=False
SKIP_RECORDED seed=13 policy=release accepted=False
SKIP_RECORDED seed=14 policy=release accepted=False
SKIP_RECORDED seed=15 policy=release accepted=False
SKIP_RECORDED seed=16 policy=release accepted=False
SKIP_RECORDED seed=17 policy=release accepted=False
SKIP_RECORDED seed=18 policy=release accepted=False
SKIP_RECORDED seed=19 policy=release accepted=False
SKIP_RECORDED seed=20 policy=release accepted=False
SKIP_RECORDED seed=21 policy=release accepted=False
SKIP_RECORDED seed=22 policy=release accepted=False
SKIP_RECORDED seed=23 policy=release accepted=False
SKIP_RECORDED seed=24 policy=release accepted=False
SKIP_RECORDED seed=25 policy=release accepted=False
SKIP_RECORDED seed=26 policy=release accepted=False
SKIP_RECORDED seed=27 policy=release accepted=False
SKIP_RECORDED seed=28 policy=release accepted=False
SKIP_RECORDED seed=29 policy=release accepted=False
SKIP_RECORDED seed=30 policy=release accepted=False
SKIP_RECORDED seed=31 policy=release accepted=False
SKIP_RECORDED seed=32 policy=release accepted=False
SKIP_RECORDED seed=33 policy=release accepted=False
SKIP_RECORDED seed=34 policy=release accepted=False
RETRY_PREVIOUS_ERROR seed=35 policy=release
S4_TEACHER seed=35 policy=release split=train
accepted=False yaw=None samples=0
reason=INVALID_EARLY_S2
S4_TEACHER seed=36 policy=release split=train
accepted=False yaw=None samples=0
reason=INVALID_EARLY_S2
S4_TEACHER seed=37 policy=release split=train
accepted=False yaw=None samples=0

```
reason=INVALID_EARLY_S2
S4_TEACHER seed=38 policy=release split=val accepted=False
yaw=None samples=0 reason=INVALID_EARLY_S2
S4_TEACHER seed=39 policy=release split=test
accepted=False yaw=None samples=0
reason=INVALID_EARLY_S2
===== S4 DATA COLLECTION ===== {"accepted": 0,
"accepted_steps": 0, "attempts": 40, "by_split": {"test": 0,
"train": 0, "val": 0}, "collection_complete": true, "isolation": "one
physical episode per fresh Python subprocess",
"num_required_episodes": 40, "success_filters_enforced": true,
"training_ready": false, "unfinished": 0}
ok

----------------------------------------------------------------------
-
Ran 20 tests in 0.510s

OK
SKIP_RECORDED seed=0 policy=release accepted=True
SKIP_RECORDED seed=1 policy=release accepted=True
SKIP_RECORDED seed=2 policy=release accepted=True
SKIP_RECORDED seed=3 policy=release accepted=True
SKIP_RECORDED seed=4 policy=release accepted=True
SKIP_RECORDED seed=5 policy=release accepted=False
SKIP_RECORDED seed=6 policy=release accepted=True
SKIP_RECORDED seed=7 policy=release accepted=True
SKIP_RECORDED seed=8 policy=release accepted=True
SKIP_RECORDED seed=9 policy=release accepted=True
SKIP_RECORDED seed=10 policy=release accepted=True
SKIP_RECORDED seed=11 policy=release accepted=True
SKIP_RECORDED seed=12 policy=release accepted=True
SKIP_RECORDED seed=13 policy=release accepted=True
SKIP_RECORDED seed=14 policy=release accepted=True
RETRY_PREVIOUS_ERROR seed=15 policy=release
S4_TEACHER seed=15 policy=release split=train
accepted=False yaw=None samples=0
reason=ERROR:ContactPlannerError:EXECUTION_REPLAN_FAIL
ED_push
===== S4 DATA COLLECTION ===== {"accepted": 32,
"accepted_steps": 4063, "attempts": 36, "by_split": {"test": 7,
"train": 19, "val": 6}, "collection_complete": false, "isolation":
"one physical episode per fresh Python subprocess",
"num_required_episodes": 40, "success_filters_enforced": true,
"training_ready": false, "unfinished": 4}
```

FATAL_ISOLATED_COLLECTION: teacher raised exception: ERROR:ContactPlannerError:EXECUTION_REPLAN_FAILED_push; log=/home/users/lemon/skill_future_viability/skill-composition-benchmark/SkillLearning_Stage1_20261031/s4_stage1_20261030_232415/dataset/worker_logs/seed0015_release_1793417592534137484.log; collection stopped, safe to --resume
(skillcompose) lemon@xlb-MS:~/skill_future_viability/skill-composition-benchmark/SkillLearning_Stage1_20261031$

思考了 2m 30s >

ChatGPT 说：

README_TEACHER_REJECT_FIX.md
文档

你说：

对啊，之前让你思考的就是创新，怎么创新呢，你可以思考10min都行，给出确切答复，至少5个创新点，还有对比的baseline算法，我是要发TRO的

思考了 1m 33s >

ChatGPT 说：

你说：

那现在到底怎么做才能发TRO，你洋洋洒洒回复这么多

ChatGPT 说：

你说：

你这样，你先分析最近三年相关在TRO上的论文，看看他们都是怎么做的，拟定出实验的基本思路和实验需求，尤其是数据集和对比的baseline一定要选择好

这次我会把范围严格限定为 **2023–2026 年 IEEE Transactions on Robotics（T-RO）正式发表的相关论文**，再单独列出虽然不是 T-RO、但必须比较的方法论文。

重点核对每篇论文的实验设计、公开数据/代码、任务长度、基线和工作量，最终给出一套适合你现有 ManiSkill/Panda 平台、单卡 4090 条件的具体实验方案。尤其会避免把“在 CoRL/ICRA 发表”误写成 T-RO，也不会为了凑 Baseline 选择根本无法公平复现的模型。

ChatGPT 说：